University of New Haven

TAGLIATELA COLLEGE OF ENGINEERING

Mechanical Engineering

Computer Aided Engineering (MECH-6627-01)

Simulation of CubeSat Electronics Assembly

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Date – 15th December 2025

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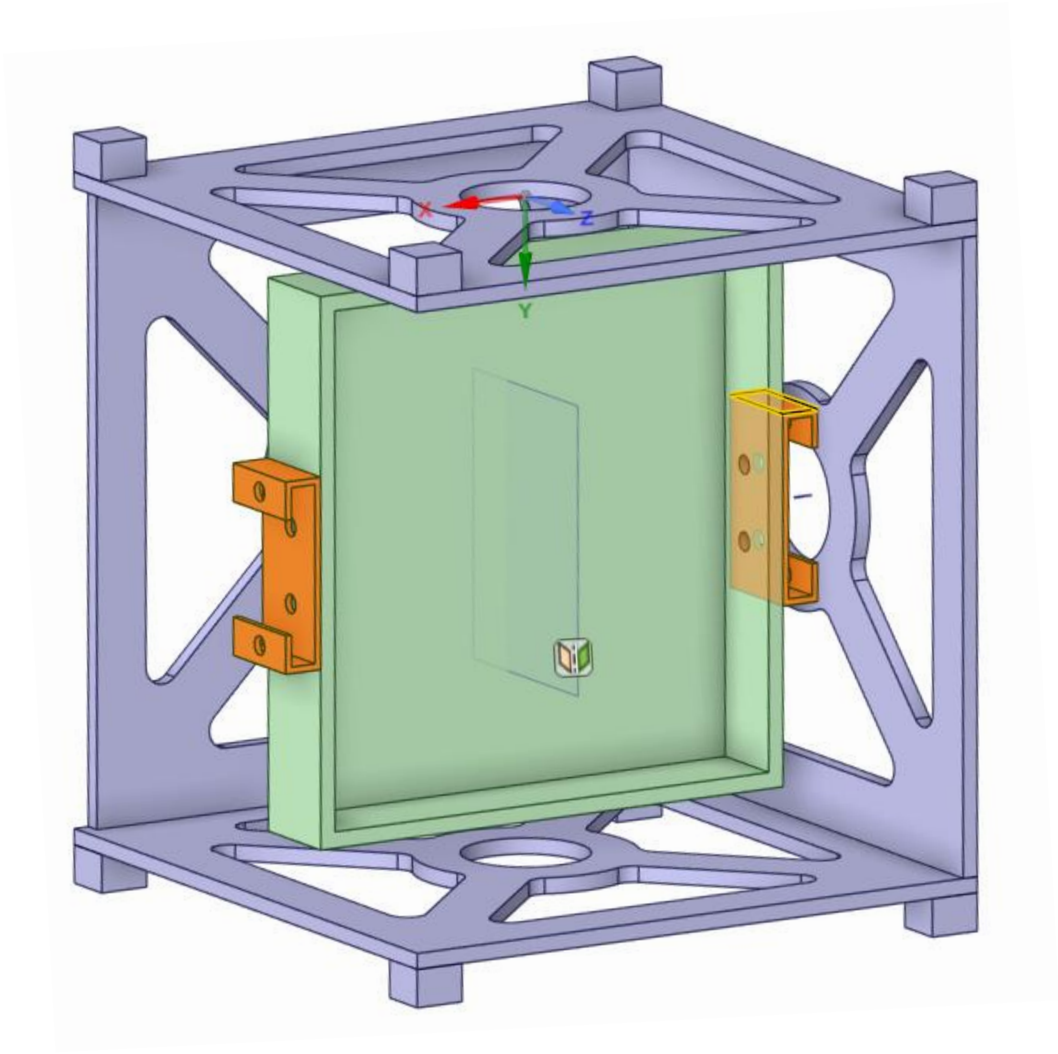
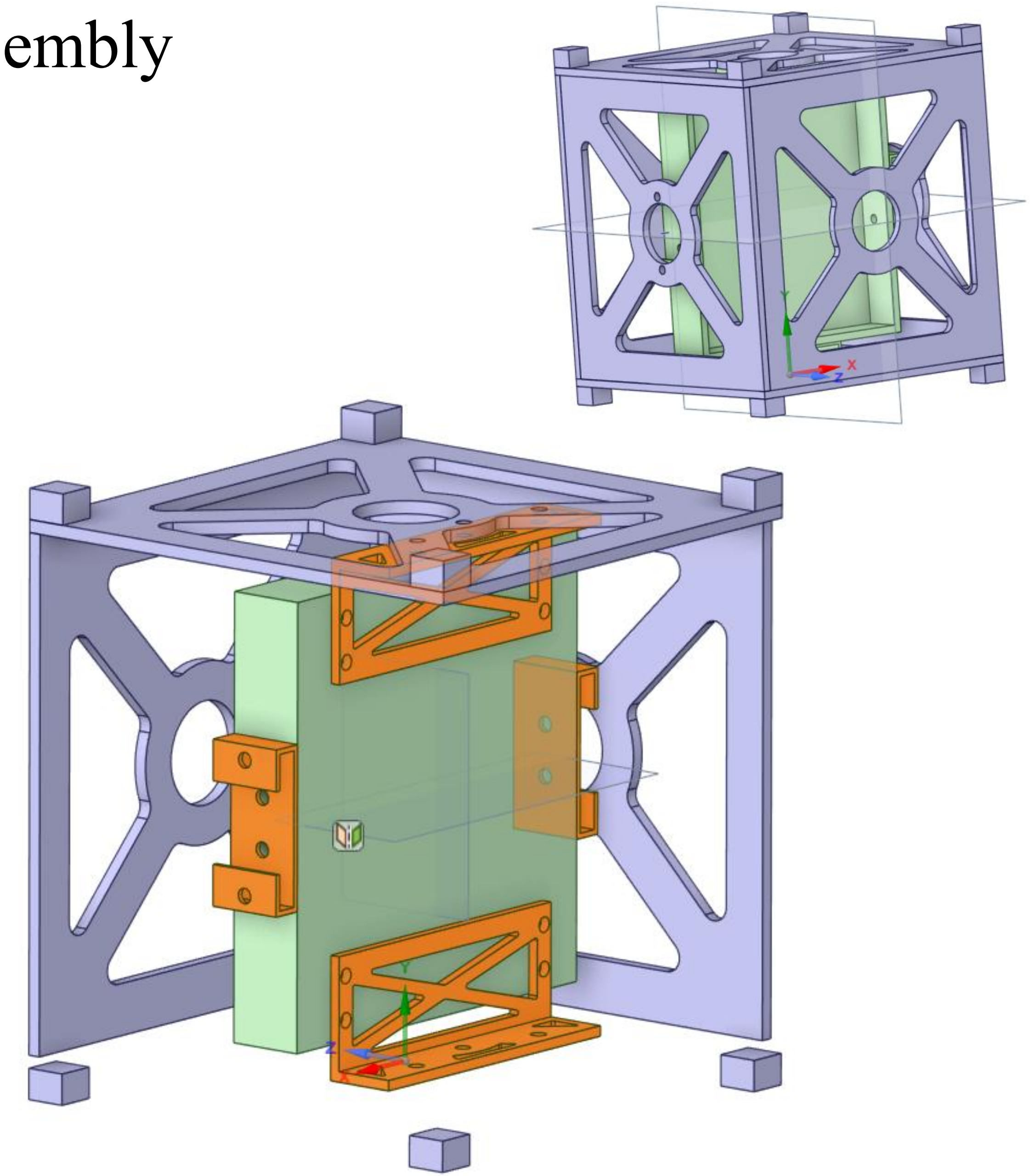


Iteration 1



Iteration 2

EXECUTIVE SUMMARY

Problem Statement & Goals

This project addresses the design and structural analysis of a custom mounting bracket assembly used to secure an electronics box within a CubeSat satellite frame. The primary goal is to develop a lightweight, high-strength bracket that ensures structural integrity during launch, where the assembly is subjected to high-g acceleration loads. Safety is the highest priority, followed by minimizing mass, while ensuring all natural frequencies remain above 65 Hz to avoid resonance with launch vehicle vibrations.

Part Description & Service Use

The bracket is a newly designed aerospace component that interfaces between the existing CubeSat and the electronics enclosure. In service, the bracket will be mounted internally within the CubeSat structure and will secure sensitive electronic payloads during spaceflight. The part is subjected to dynamic launch loads across all axes and must maintain alignment and contact under vibration. The size of the bracket for **Iteration 1** is shown in (Fig.1 - 5) and it attaches to both the CubeSat side panels and the electronics box using fastened connections.

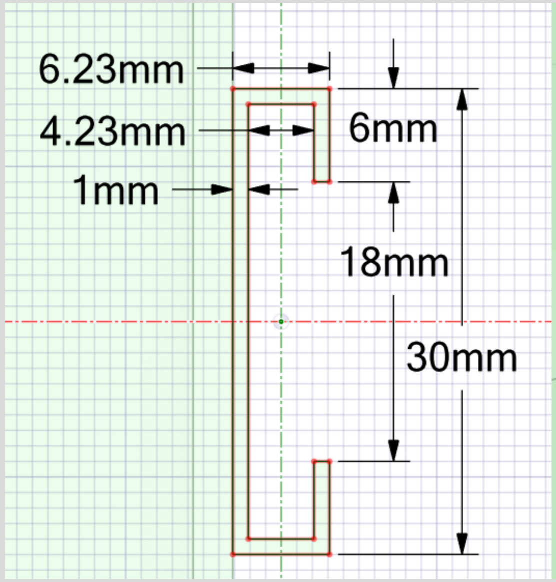


Fig.1

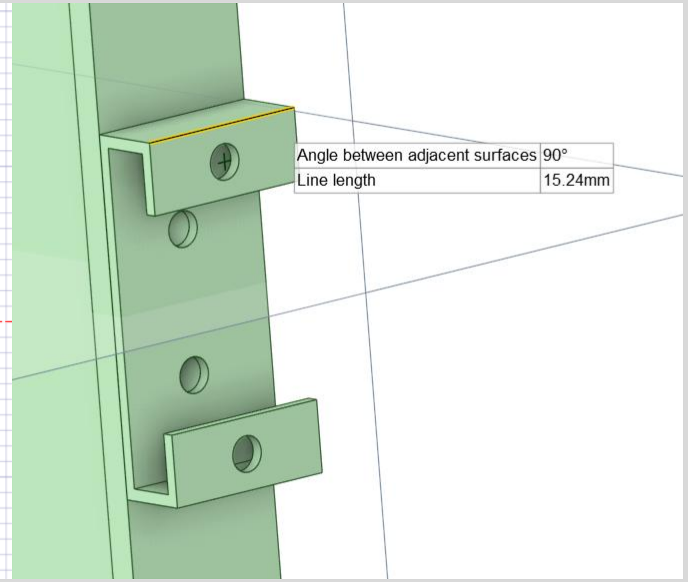


Fig.2

PART DIMENSIONS- ITERATION 1

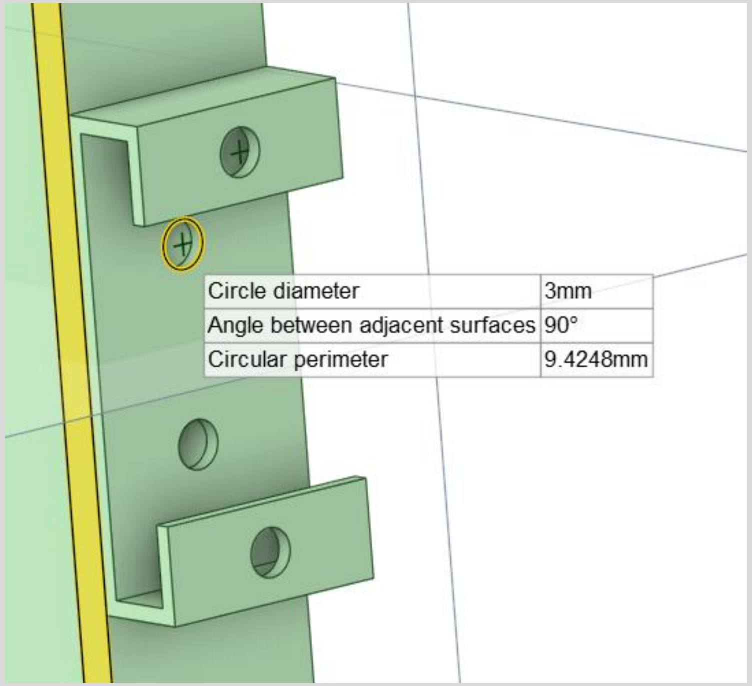


Fig.3

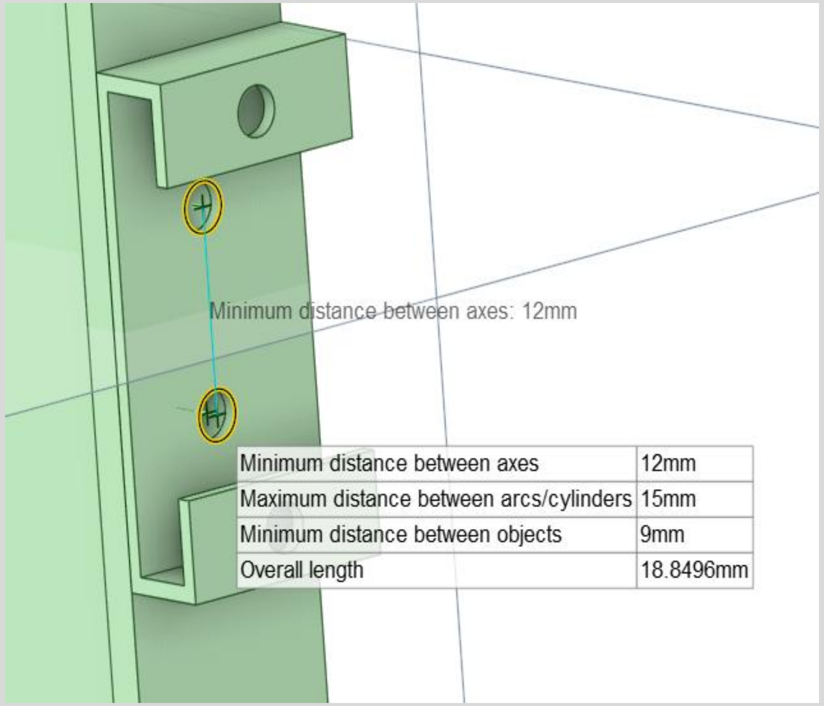


Fig.4

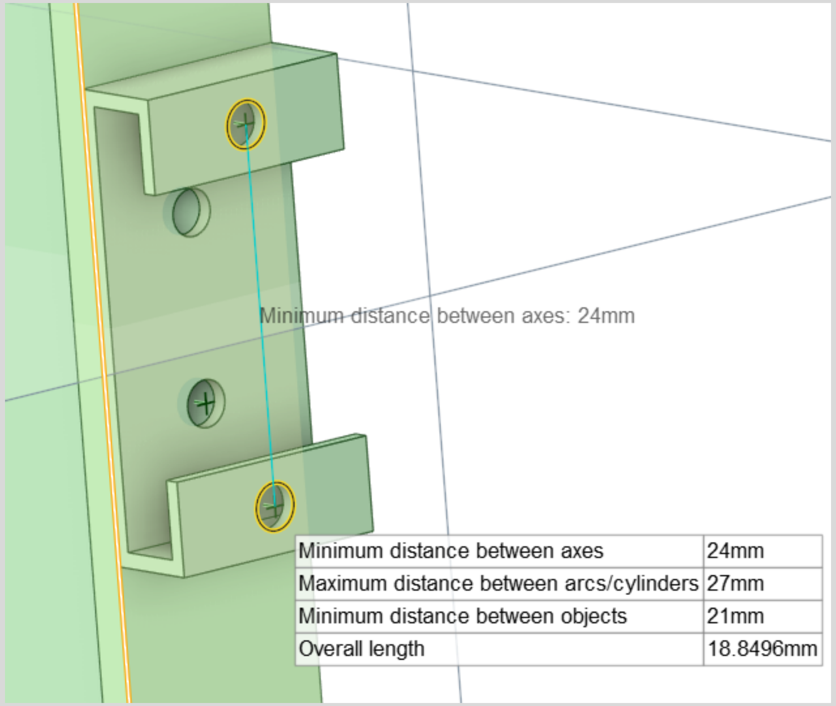


Fig.5

Model Creation & Materials

A parametric 3D CAD model of the bracket was created in **Ansys Space-Claim**, allowing for dimensional adjustments through two key design parameters. The model was imported into Ansys Workbench for finite element analysis. The primary material selected for the final design iteration was **Titanium Ti-6Al-4V for Iteration 1**, chosen for its high strength-to-weight ratio, yield strength of 880 MPa, and suitability for aerospace applications. Material properties were sourced from Mat-Web to ensure accuracy.

ENGINEERING DATA - ITERATION 1

4	Structural Steel	G	Fatigue Data at zero mean stress comes from 1998 ASME BPV Code, Section 8, Div 2, Table 5-110.1
5	Ti-6Al-4V	C	
6	Titanium Alloy	G	

Properties of Outline Row 5: Ti-6Al-4V

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	4.43E-06	kg mm^-3		
4	Isotropic Secant Coefficient of Thermal Expansion				
5	Coefficient of Thermal Expansion	8.6E-06	C^-1		
6	Isotropic Elasticity				
7	Derive from	Young's Modulus an...			
8	Young's Modulus	1.138E+05	MPa		
9	Poisson's Ratio	0.34			
10	Bulk Modulus	1.1854E+05	MPa		
11	Shear Modulus	42463	MPa		
12	Tensile Yield Strength	880	MPa		
13	Compressive Yield Strength	880	MPa		
14	Tensile Ultimate Strength	970	MPa		
15	Compressive Ultimate Strength	0	MPa		

ANALYSES PERFORMED - ITERATION 1

The following analyses were conducted :

1. Nonlinear Static Structural Analysis with frictional contact ($\mu = 0.2$) and large deflection effects.
2. Three Acceleration Load Cases: 10g in the X, Y, and Z directions.
3. Modal Analysis to determine natural frequencies and mode shapes.
4.Two Design Iterations: In this first **- Iteration** using Titanium Ti-6Al-4V.

Results & Conclusions

The final titanium bracket design successfully met all safety and performance criteria:

•**Stress margins** remained positive under all load cases, with maximum stresses well below the yield strength of Ti-6Al-4V.

•**Mass reduction** was achieved compared to the aluminum baseline.

•**Natural frequencies** all exceeded 65 Hz, avoiding potential resonance during launch.

•**Model validation** was confirmed through reaction force summation and mesh convergence studies.

MESHING DESCRIPTION - ITERATION 1

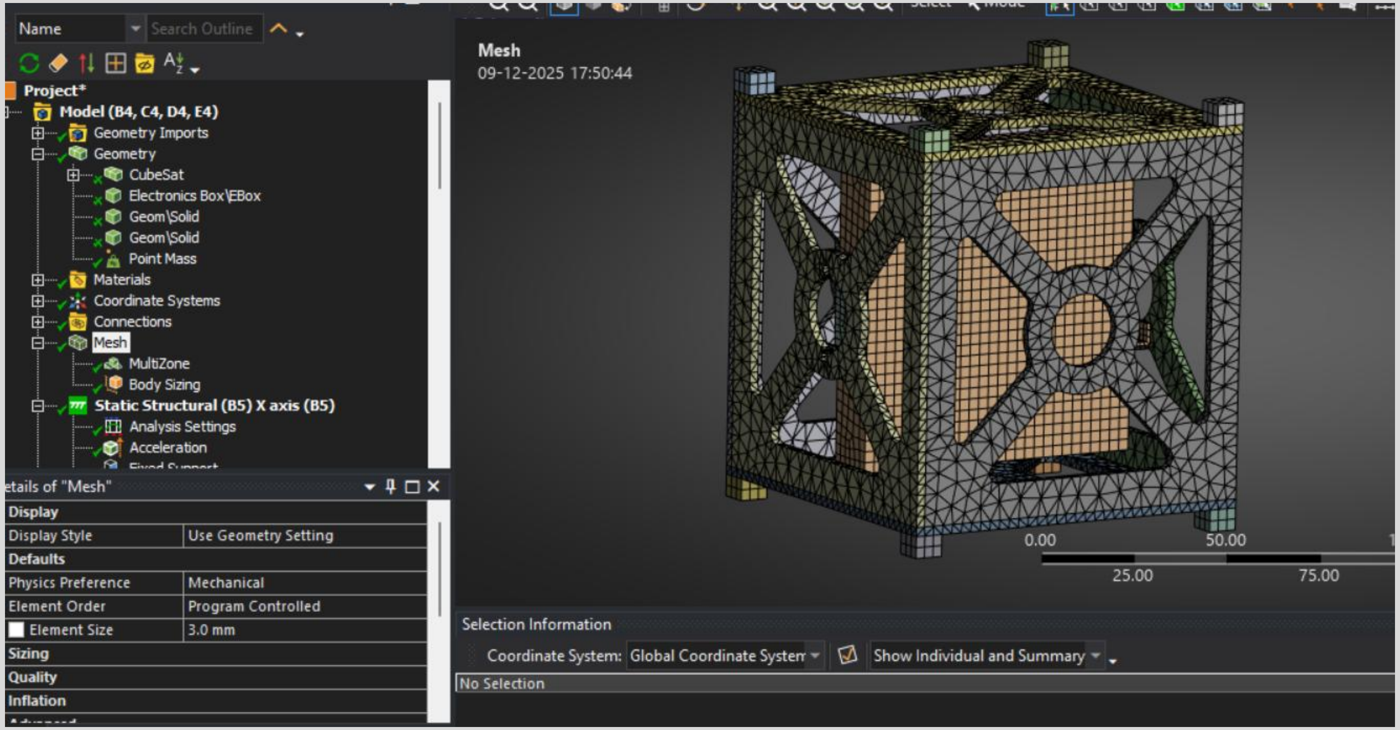
•**Method:** Multi Zone hexahedral mesh for structured elements.

•**Mesh Sizing used at brackets:** 2.0 mm element size

•**Element Size:** 3.0 mm global sizing, with local refinement near holes and fillets.

•**Total Elements:** ~ 16956

•**Total Nodes:** ~ 51679



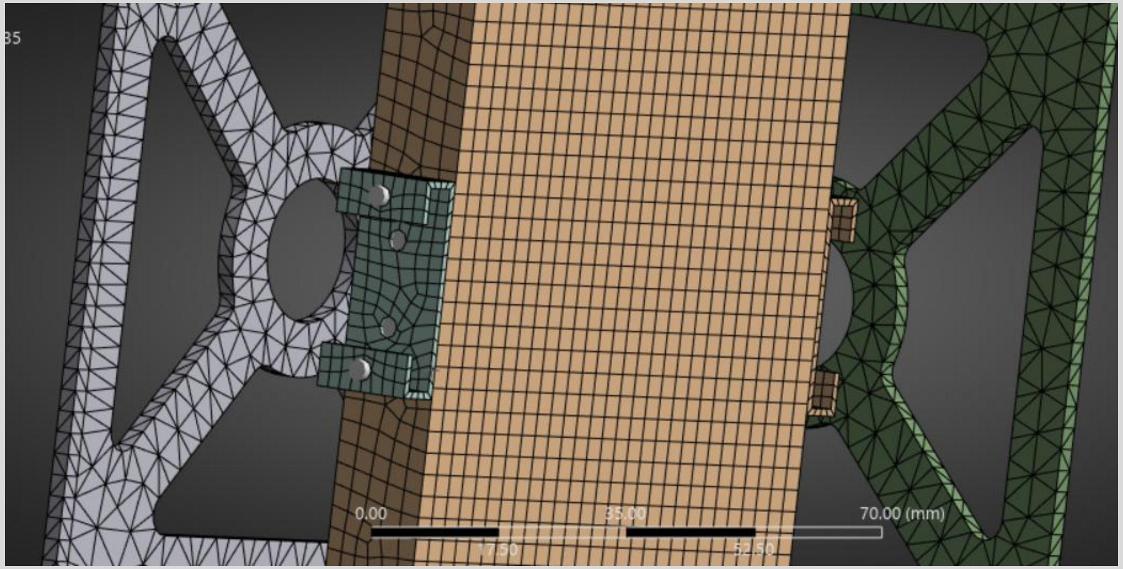


Fig.6 Mesh Details

BOUNDARY CONDITIONS - ITERATION 1

- **Point Mass:** Mass of 3 kg, attached to the electronics box surface. The mass position in central to the box and 10mm off the inside surface.

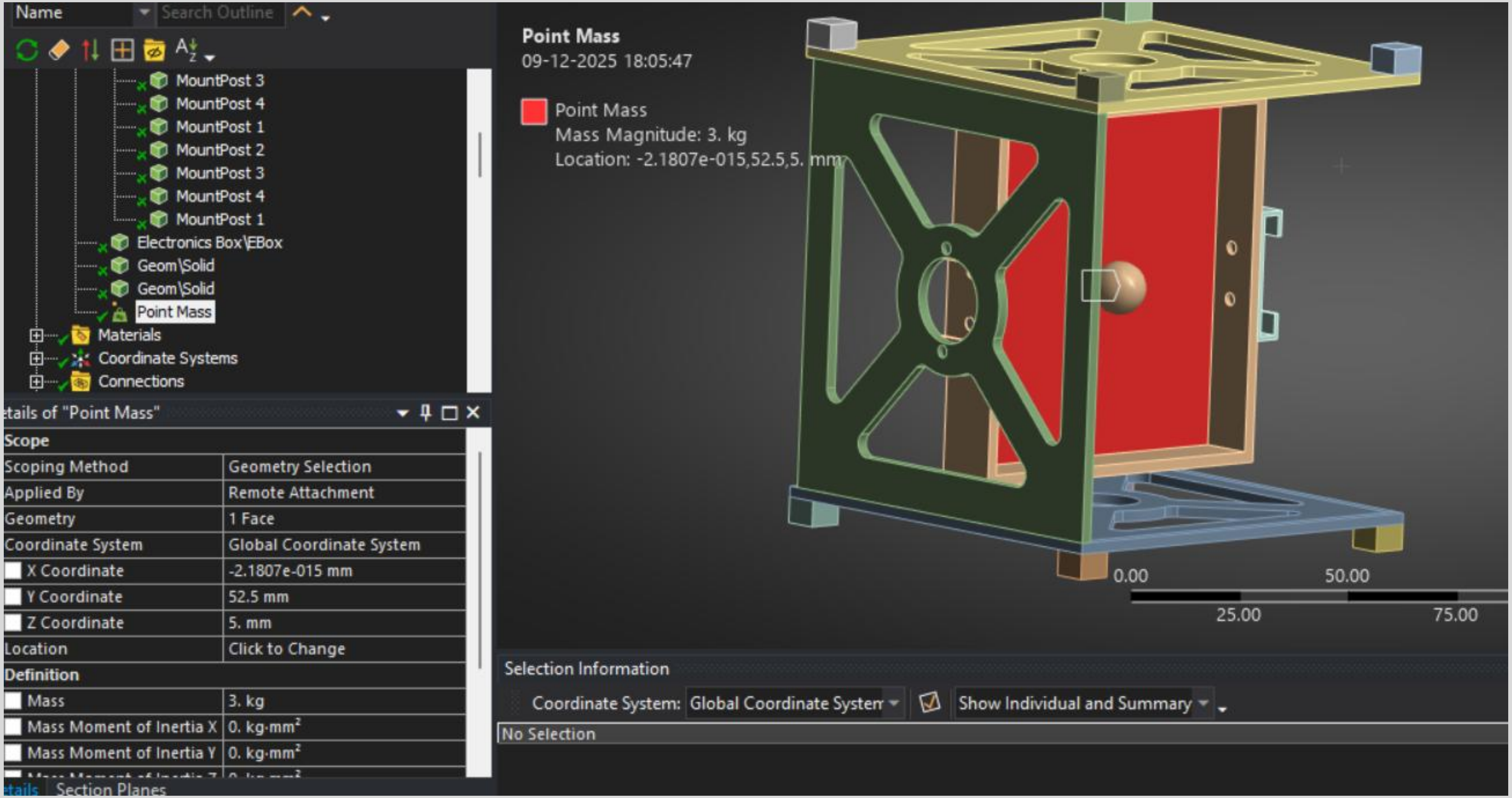


Fig.7 Point Mass

CONNECTIONS AND CONTACT - ITERATION 1

- **Frictional Contact:** Coefficient = 0.2 between all contacting surfaces.

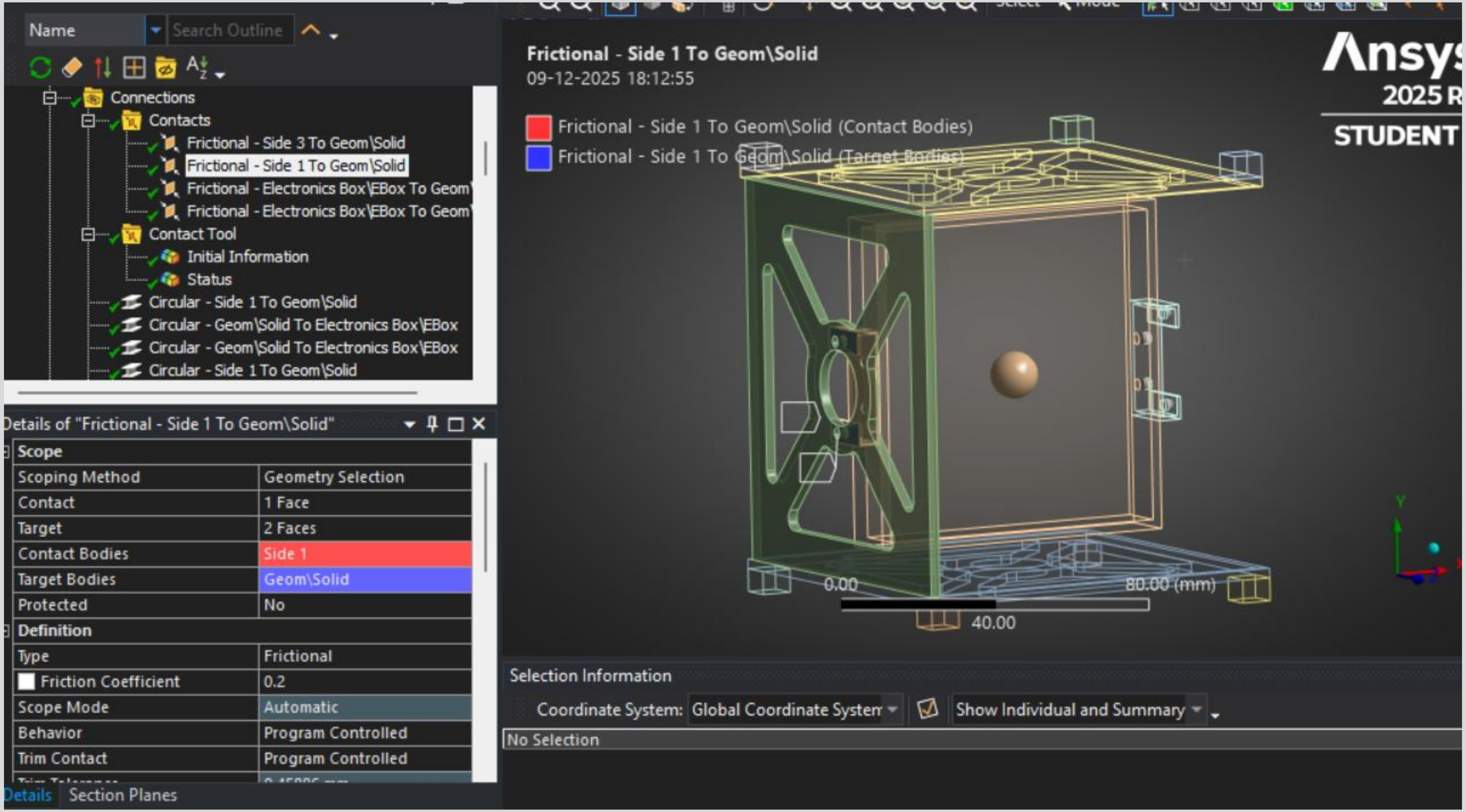


Fig.8 Contact Details

CONNECTIONS AND CONTACT - ITERATION 1

•**Beam body to body Connections:** Used for all fastener representations of 1.5mm Radius and 1.75 mm length of beam.

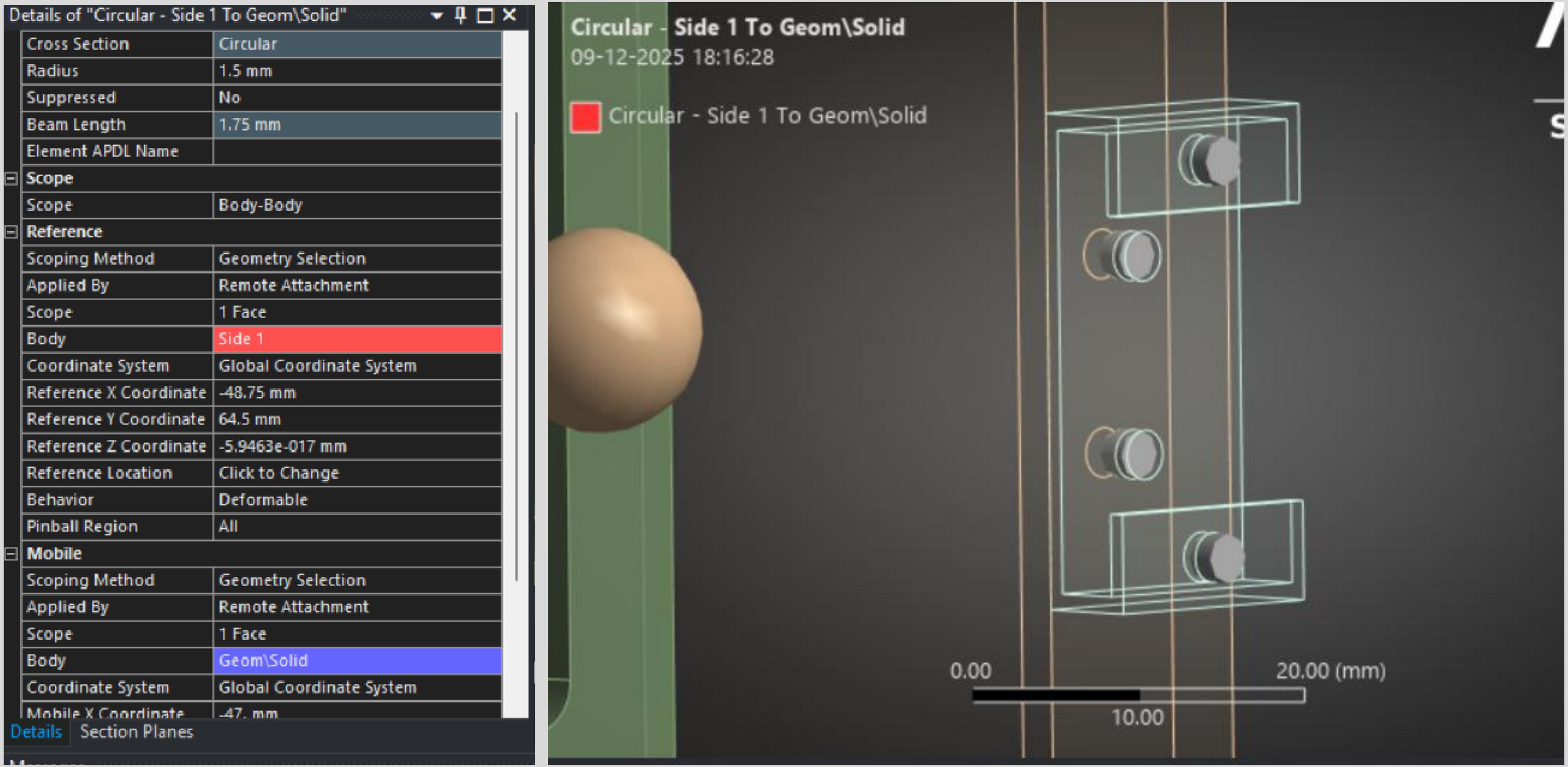


Fig.9 Beam Connection

CONTACT TOOL - ITERATION 1

- **Analysis Type:** Nonlinear with large deflection enabled.

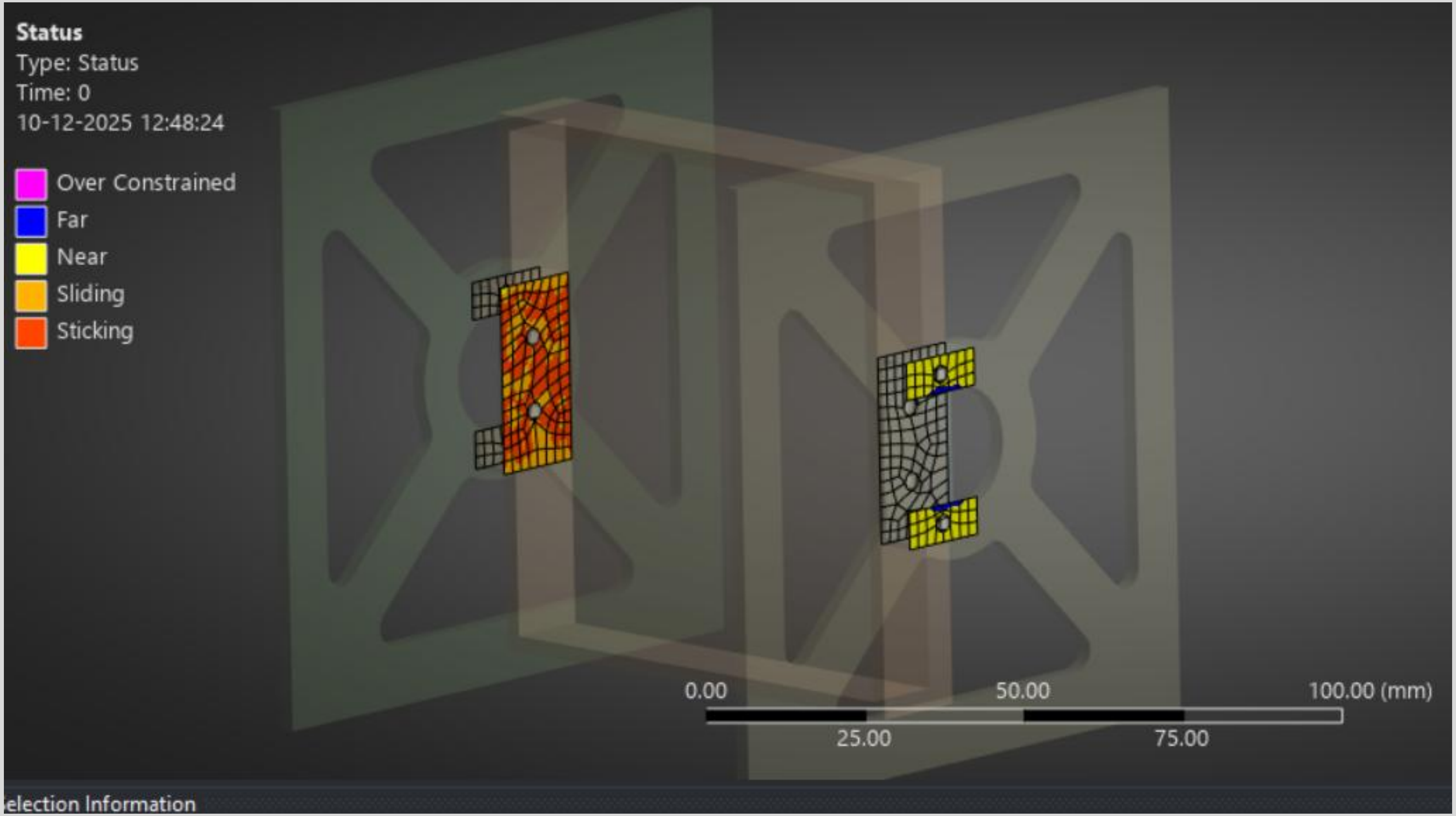
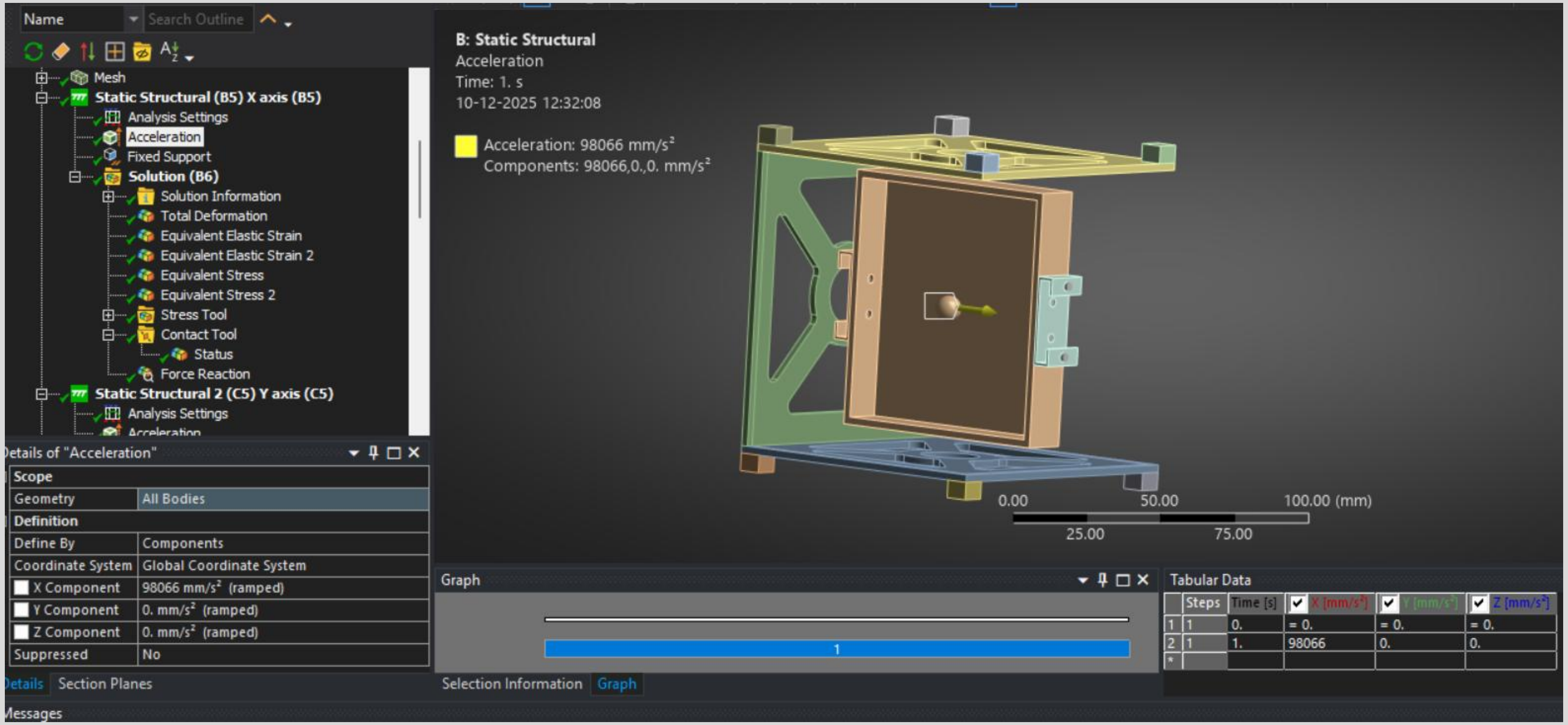
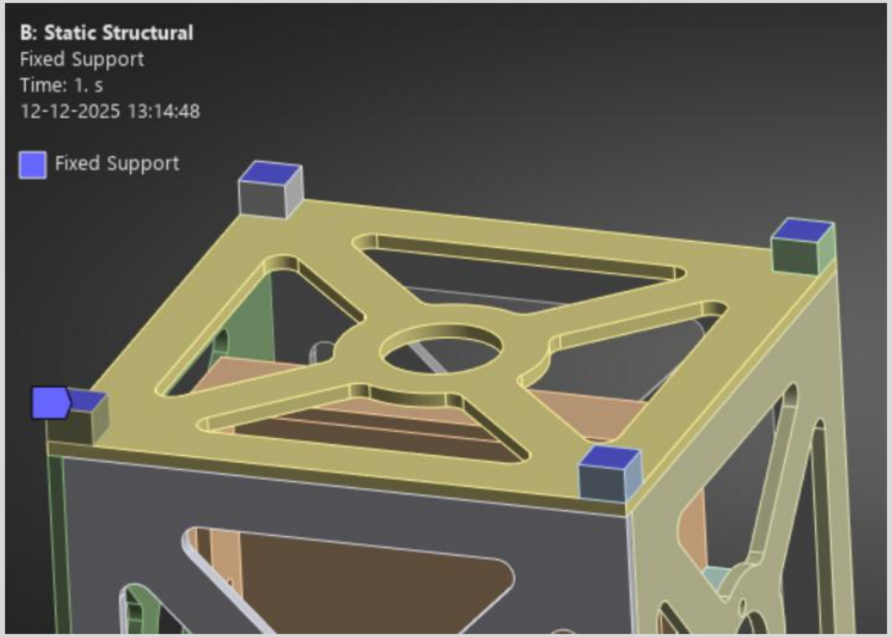


Fig.10 Status Result

ANALYSIS AND RESULTS - ITERATION 1

- **Acceleration Loads:** 98,066 mm/s² applied separately in X global directions.
- **Load Application:** Applied to all bodies in the assembly.





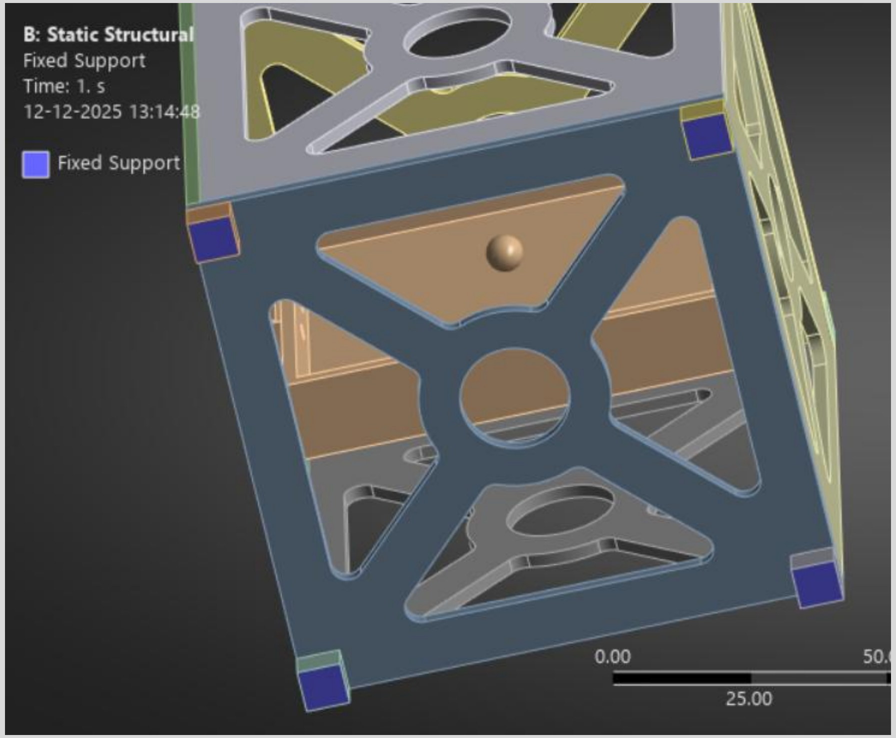


Fig.11 Acceleration Loads at X – Axis / Fixed Supports

ANALYSIS AND RESULTS - ITERATION 1

•The force convergence plot shows the solution residuals (in Newtons) versus iteration count for the nonlinear static analysis with frictional contact under X-direction 10g acceleration.

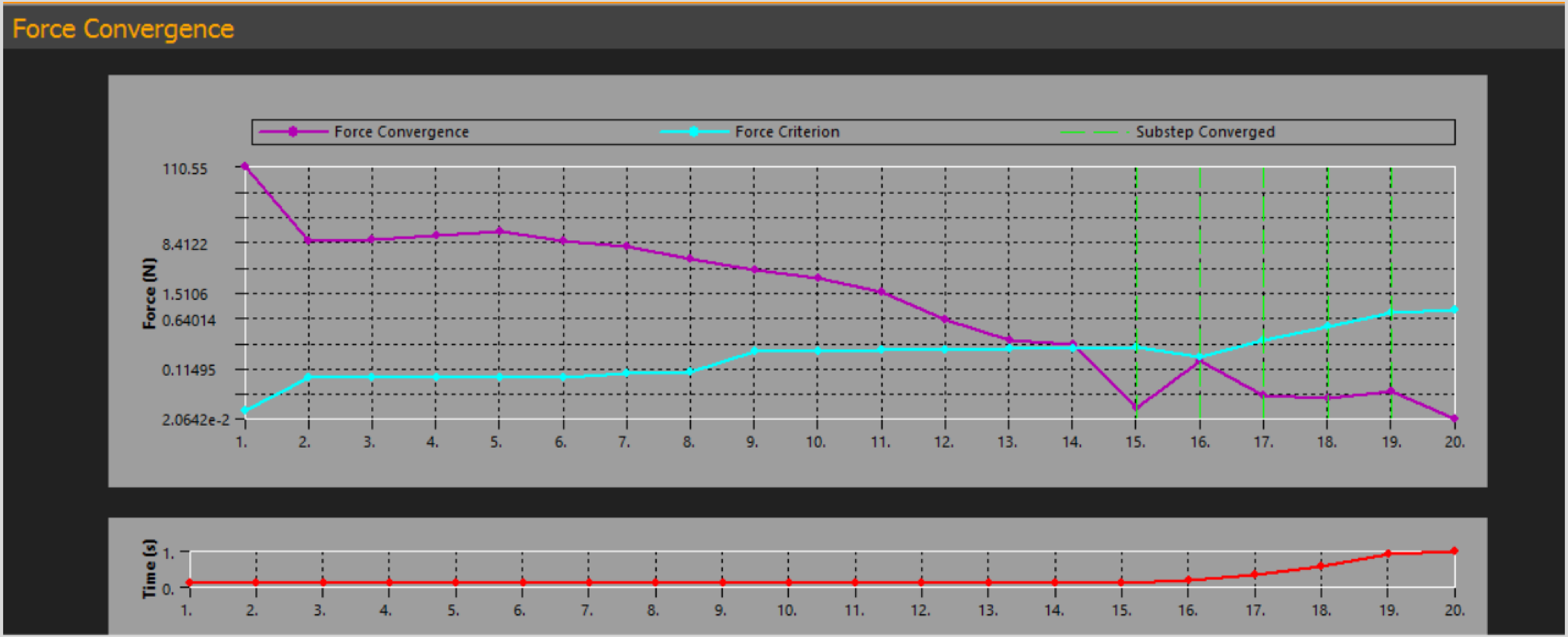
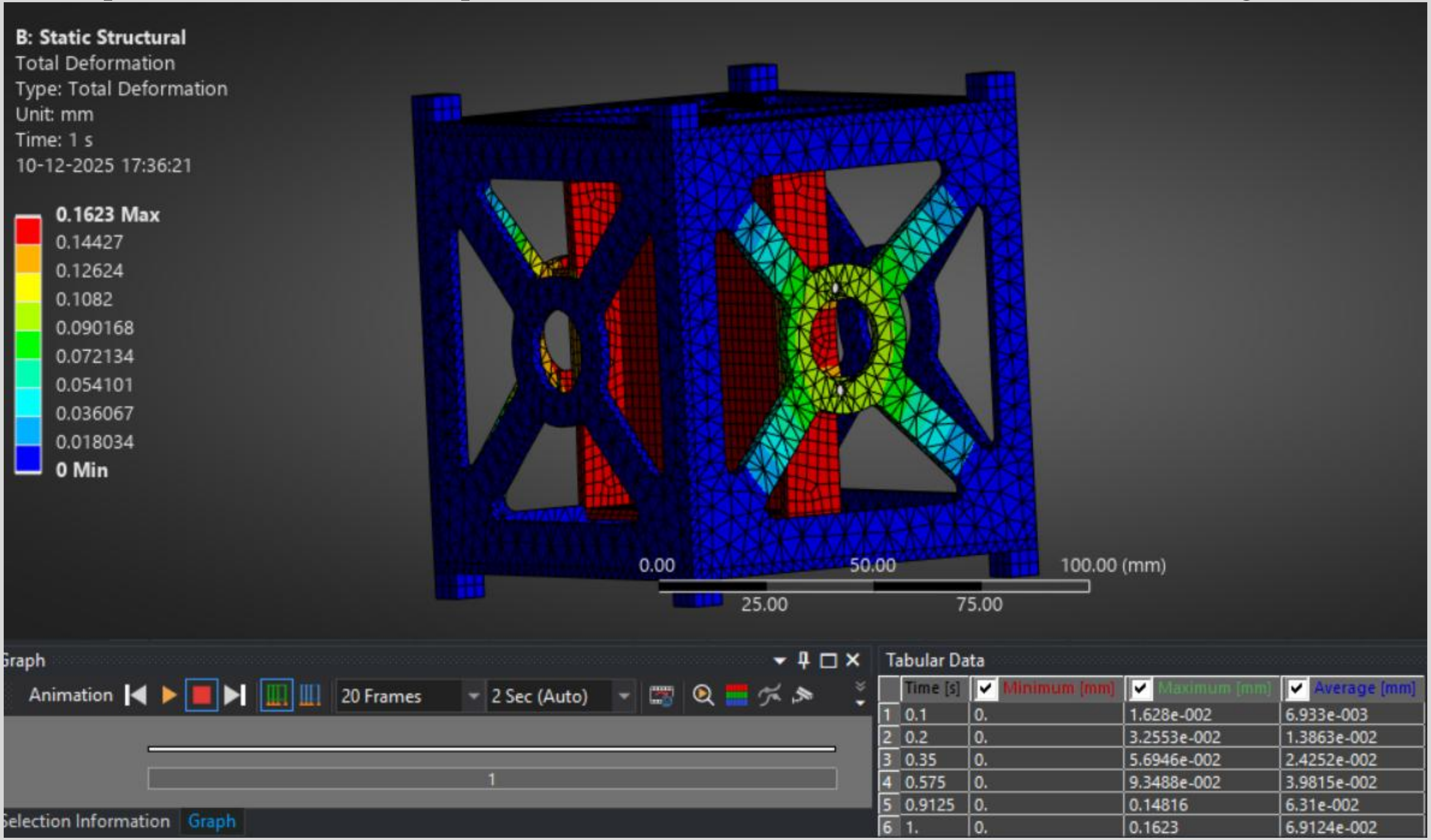


Fig.12 Force Convergence Plot for X - Axis

DEFORMATION RESULTS - ITERATION 1 X-AXIS

•Total deformation plot shows maximum displacement at the free end of the bracket arm. Scale ranges from 0 to 0.1623 mm.



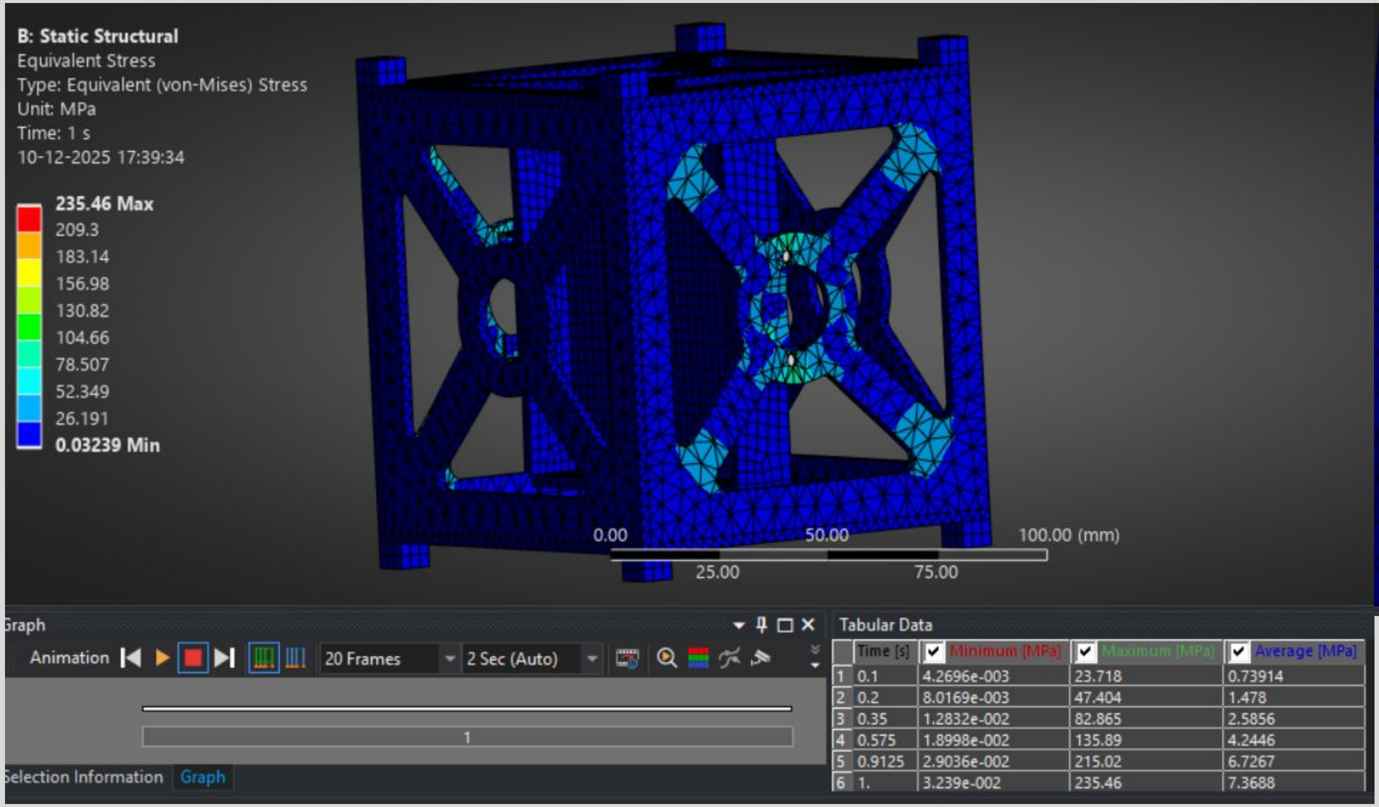
	Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1	0.1	0.	1.628e-002	6.933e-003
2	0.2	0.	3.2553e-002	1.3863e-002
3	0.35	0.	5.6946e-002	2.4252e-002
4	0.575	0.	9.3488e-002	3.9815e-002
5	0.9125	0.	0.14816	6.31e-002
6	1.	0.	0.1623	6.9124e-002

Fig.13 Total Deformation Plot for X - Axis

STRESS RESULTS - ITERATION 1 X-AXIS

•Equivalent stress plot (averaged) shows stress concentration at bracket root. Scale: 0.03239 to 235.46 MPa.

•**Max Stress:** 235.46 MPa



	Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1	0.1	4.2696e-003	23.718	0.73914
2	0.2	8.0169e-003	47.404	1.478
3	0.35	1.2832e-002	82.865	2.5856
4	0.575	1.8998e-002	135.89	4.2446
5	0.9125	2.9036e-002	215.02	6.7267
6	1.	3.239e-002	235.46	7.3688

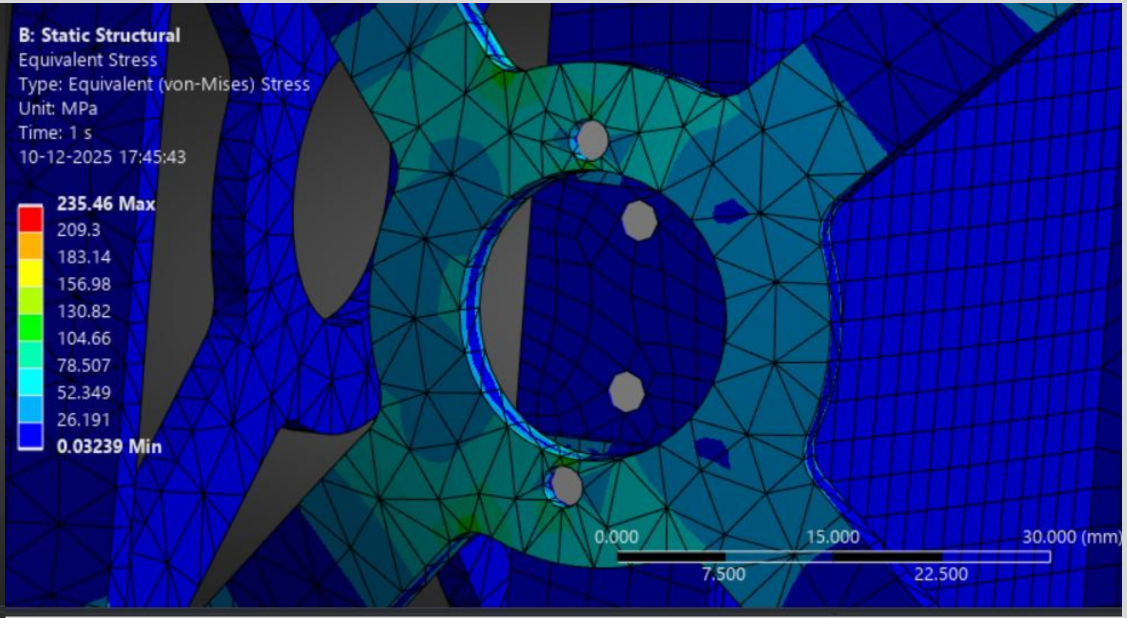
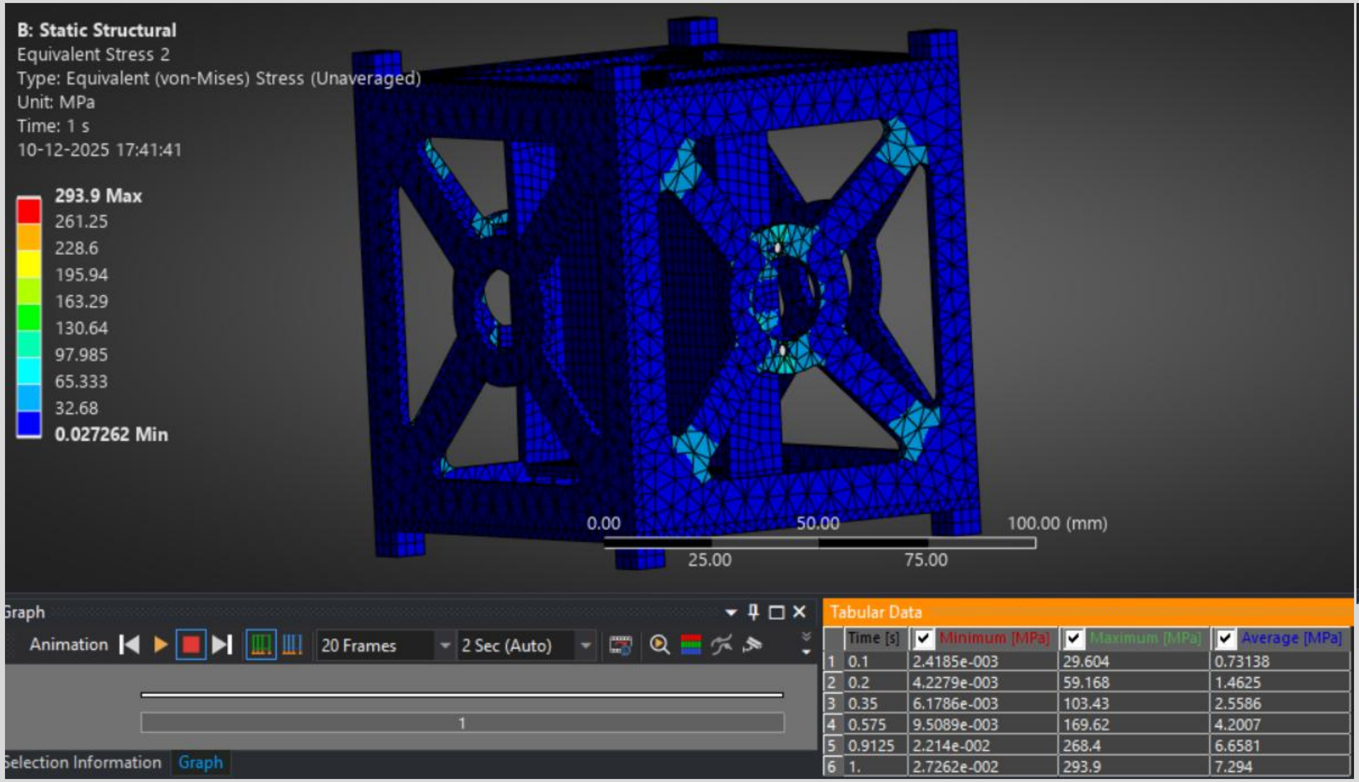


Fig.15 close-up stress plots of critical region

Fig.14 Equivalent stress plot (averaged) for X - Axis

STRESS RESULTS - ITERATION 1 X-AXIS

- Equivalent (Unaveraged) stress plot highlights sharp gradients at hole edges. Scale: 0.02726 to 293.9 MPa.
- **Max Stress:** 293.9 MPa



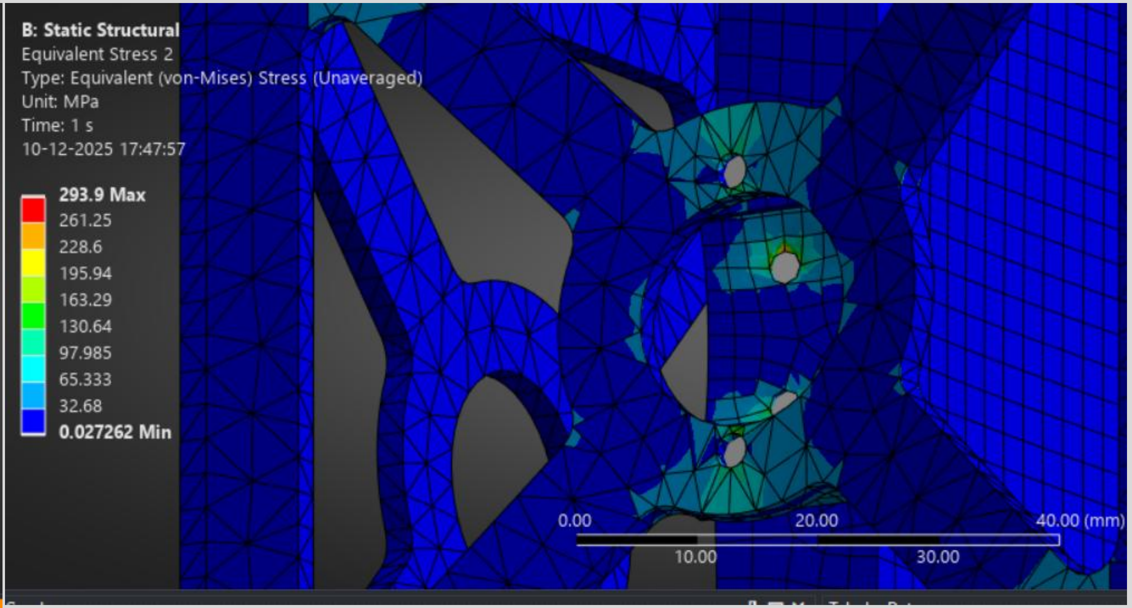


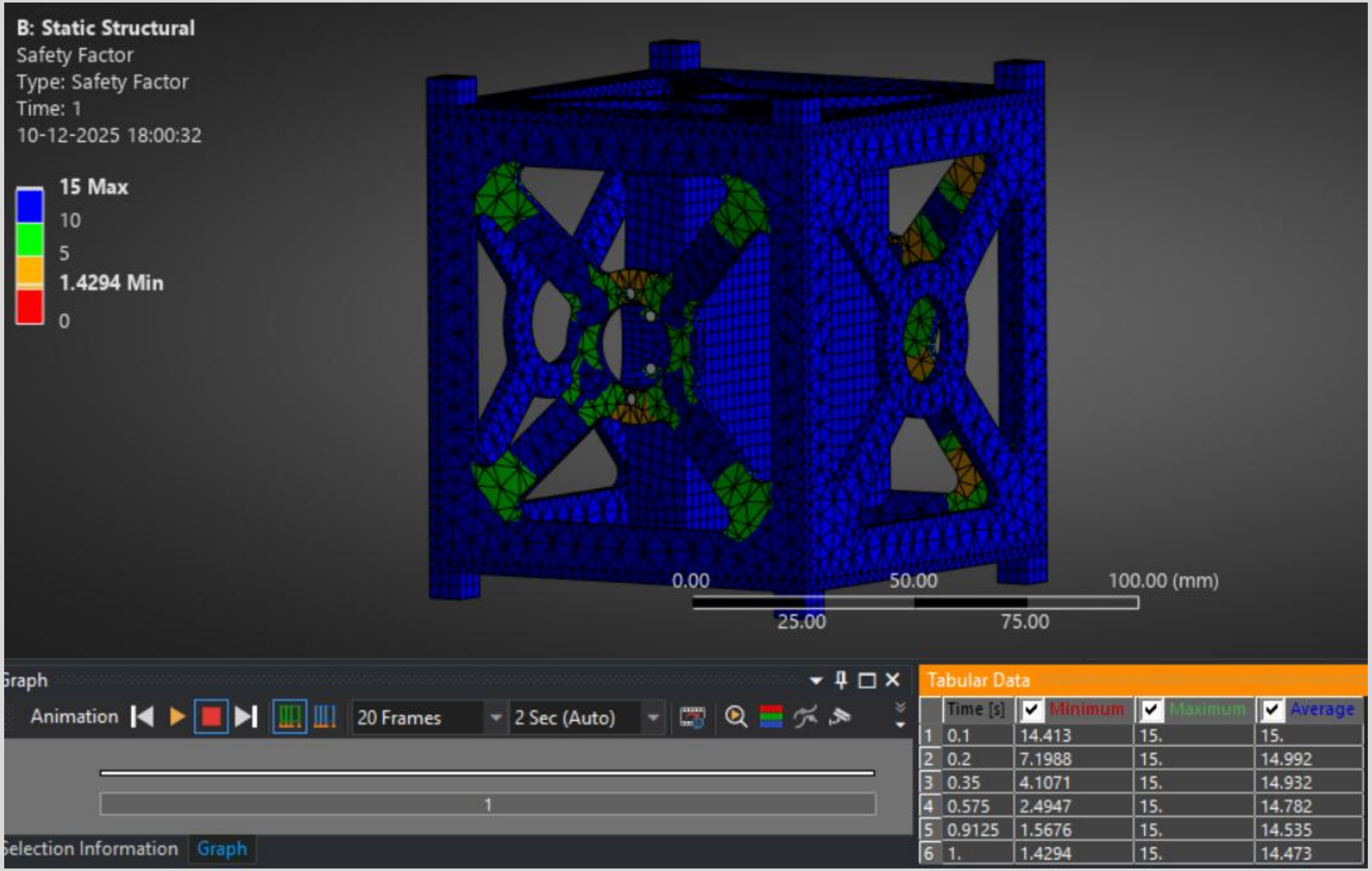
Fig.17 close-up stress plots of critical region (unaveraged) for X - Axis

Fig.16 Equivalent stress plot (unaveraged) for X - Axis

MARGINS OF SAFETY - ITERATION 1 X-AXIS

Min Safety Factor: 1.4294 (X-load case)

Safety factor contour plot shows values from 1.43 (min) to 15 (max). Critical region shown in red.



	Time [s]	Minimum	Maximum	Average
1	0.1	14.413	15.	15.
2	0.2	7.1988	15.	14.992
3	0.35	4.1071	15.	14.932
4	0.575	2.4947	15.	14.782
5	0.9125	1.5676	15.	14.535
6	1.	1.4294	15.	14.473

Fig.18 Margins of Safety Plot for X - Axis

FORCE REACTION RESULTS & HAND CALCULATION COMPARISON- ITERATION 1 : X-AXIS

Applied Load: 300 N (3 kg × 10g)
Reaction Force Sum: ~ 370 **Error:** < 23 %

Force reaction table shows forces at fixed supports matching applied load.

- **No Separation/Penetration:** Confirmed in both iterations.
- **Contact Pressure:** Within acceptable limits (<50 MPa).
- Contact tool plot shows pressure distribution at interfaces, with no gaps or penetration.

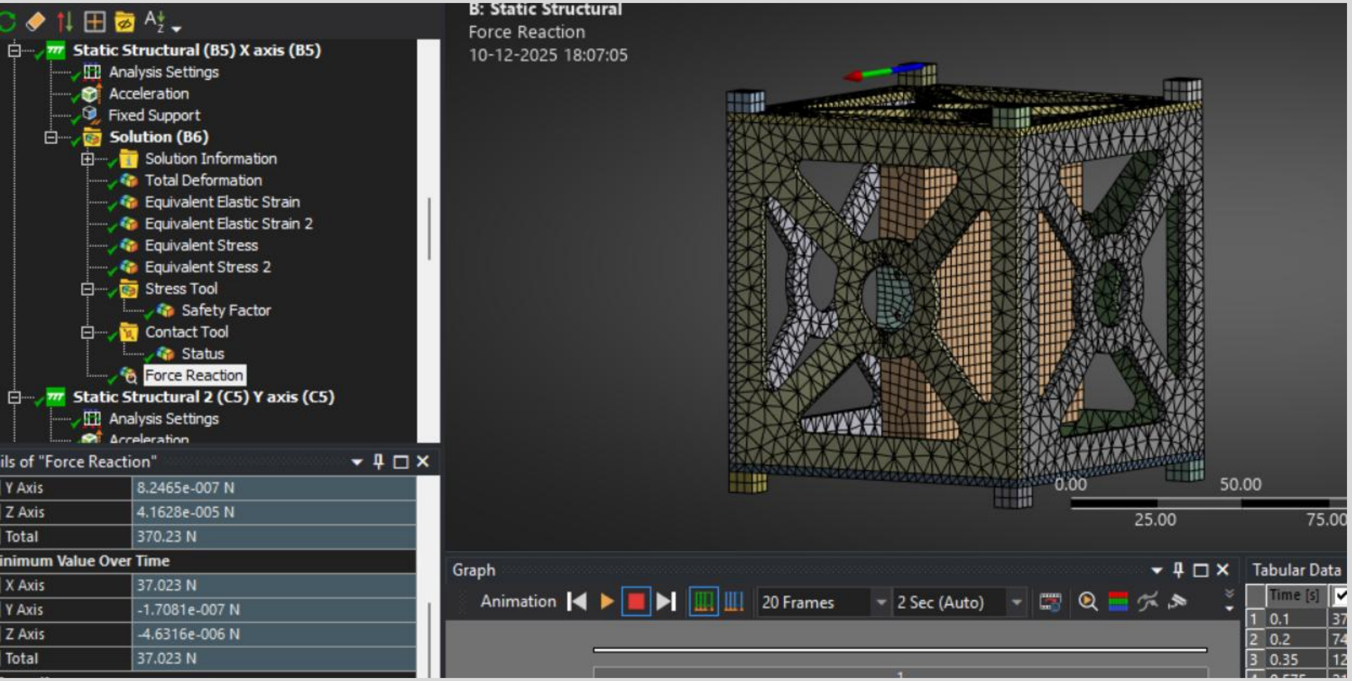


Fig.19 Force Reaction Results Plot for X - Axis

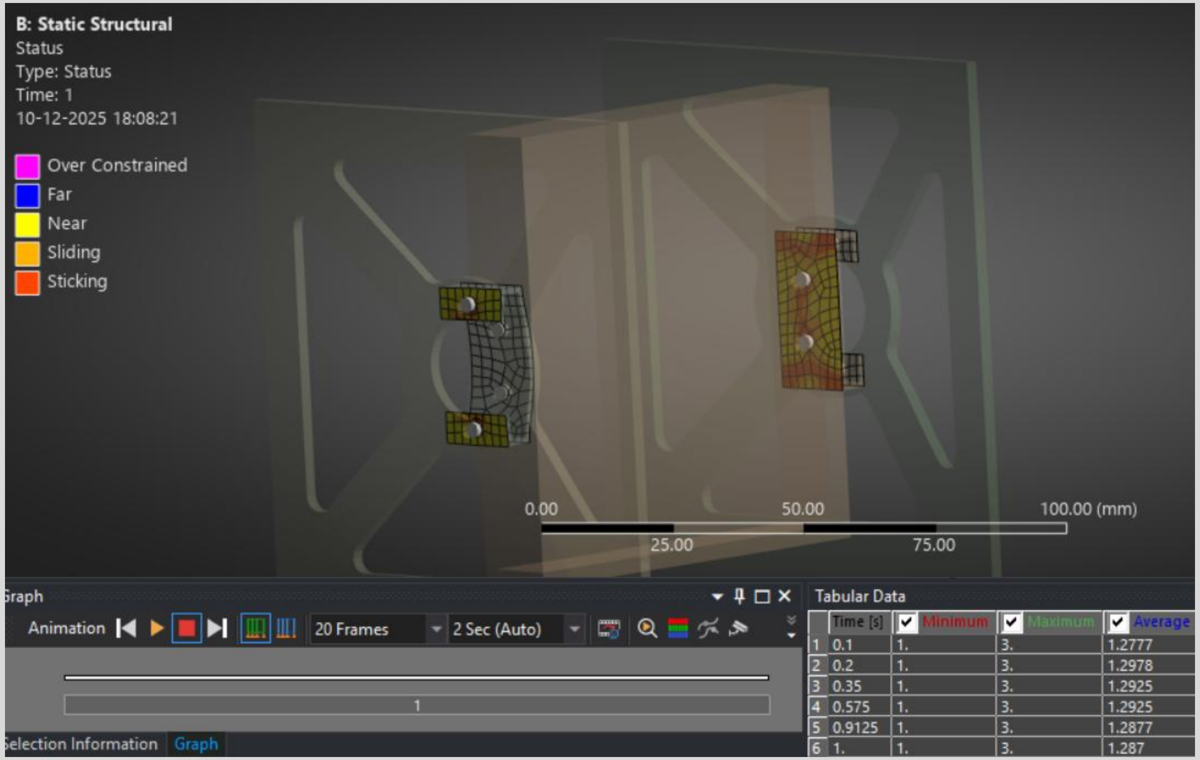


Fig.20 Contact Result for X - Axis

ANALYSIS AND RESULTS - ITERATION 1 Y-AXIS

•**Acceleration Loads:** 98,066 mm/s² applied separately in Y global directions.

•**Load Application:** Applied to all bodies in the assembly.

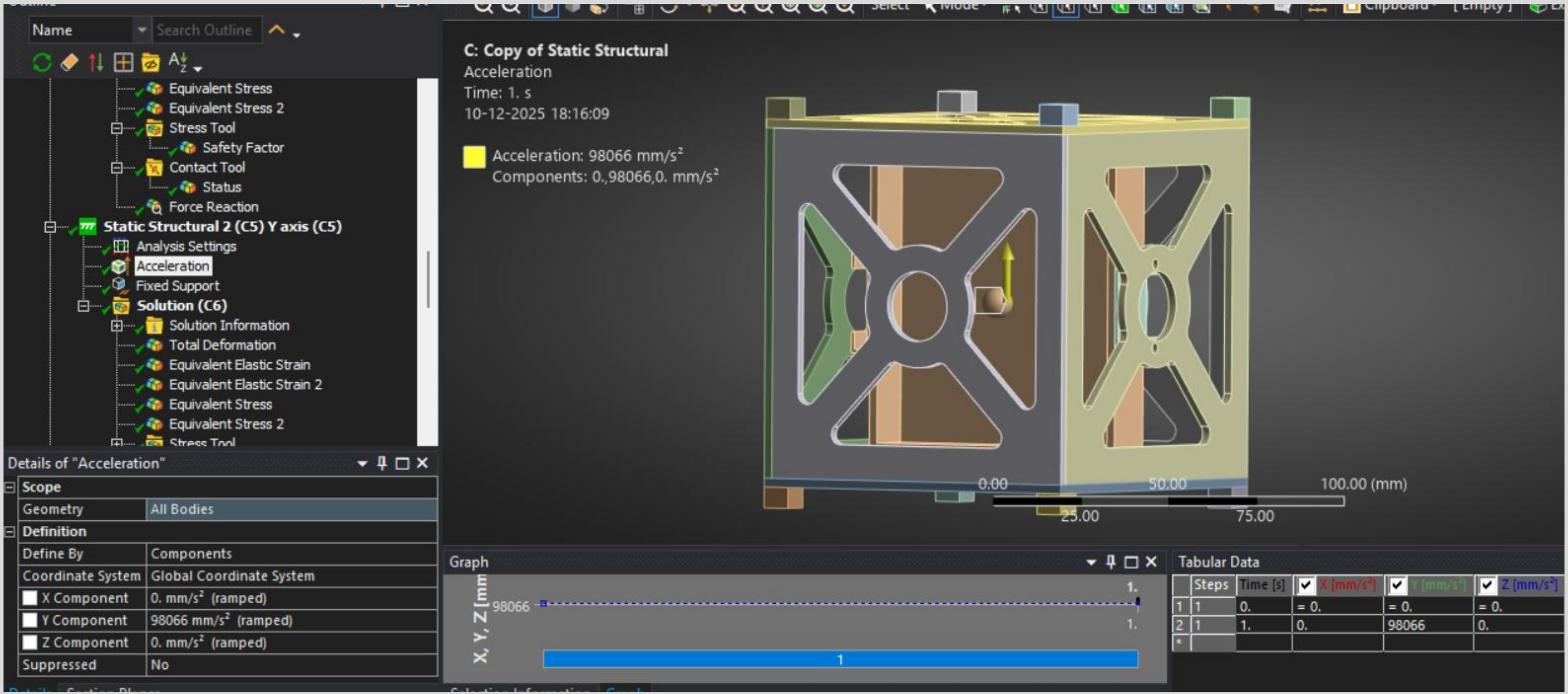
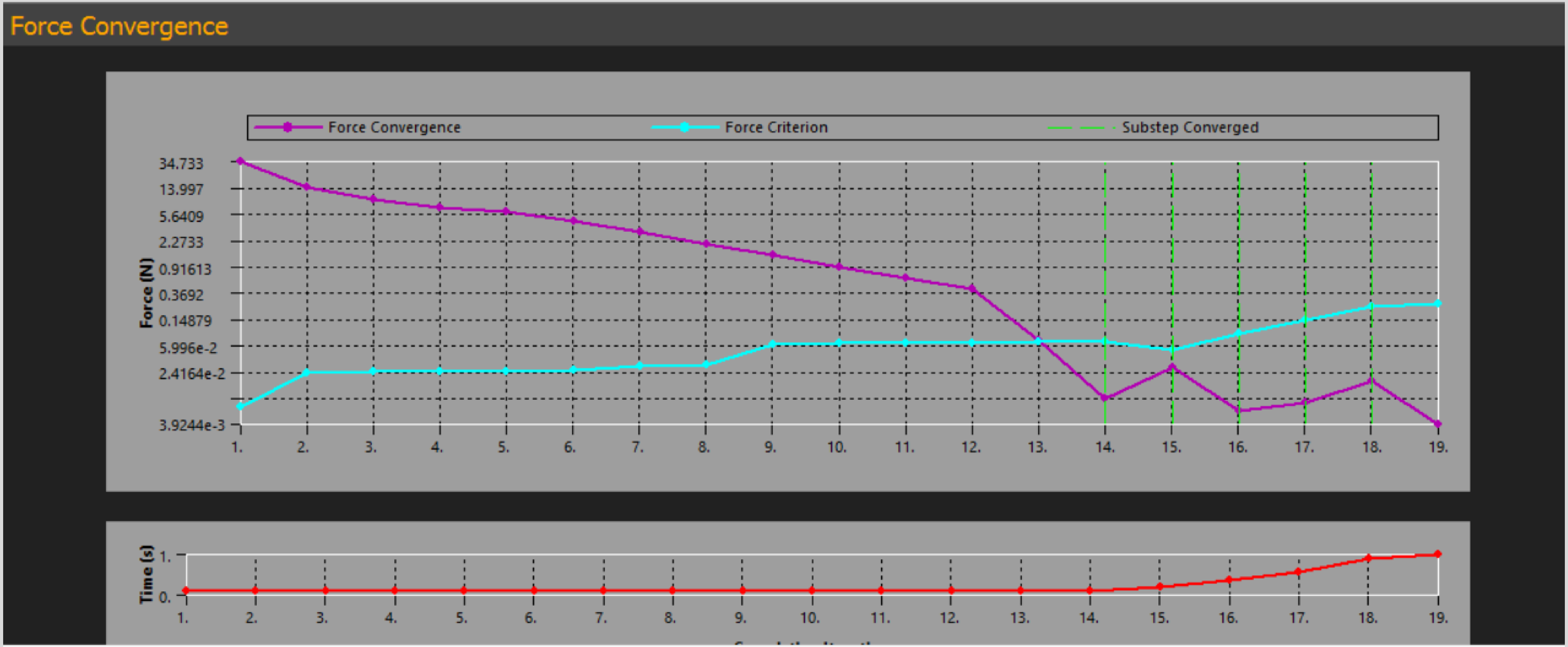


Fig.21 Acceleration Loads at Y - Axis

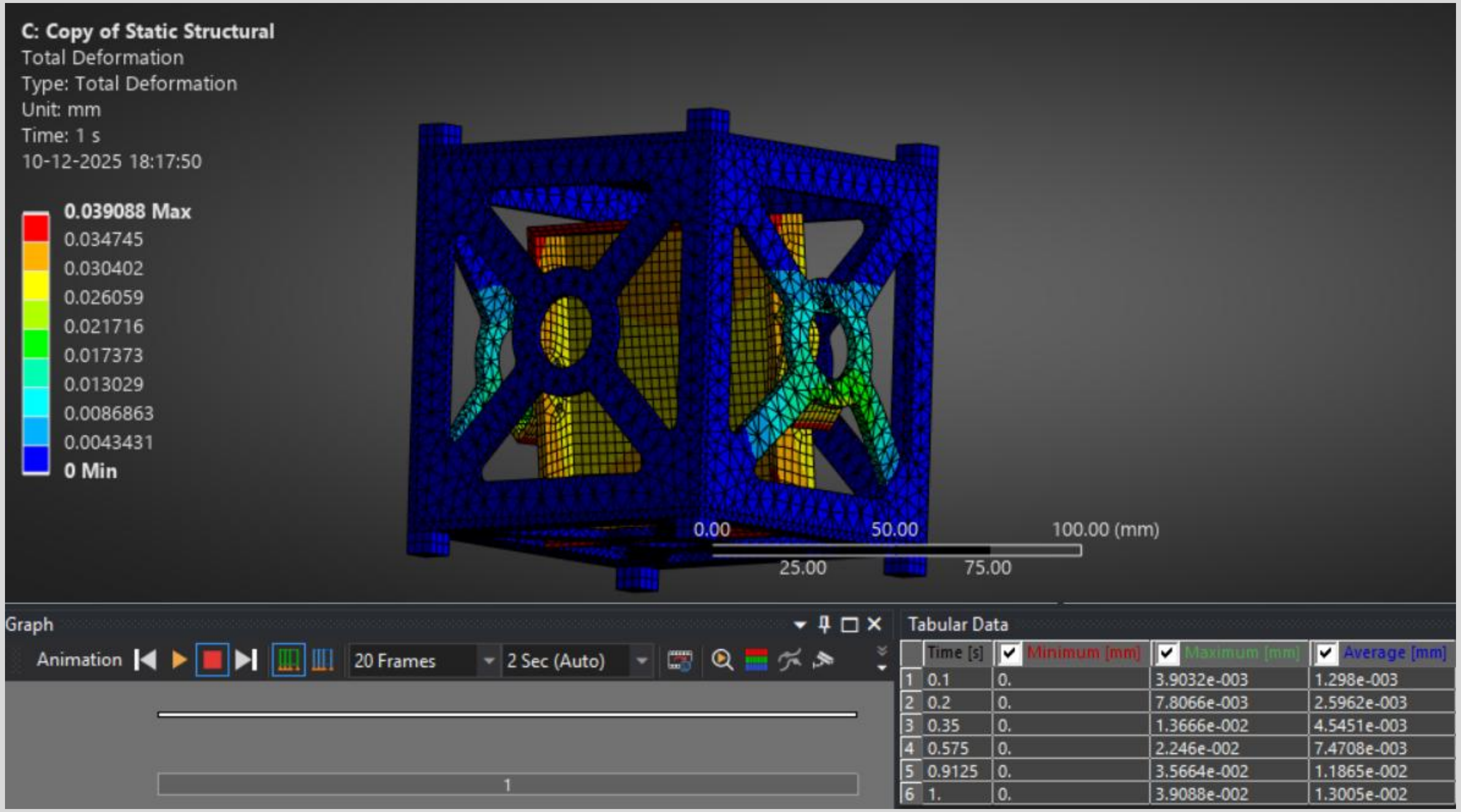
ANALYSIS AND RESULTS - ITERATION 1 Y-AXIS

•The force convergence plot shows the solution residuals (in Newtons) versus iteration count for the nonlinear static analysis with frictional contact under Y-direction 10g acceleration.



DEFORMATION RESULTS - ITERATION 1 Y-AXIS

- **Max Deformation:** 0.039088 mm
- Deformation plot shows minimal displacement, with highest values near the mounting interface. Scale: 0 to 0.039088 mm.



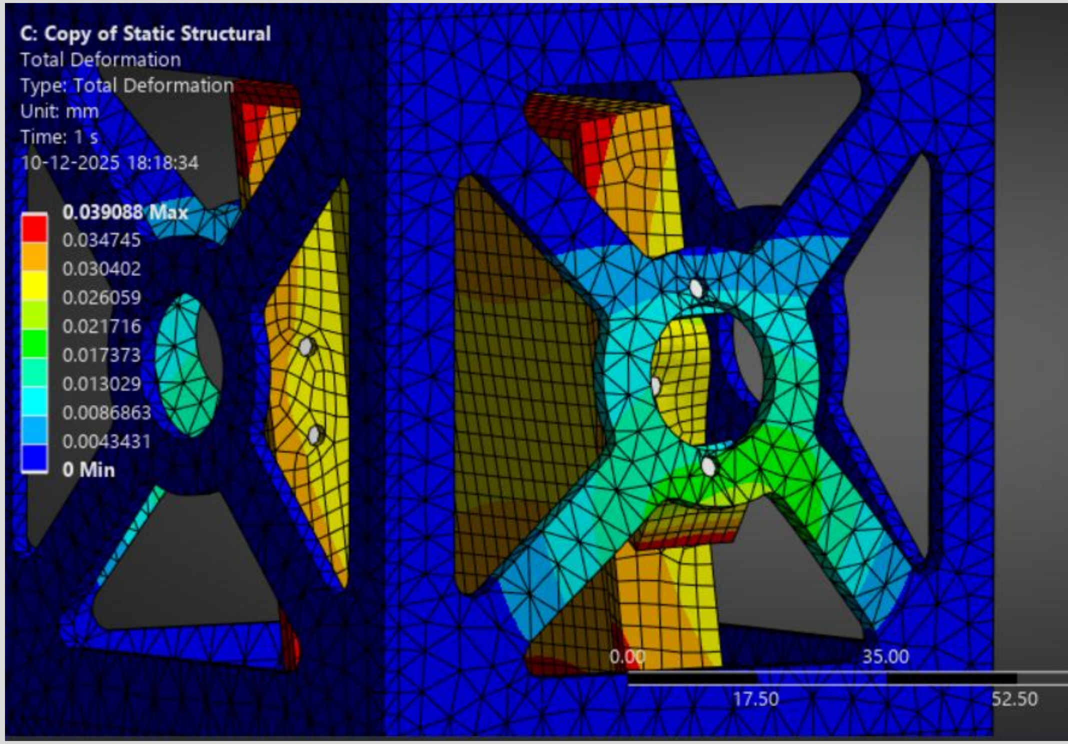
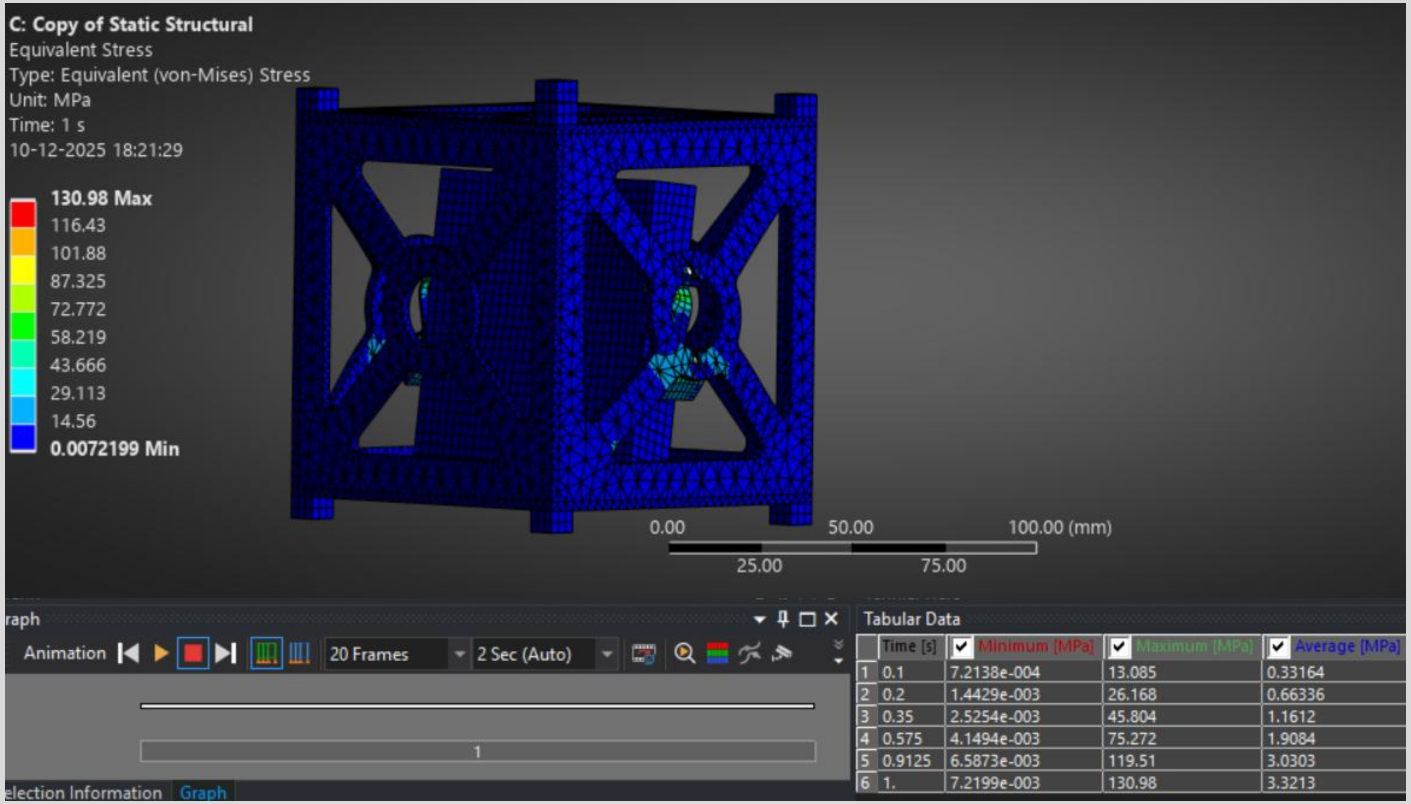


Fig.22 Total Deformation Plot for Y - Axis

STRESS RESULTS - ITERATION 1 Y-AXIS

•Equivalent stress plot (averaged) shows stress concentration at bracket root. Scale: 0.0072199 to 130.98 MPa.



	Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1	0.1	7.2138e-004	13.085	0.33164
2	0.2	1.4429e-003	26.168	0.66336
3	0.35	2.5254e-003	45.804	1.1612
4	0.575	4.1494e-003	75.272	1.9084
5	0.9125	6.5873e-003	119.51	3.0303
6	1.	7.2199e-003	130.98	3.3213

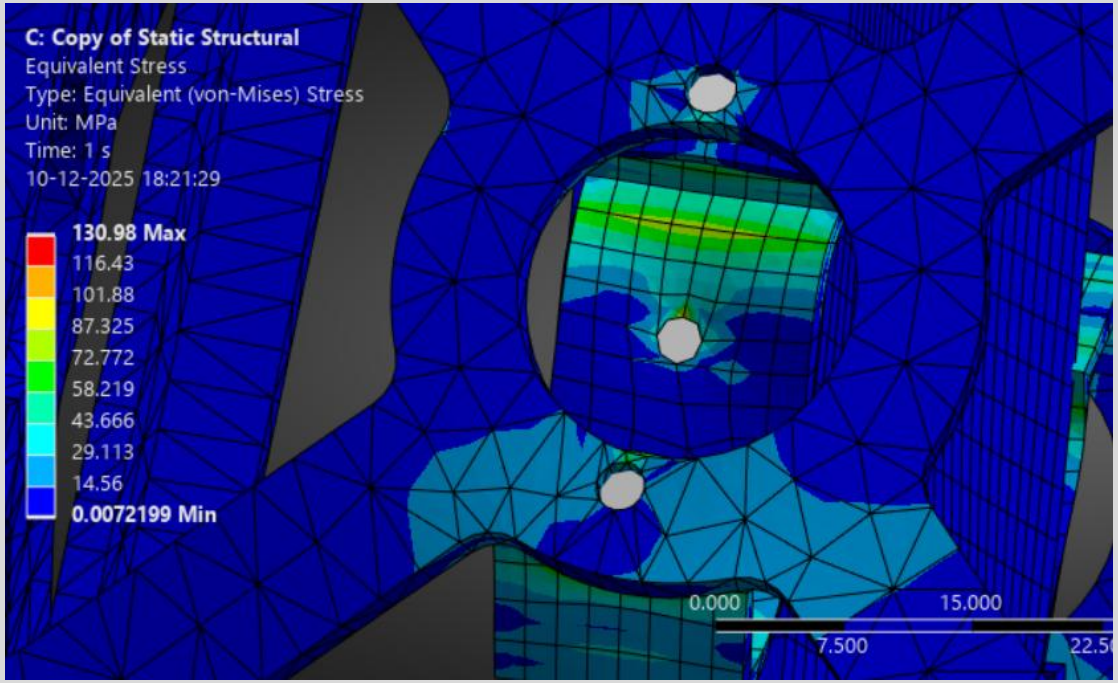


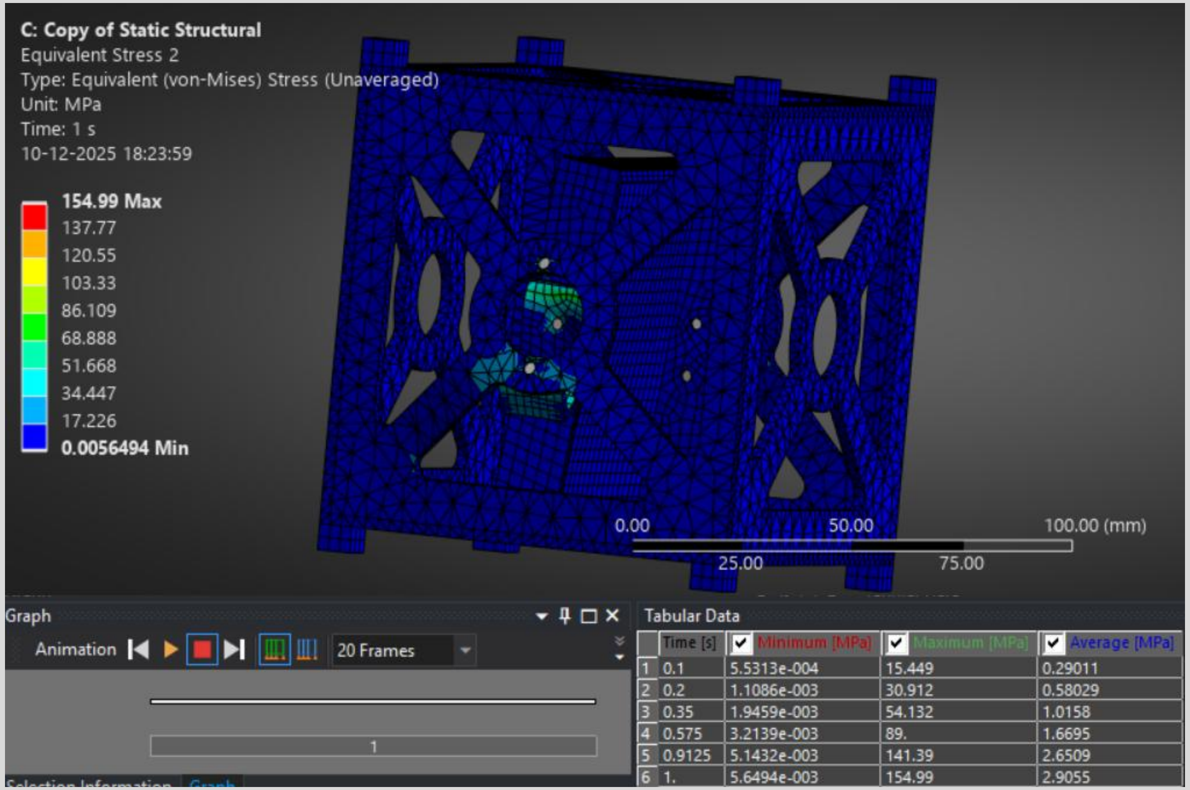
Fig.24 close-up stress plots of critical region

Fig.23 Equivalent stress plot (averaged) for Y - Axis

STRESS RESULTS - ITERATION 1 Y-AXIS

•Equivalent (Unaveraged) stress plot highlights sharp gradients at hole edges. Scale: 0.0056494 to 154.99 MPa.

•**Max Stress:** 154.99 MPa



	Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1	0.1	5.5313e-004	15.449	0.29011
2	0.2	1.1086e-003	30.912	0.58029
3	0.35	1.9459e-003	54.132	1.0158
4	0.575	3.2139e-003	89.	1.6695
5	0.9125	5.1432e-003	141.39	2.6509
6	1.	5.6494e-003	154.99	2.9055

Fig.25 Equivalent stress plot (unaveraged) for Y - Axis

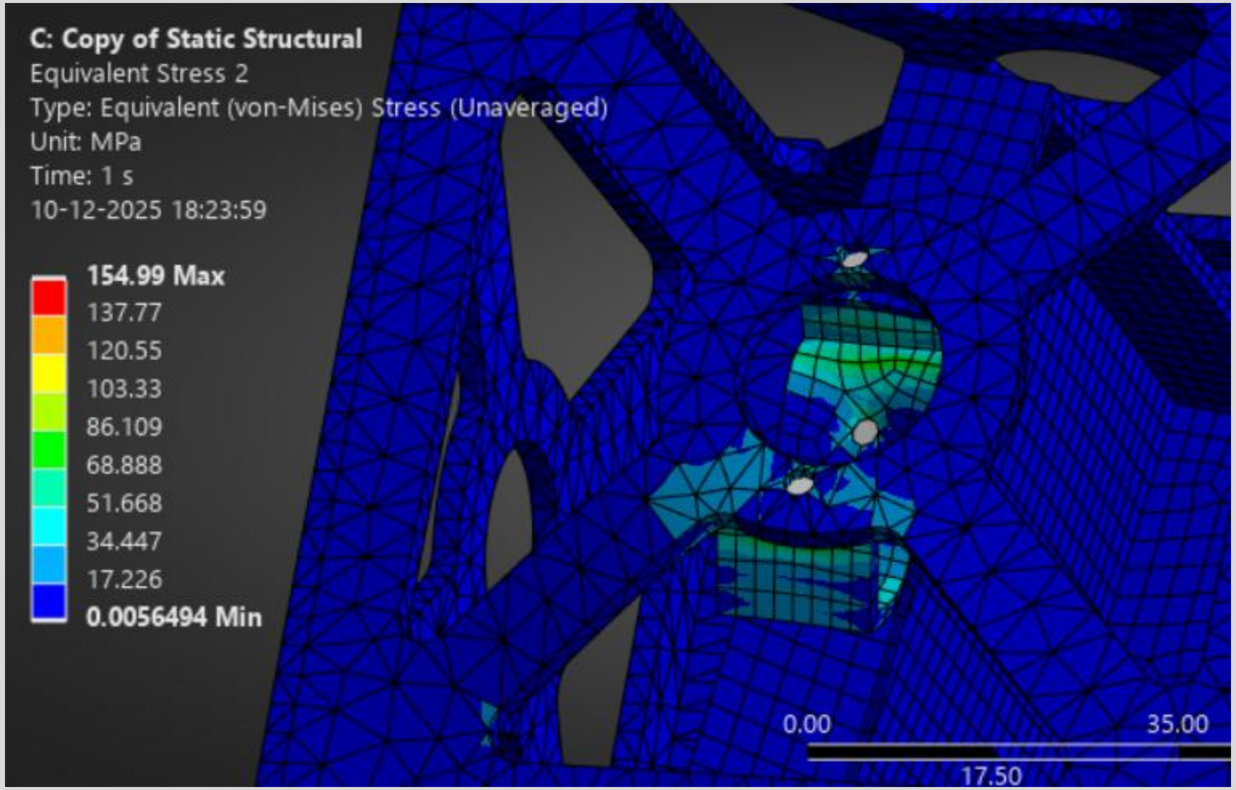


Fig.26 close-up stress plots of critical region (unaveraged) for Y - Axis

MARGINS OF SAFETY - ITERATION 1 Y-AXIS

Min Safety Factor: 3.76 (Y-load case)

Safety factor contour plot shows values from 3.76 (min) to 15 (max).

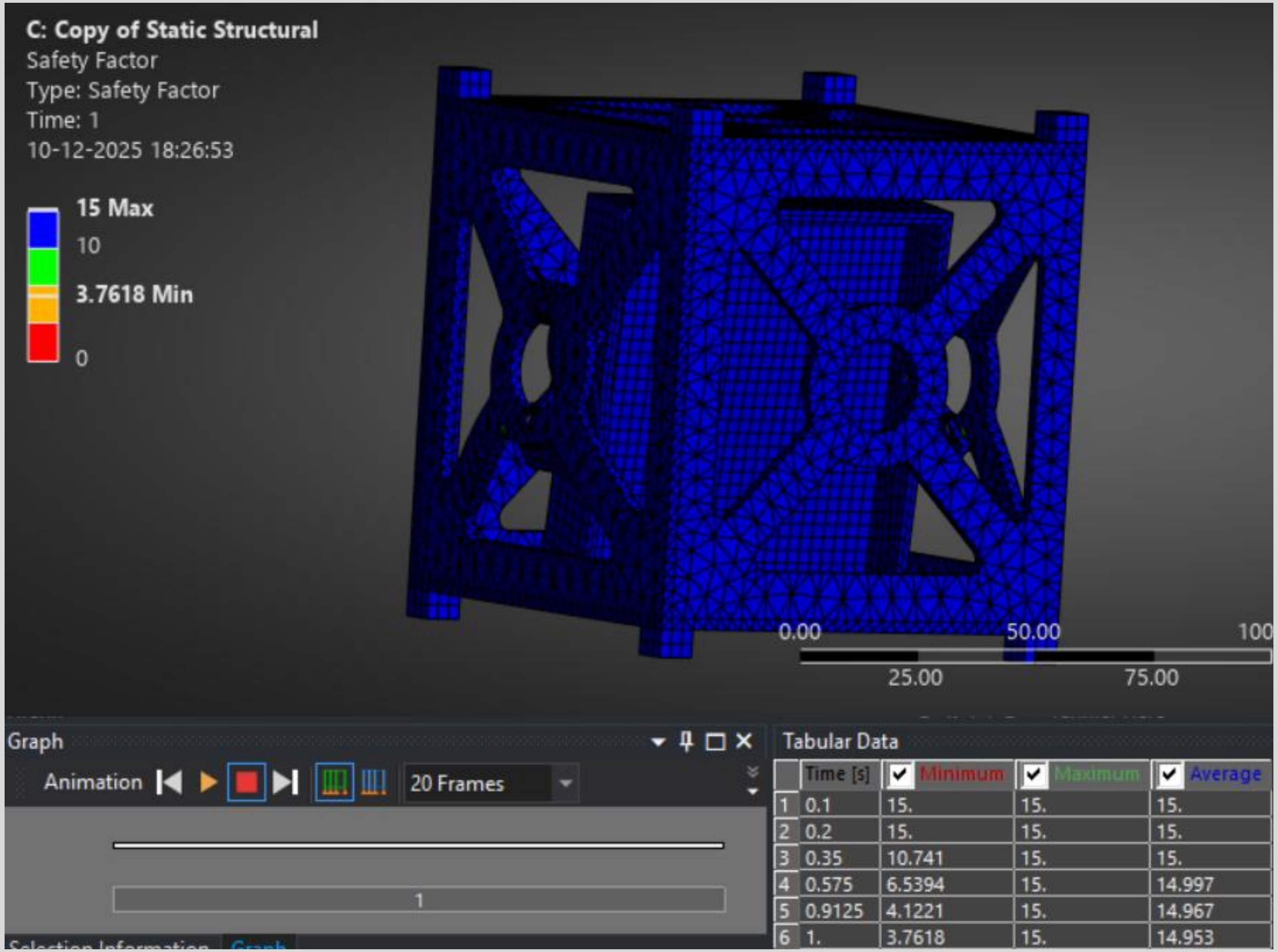


Fig.27 Margins of Safety Plot for Y - Axis

FORCE REACTION RESULTS & HAND CALCULATION COMPARISON- ITERATION 1 Y-AXIS

Applied Load: 300 N (3 kg × 10g)
Reaction Force Sum: ~ 370 **Error:** < 23 %

Force reaction table shows forces at fixed supports matching applied load.

- **No Separation/Penetration:** Confirmed in both iterations.
- **Contact Pressure:** Within acceptable limits.

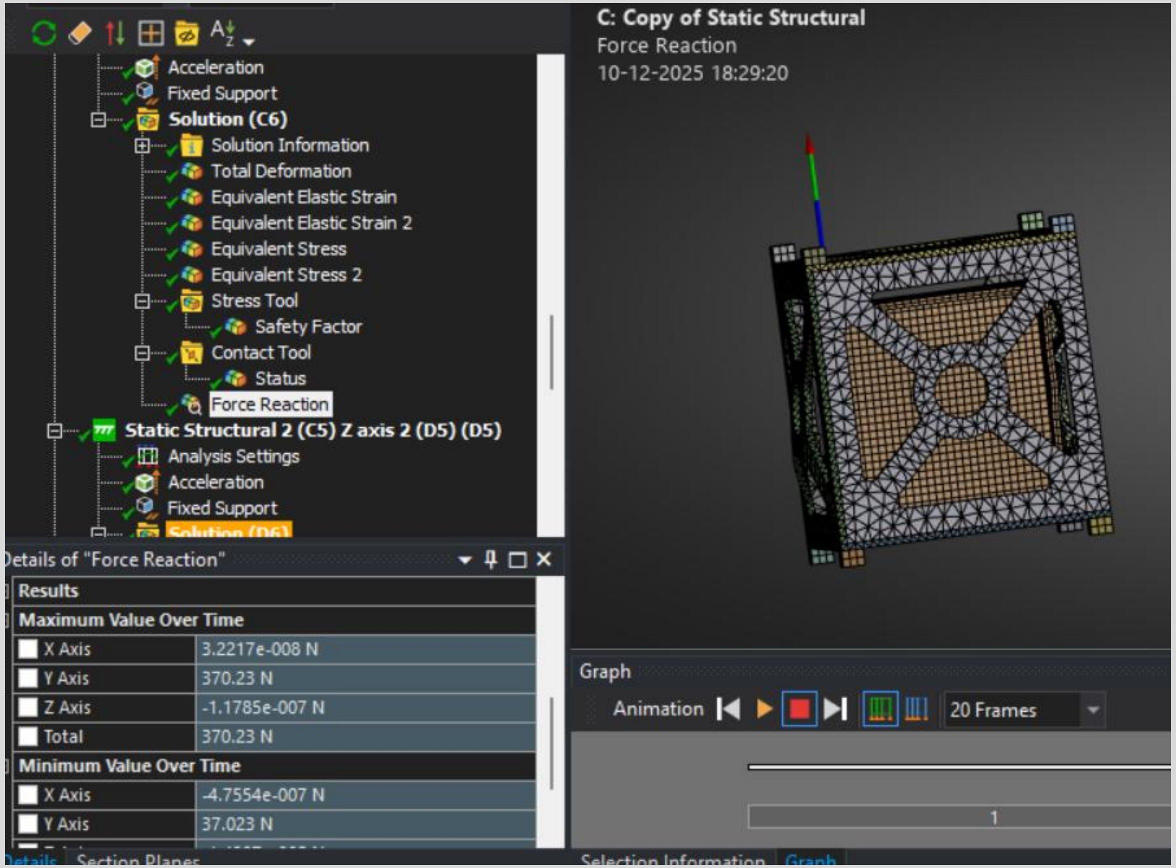
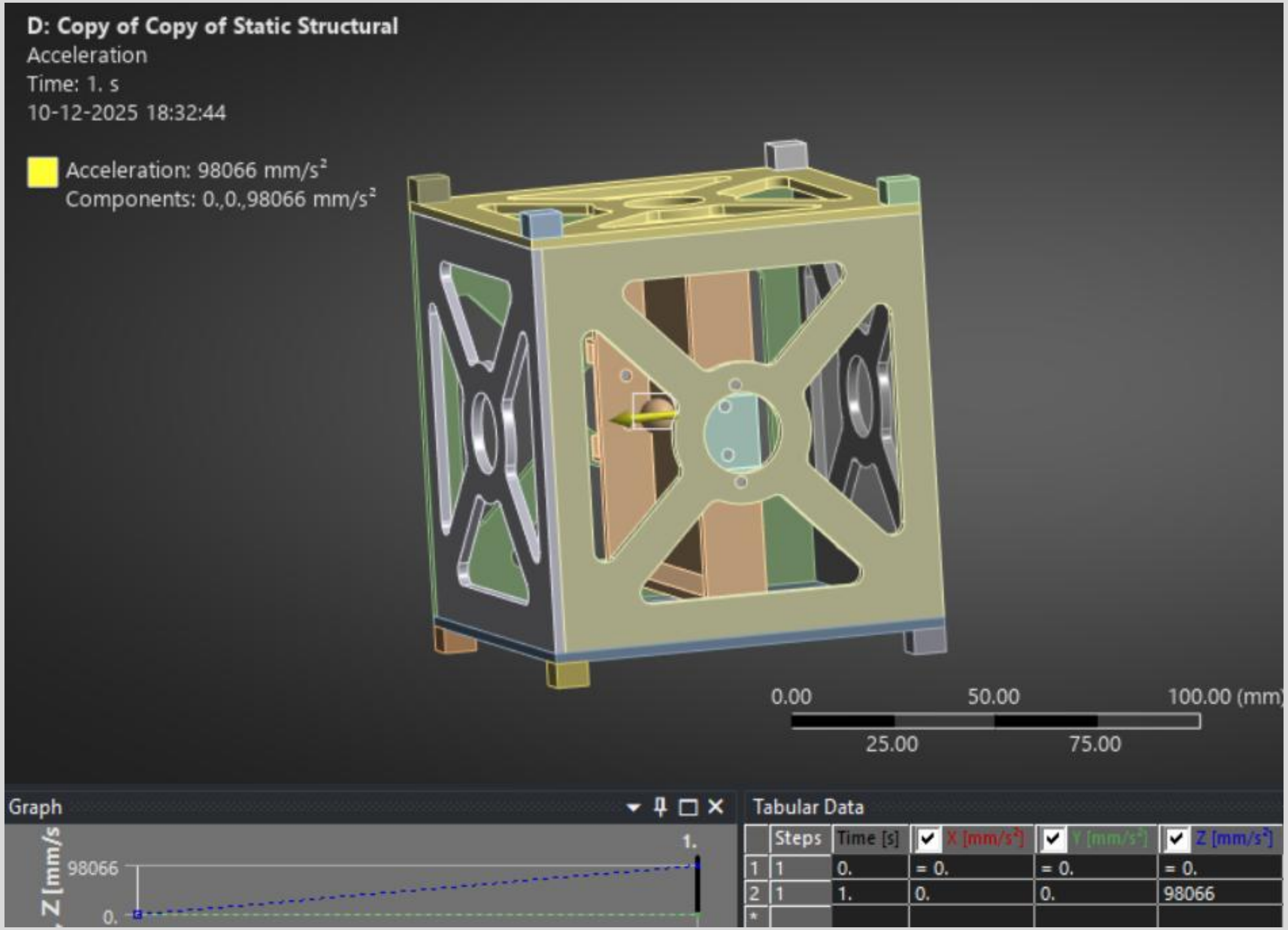


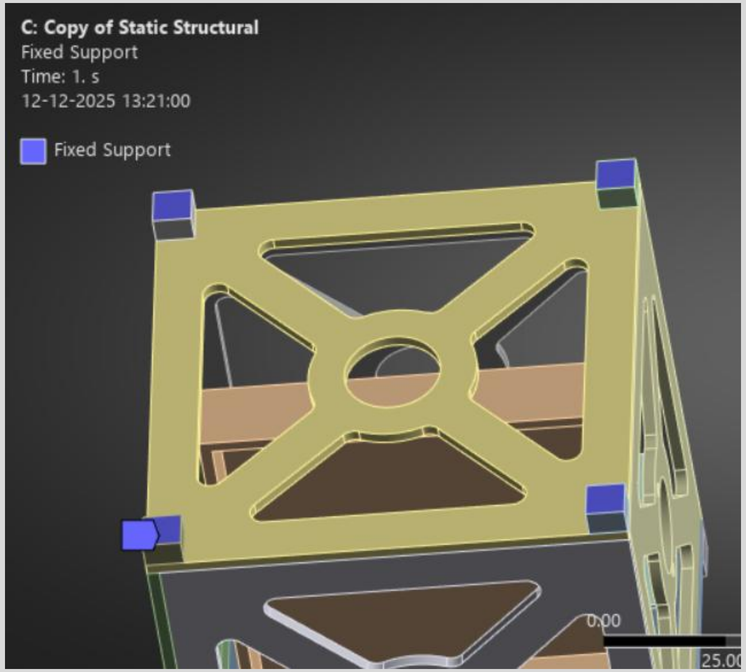
Fig.28 Force Reaction Results Plot for Y - Axis

Fig.29 Contact Result for Y - Axis

ANALYSIS AND RESULTS - ITERATION 1 Z-AXIS

- **Acceleration Loads:** 98,066 mm/s^2 applied separately in Z global directions.
- **Load Application:** Applied to all bodies in the assembly.





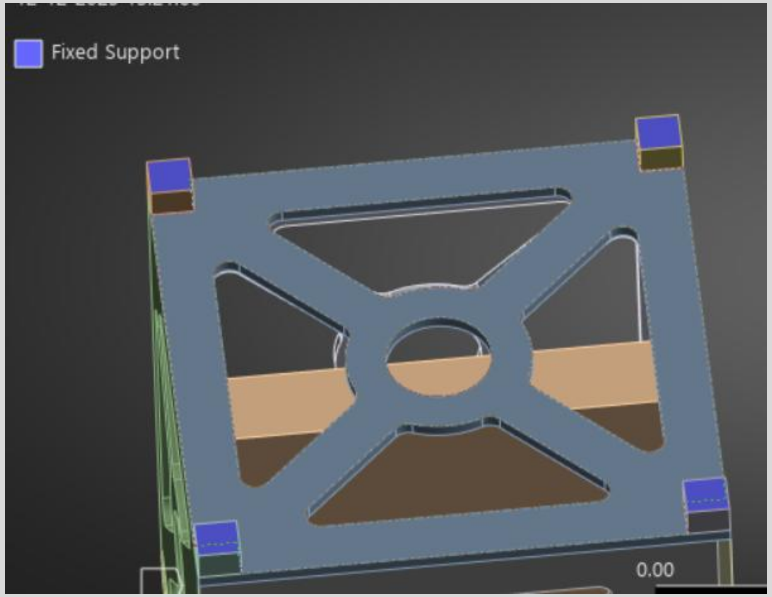
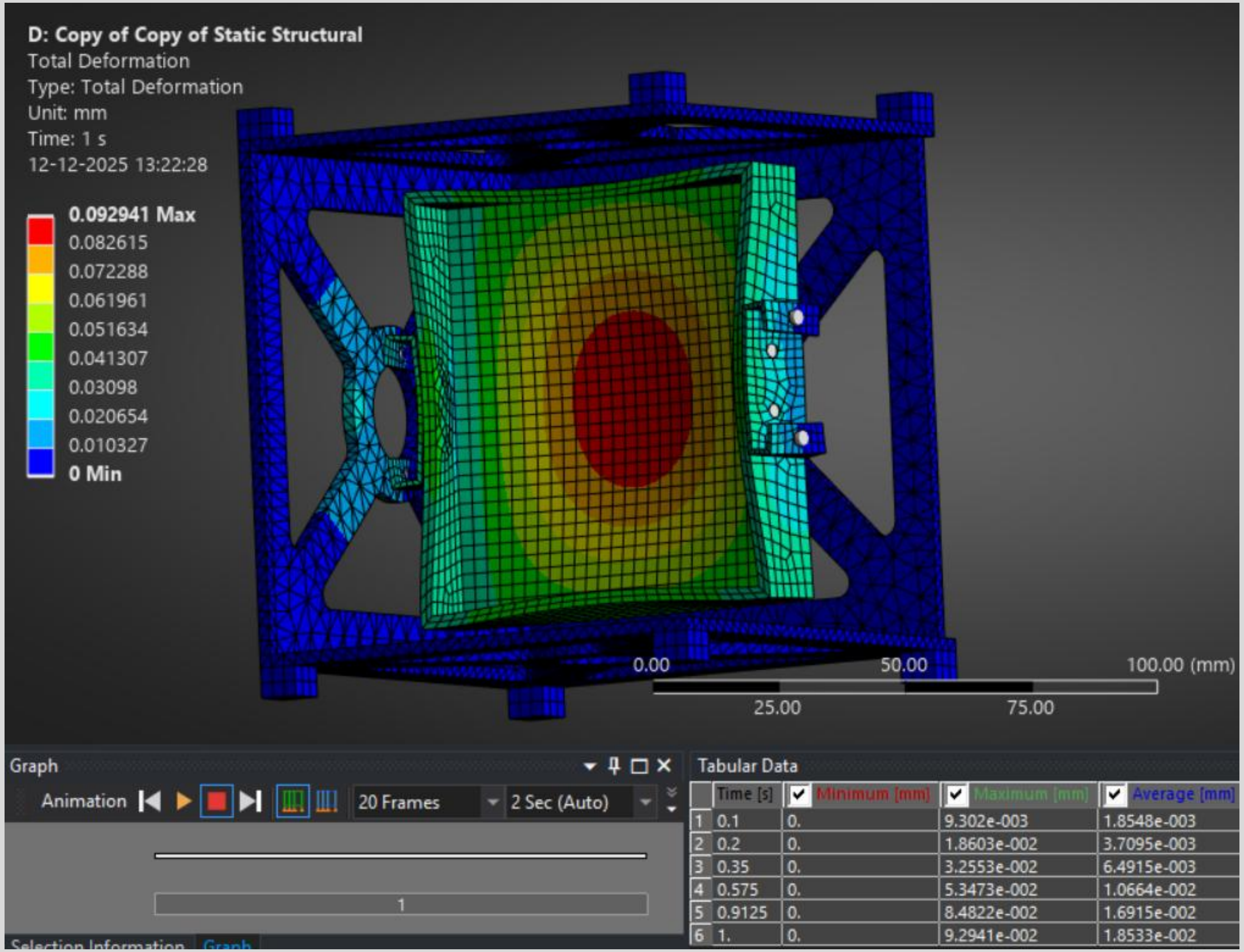
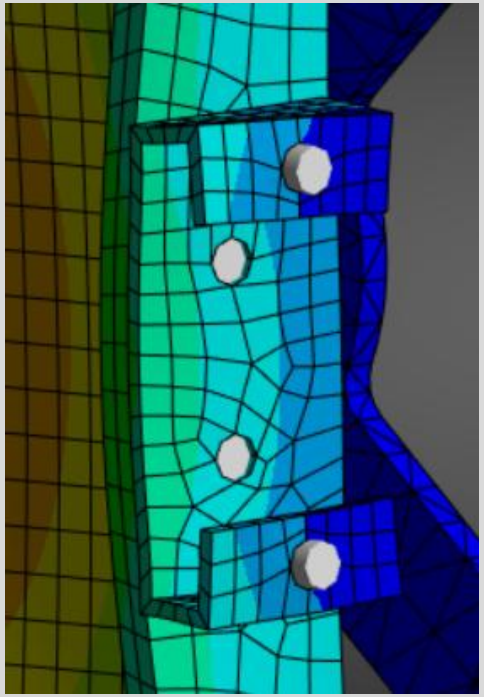


Fig.30 Acceleration Loads at Z - Axis

DEFORMATION RESULTS - ITERATION 1 Z-AXIS

- **Max Deformation:** 0.092941 mm
- Deformation plot indicates highest deformation compare to X and Y. Scale: 0 to 0.092941 mm.





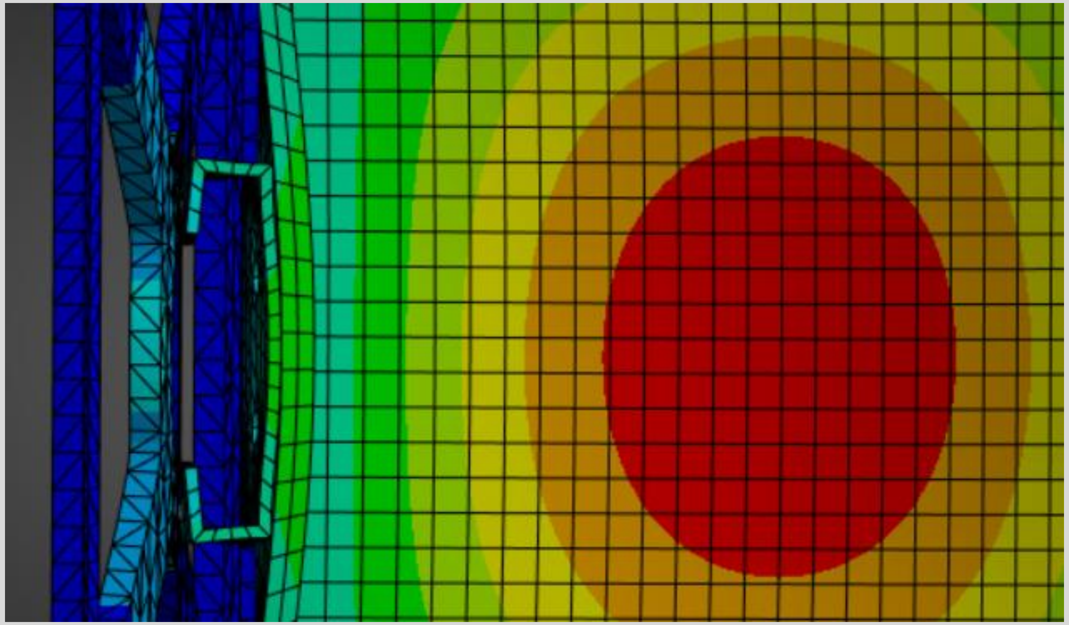
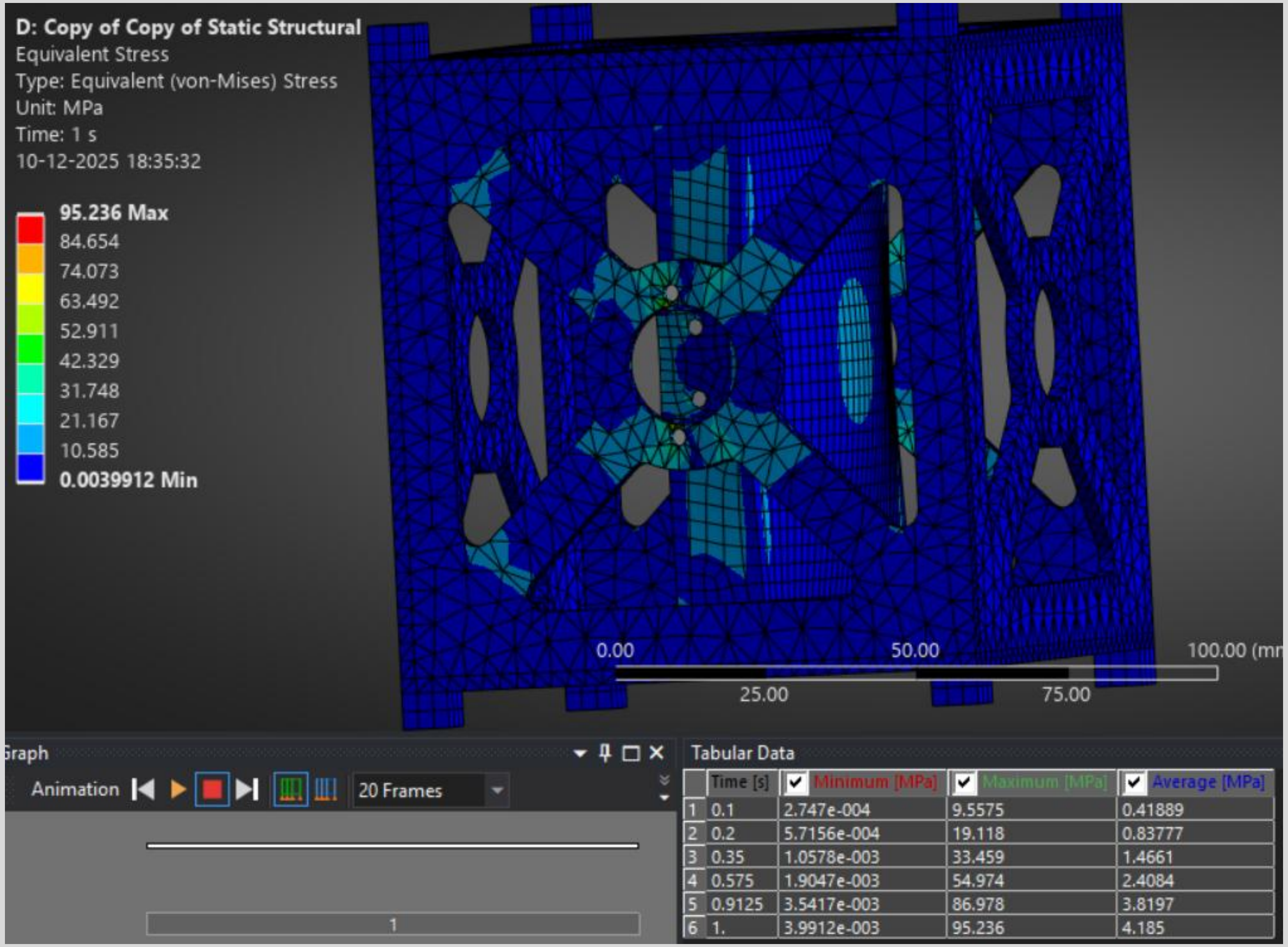


Fig.71 Total Deformation Plot for Z - Axis

STRESS RESULTS - ITERATION 1 Z-AXIS

- Z-load stress plot shows lower stress levels compared to X and Y. Scale: 0.00399 to 95.33 MPa (averaged)



	Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1	0.1	2.747e-004	9.5575	0.41889
2	0.2	5.7156e-004	19.118	0.83777
3	0.35	1.0578e-003	33.459	1.4661
4	0.575	1.9047e-003	54.974	2.4084
5	0.9125	3.5417e-003	86.978	3.8197
6	1.	3.9912e-003	95.236	4.185

Fig.72 Equivalent stress plot (averaged) for Z - Axis

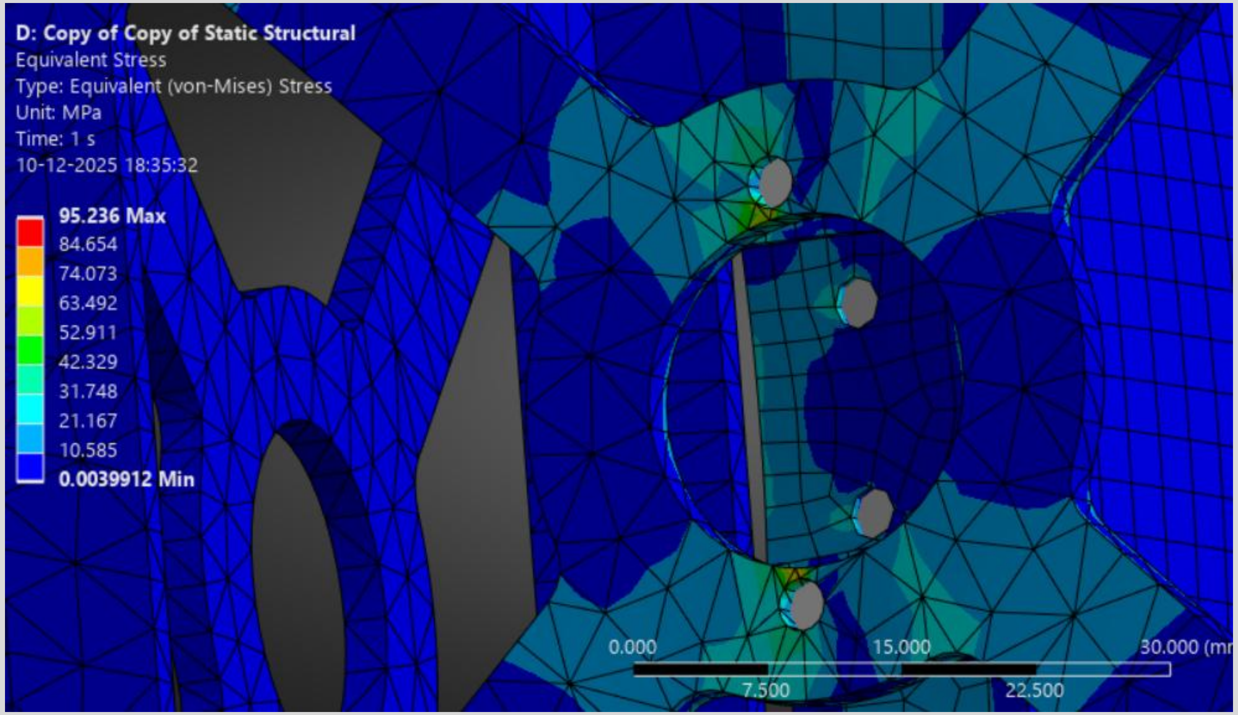


Fig.73 close-up stress plots of critical region

STRESS RESULTS - ITERATION 1 Z-AXIS

•Z-load stress plot shows lower stress levels compared to X and Y. Scale: 0.00257 to 126.49 MPa (unaveraged).

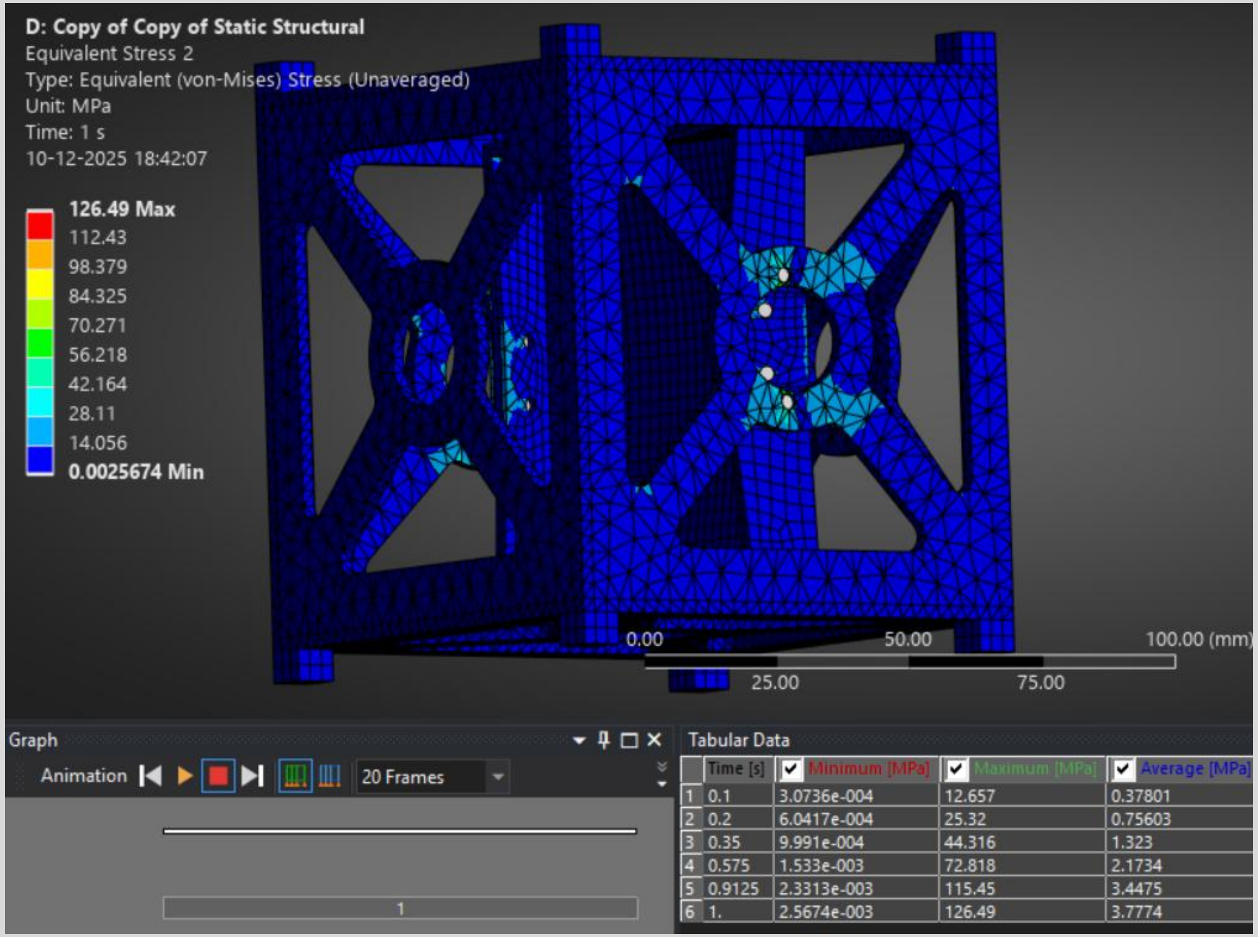


Fig.34 Equivalent stress plot (unaveraged) for Z - Axis

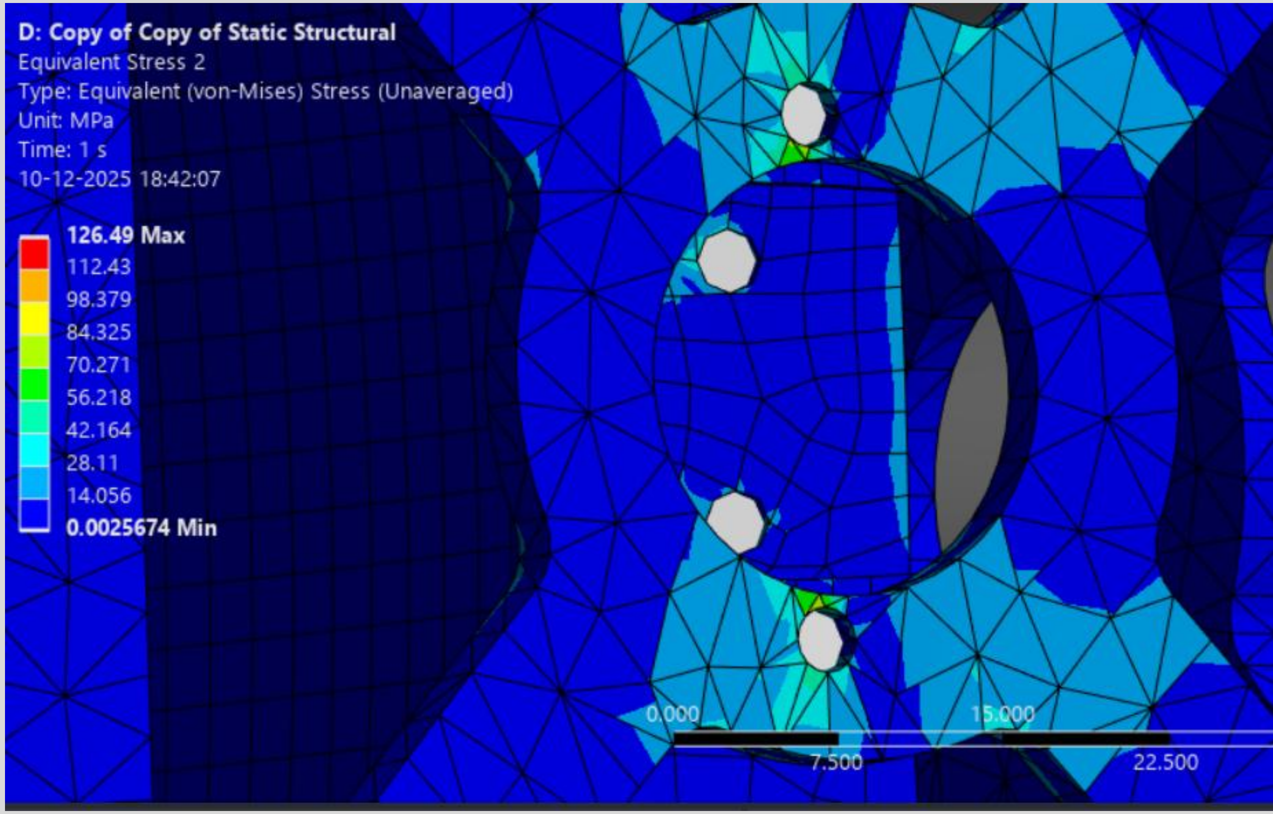
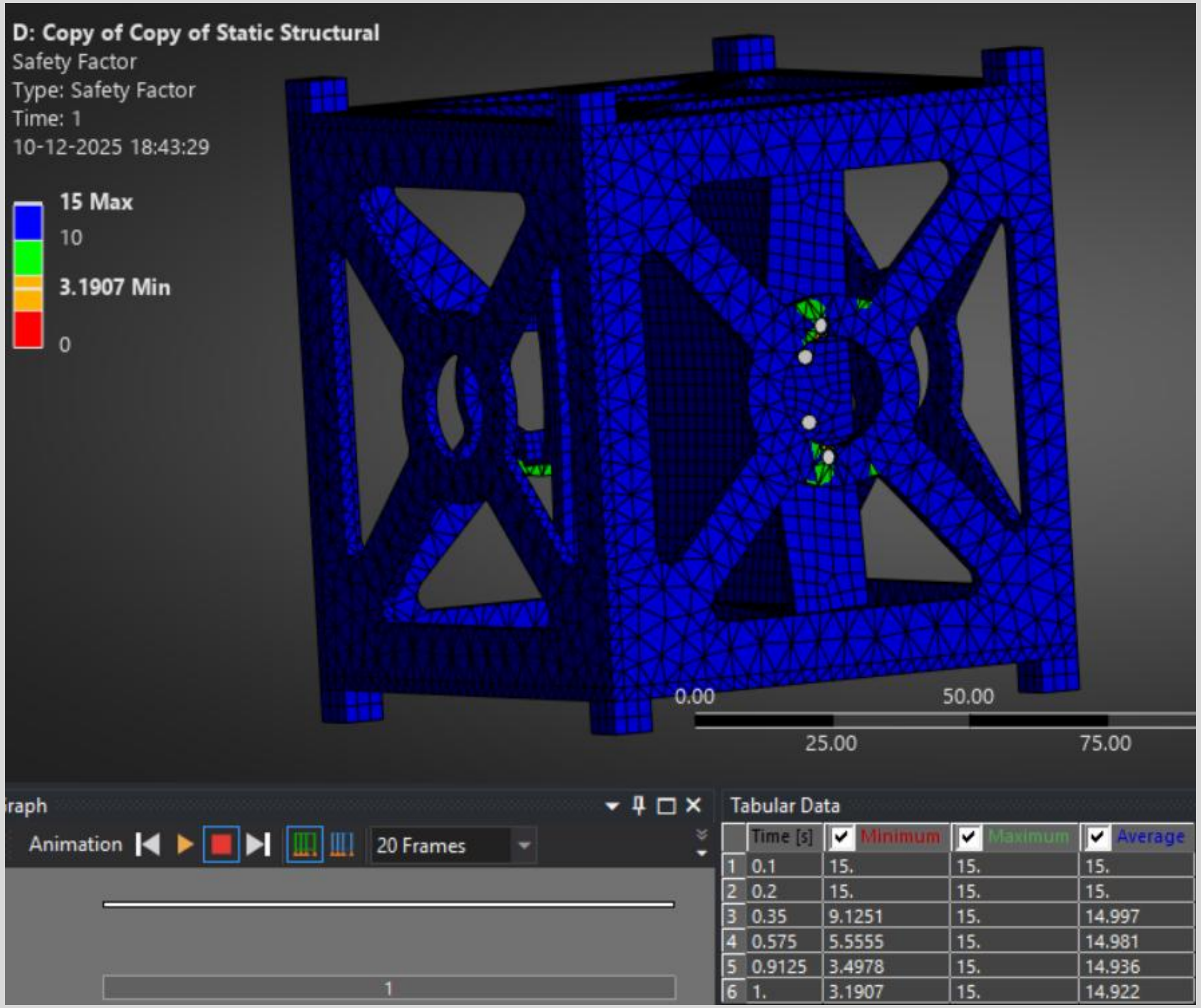


Fig.35 close-up stress plots of critical region (unaveraged) for Z - Axis

MARGINS OF SAFETY - ITERATION 1 Z-AXIS

Min Safety Factor: 3.1907 (Z-load case)

Safety factor contour plot shows values from 3.1907 (min) to 15 (max).



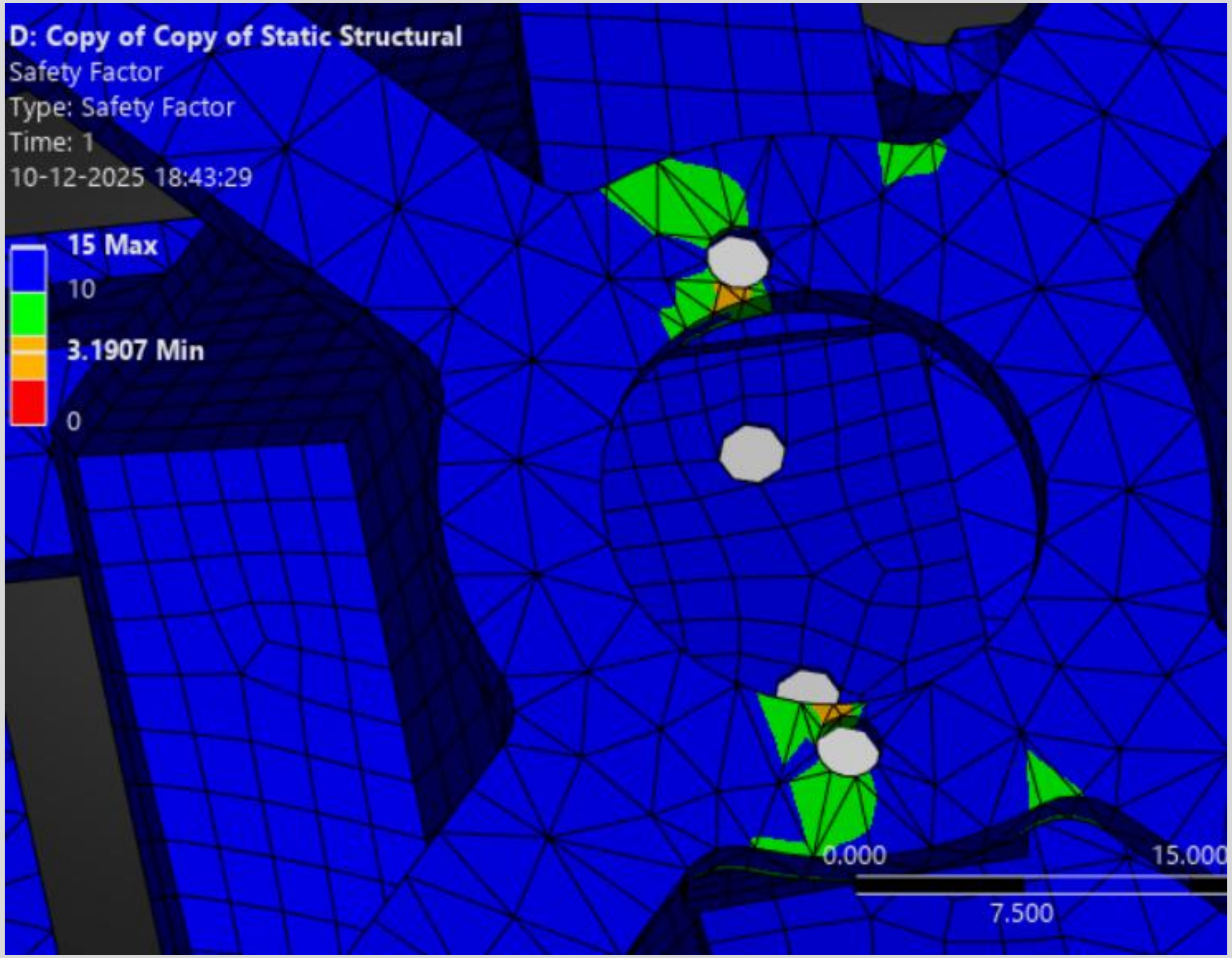


Fig.36 Margins of Safety Plot for Z - Axis

FORCE REACTION RESULTS & HAND CALCULATION COMPARISON- ITERATION 1 Z-AXIS

Applied Load: 300 N (3 kg × 10g)
Reaction Force Sum: ~ 370 **Error:** < 23 %

Force reaction table shows forces at fixed supports matching applied load.

- **No Separation/Penetration:** Confirmed in both iterations.
- **Contact Pressure:** Within acceptable limits.

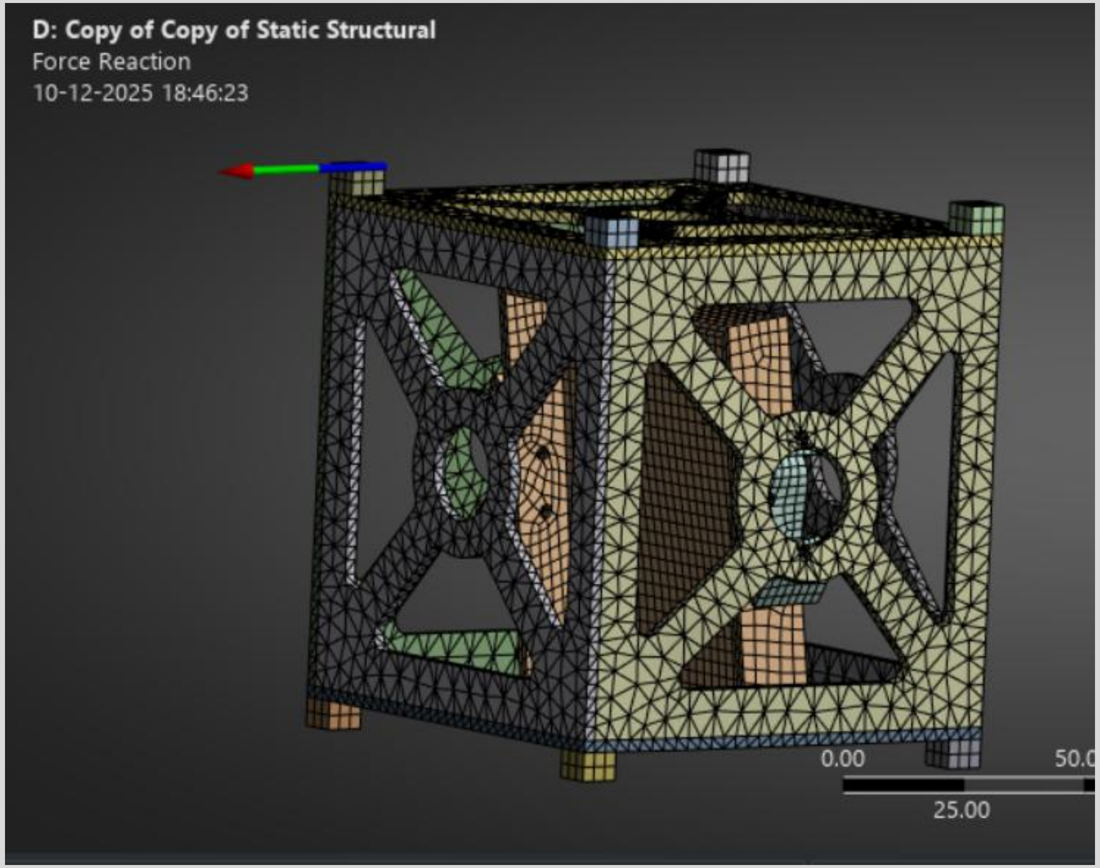
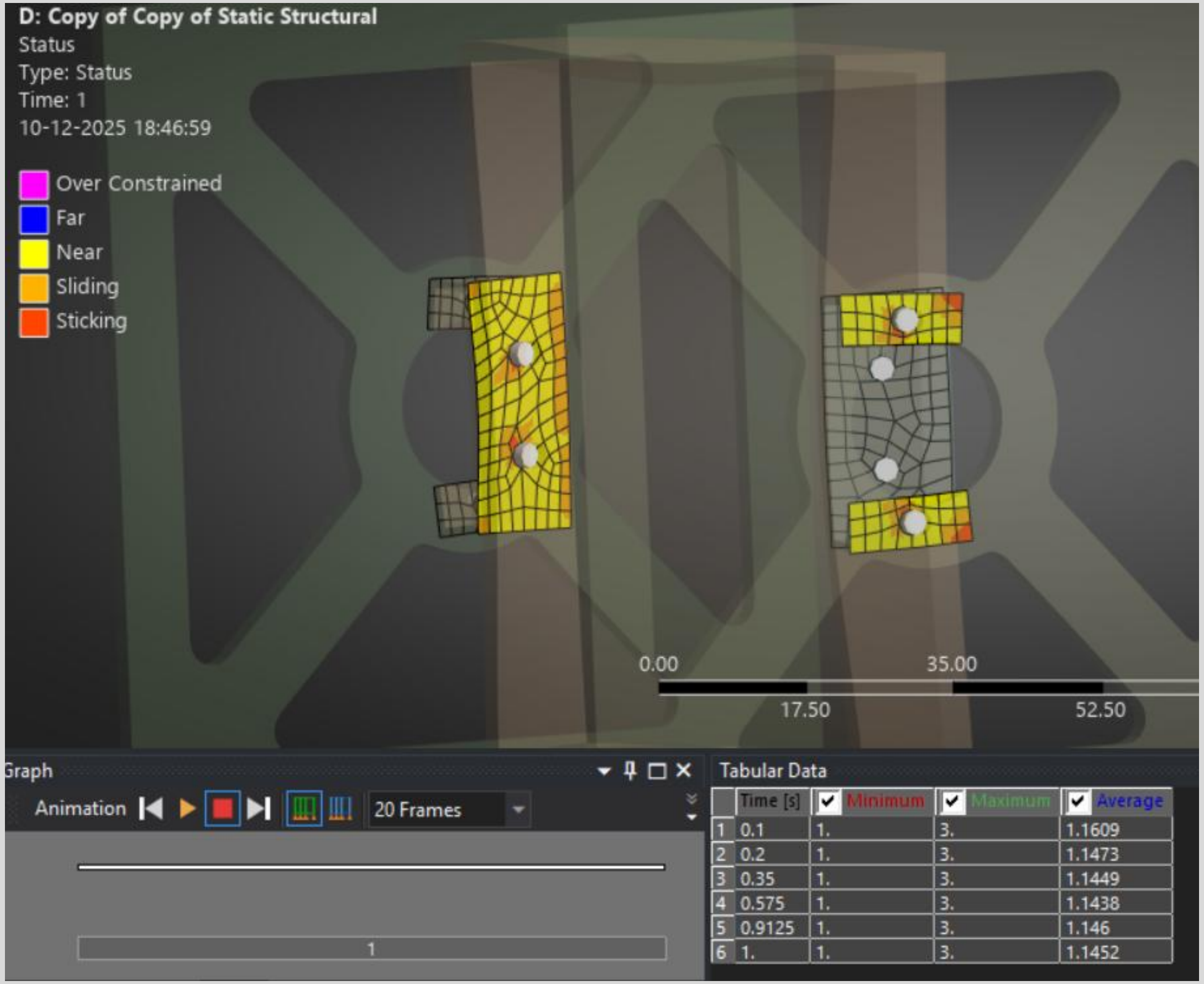


Fig.37 Force Reaction Results Plot for Z - Axis

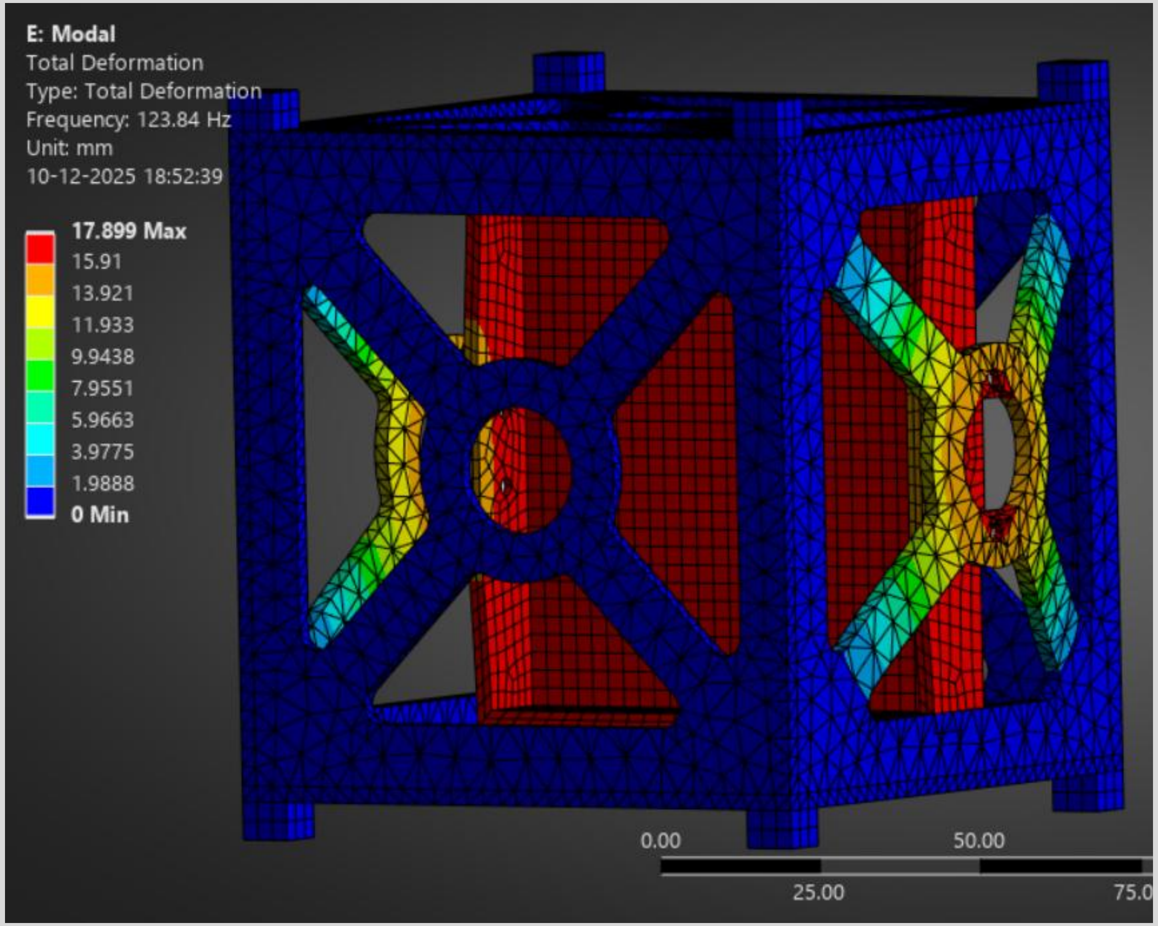


	Time [s]	Minimum	Maximum	Average
1	0.1	1.	3.	1.1609
2	0.2	1.	3.	1.1473
3	0.35	1.	3.	1.1449
4	0.575	1.	3.	1.1438
5	0.9125	1.	3.	1.146
6	1.	1.	3.	1.1452

Fig.38 Contact Result for Z - Axis

MODAL ANALYSIS RESULTS - ITERATION 1

- **First Natural Frequency:** 123 Hz
- **Mode Shape:** First bending mode of bracket arms.
- **Compliance:** All frequencies >65 Hz.



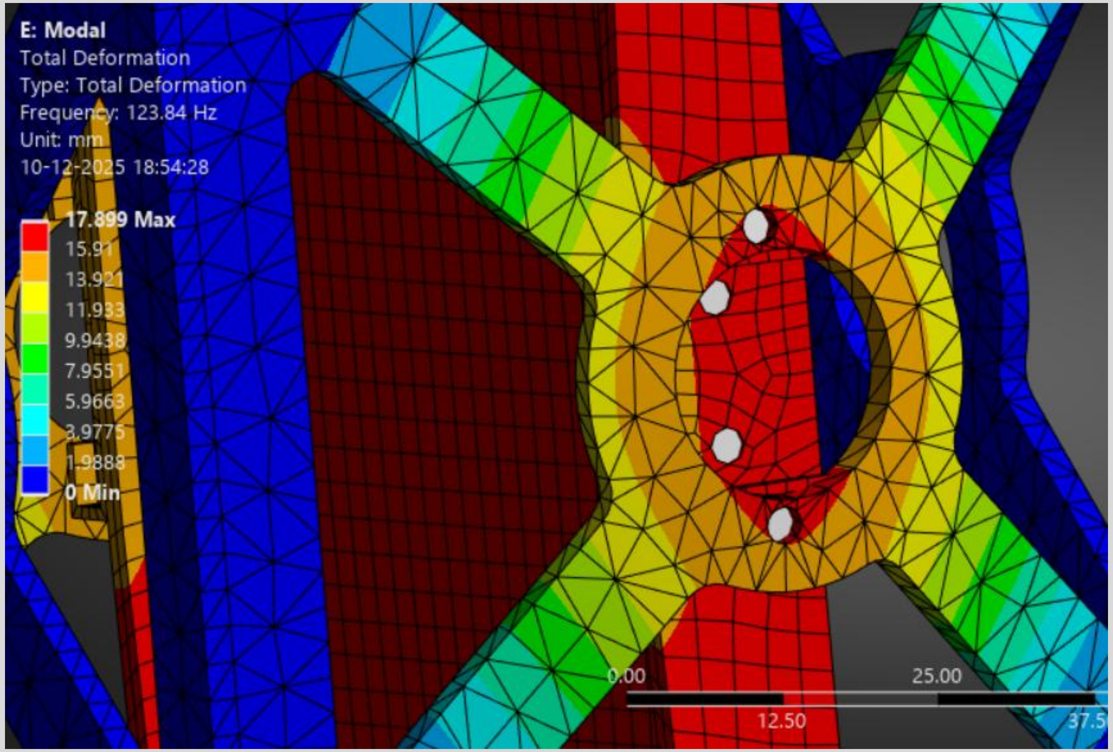


Fig.39 Mode shape plot shows deformation pattern of first mode with frequency labeled

PART DIMENSIONS- ITERATION 2

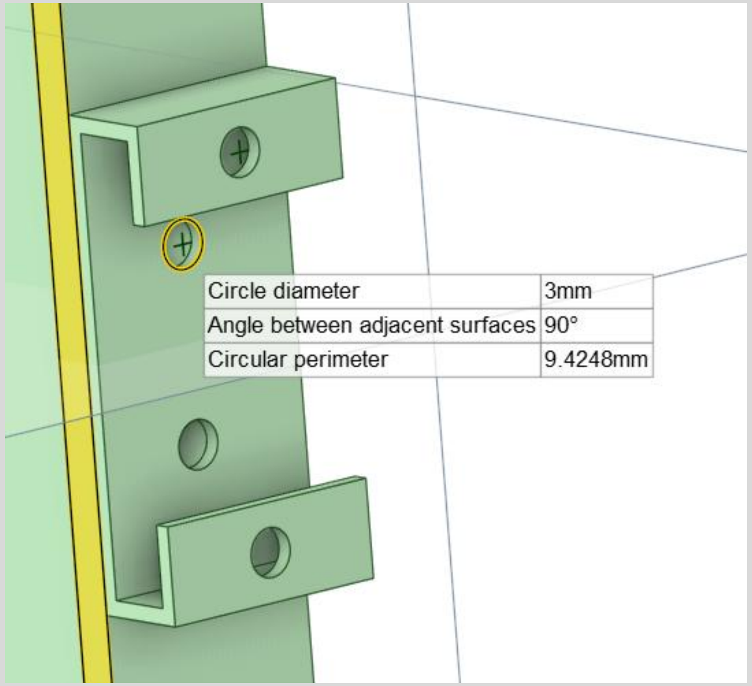


Fig.40

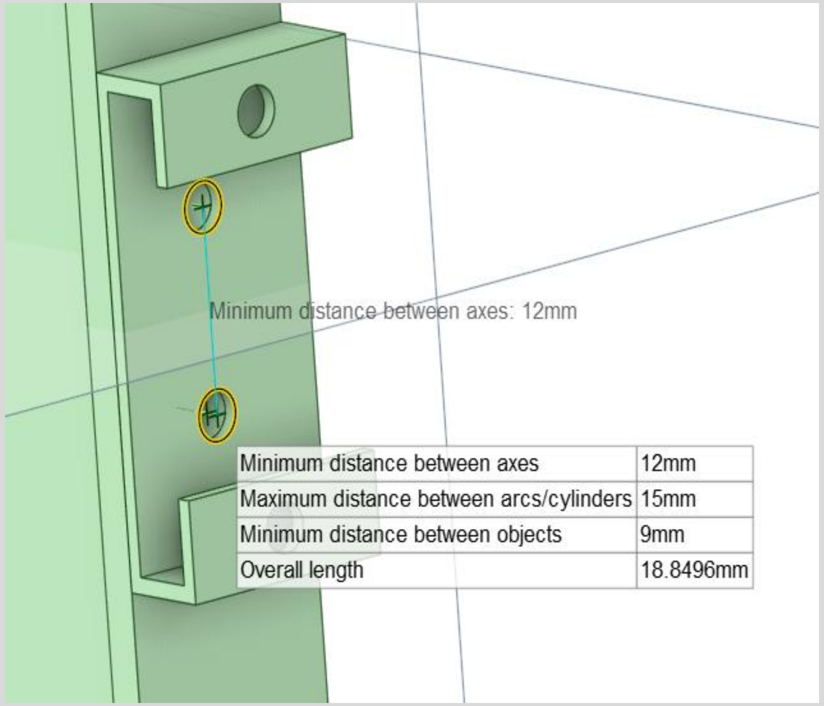


Fig.41

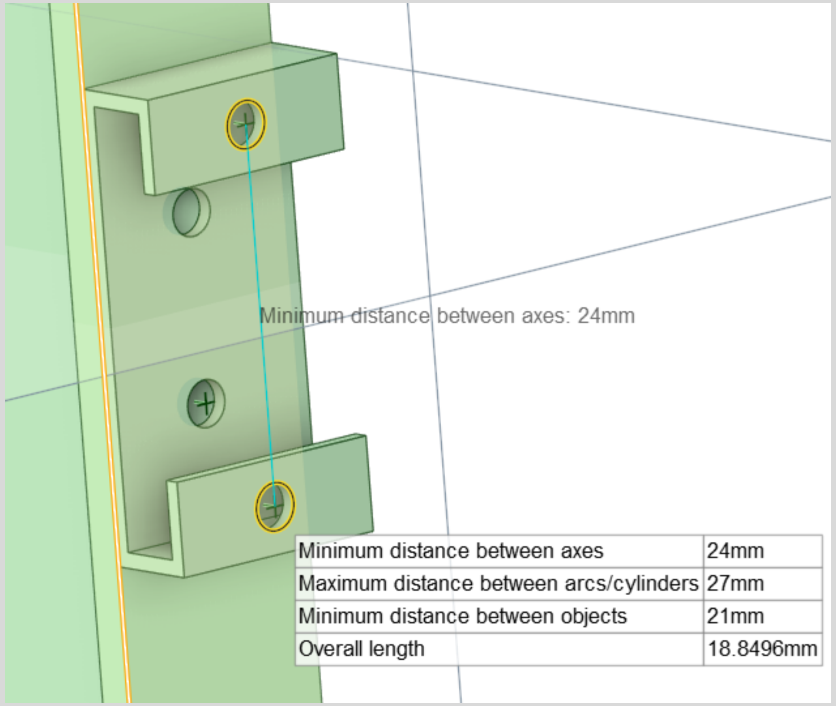


Fig.42

Model Creation & Materials

A refined 3D CAD model of the bracket for **Iteration 2** was developed in **Ansys Space-Claim** using parametric design principles. While maintaining the same two key dimensional parameters for consistency, geometric enhancements were implemented. The primary material selected for this optimized iteration was **AISI 303 Stainless Steel**, chosen for its high strength, excellent corrosion resistance, and ease of machining, making it suitable for aerospace fastener and bracket applications. All material properties, including yield strength (≥205 MPa) and modulus of elasticity (193–200 GPa), were sourced from **Mat Web** to ensure reliable and traceable data.

PART DIMENSIONS- ITERATION 2

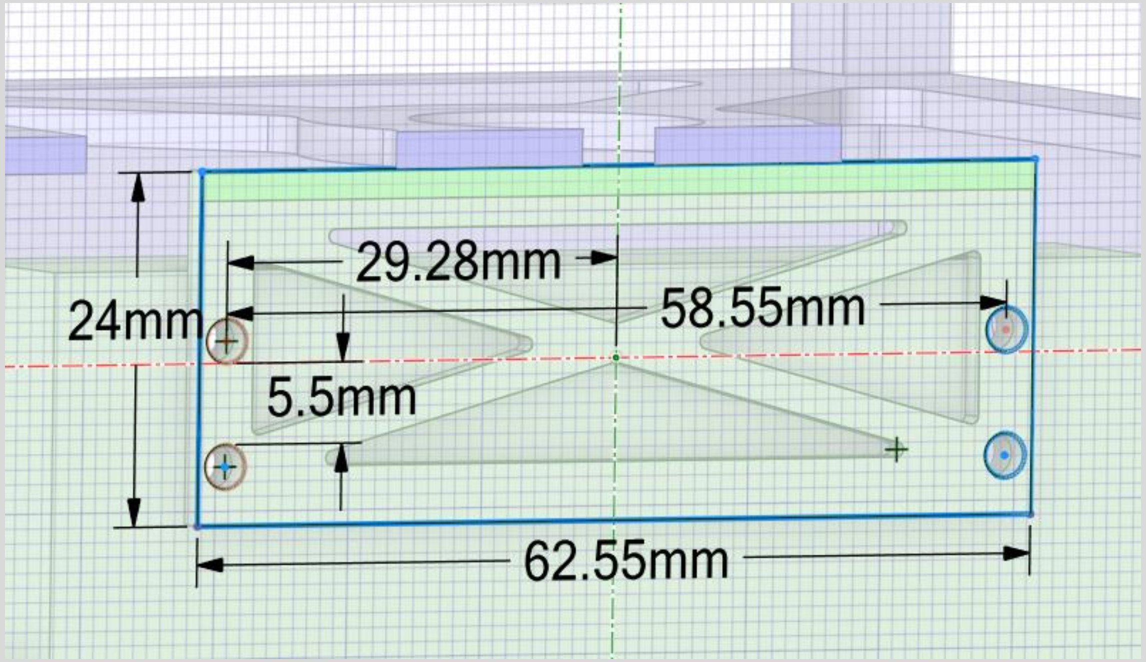


Fig.43

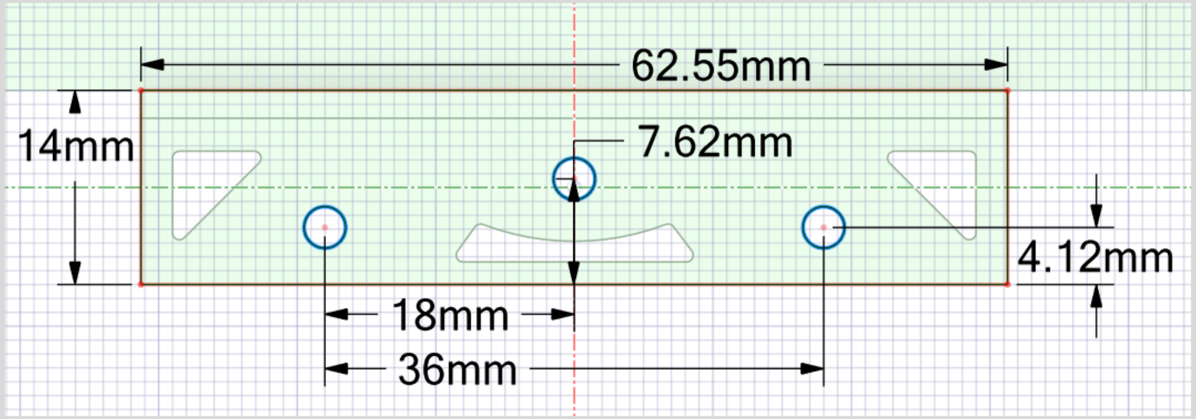
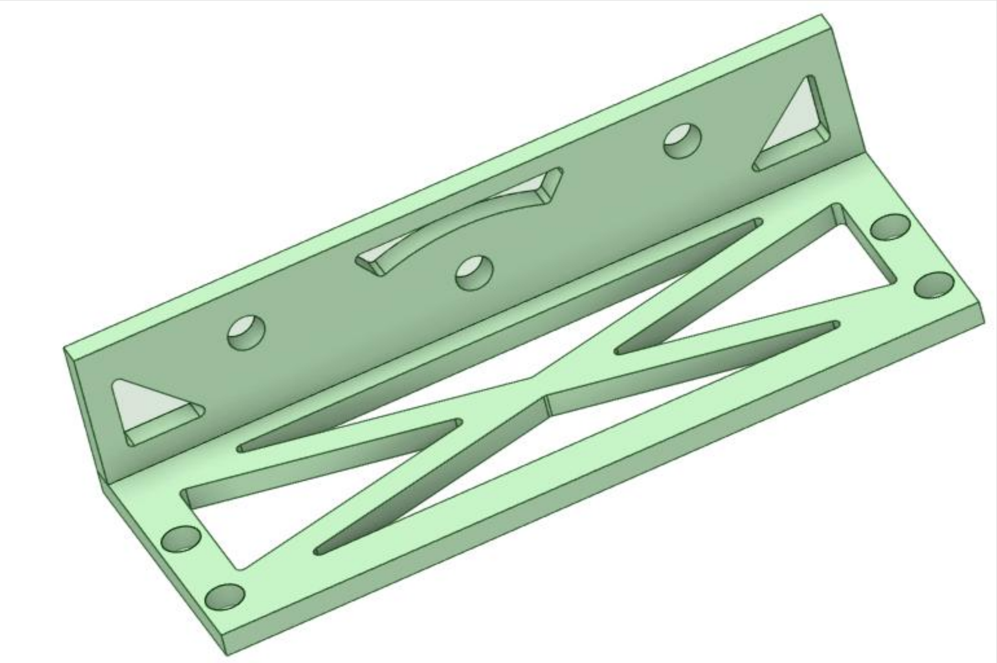


Fig.44



Isometric View

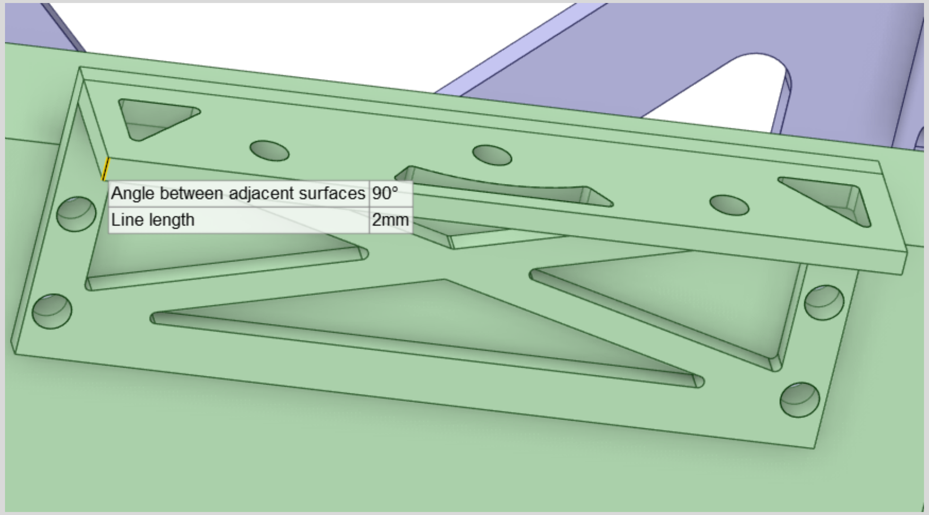


Fig.45

ENGINEERING DATA - ITERATION 2

Outline of Schematic B2, C2, D2, E2: Engineering Data

	A	B	C	D	E
1	Contents of Engineering Data			Source	Description
2	Material				
3	AISI 303 Stainless Steel			C	

Properties of Outline Row 3: AISI 303 Stainless Steel

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	8.03E-06	kg mm^-3		
4	Isotropic Secant Coefficient of Thermal Expansion				
5	Coefficient of Thermal Expansion	1.72E-05	C^-1		
6	Isotropic Elasticity				
7	Derive from	Young's Modulus a...			
8	Young's Modulus	1.93E+05	MPa		
9	Poisson's Ratio	0.29			
10	Bulk Modulus	1.5317E+05	MPa		
11	Shear Modulus	74806	MPa		
12	Tensile Yield Strength	240	MPa		
13	Compressive Yield Strength	240	MPa		
14	Tensile Ultimate Strength	620	MPa		
15	Compressive Ultimate Strength	0	MPa		

Analysis and Results – Iteration 2

Same Boundary conditions used for Static Structural Analysis

MESHING DESCRIPTION - ITERATION 2

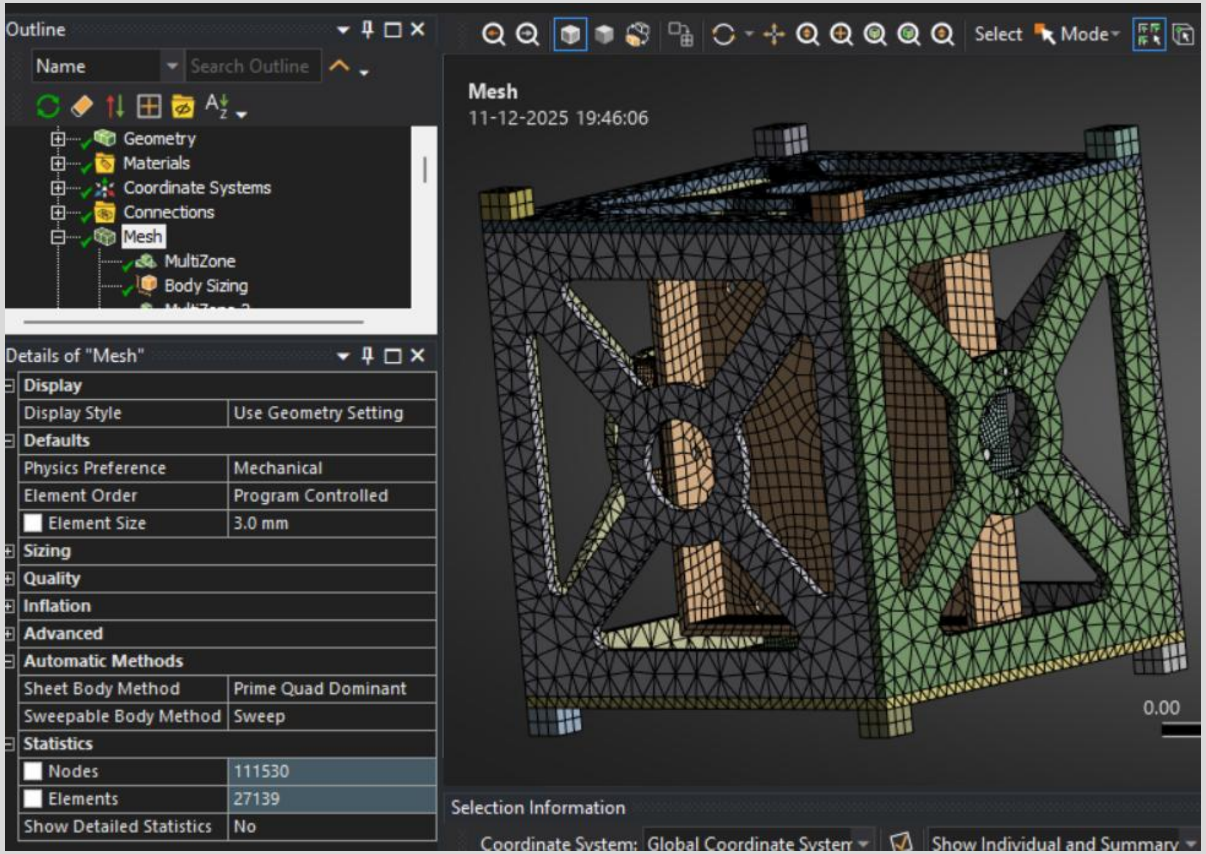
•**Method:** Multi Zone hexahedral meshing Method for structured elements.

•**Mesh Sizing used at brackets:** 1.0 mm element size.

•**Element Size:** 3.0 mm global sizing.

•**Total Elements:** ~ 27139

•**Total Nodes:** ~ 111530



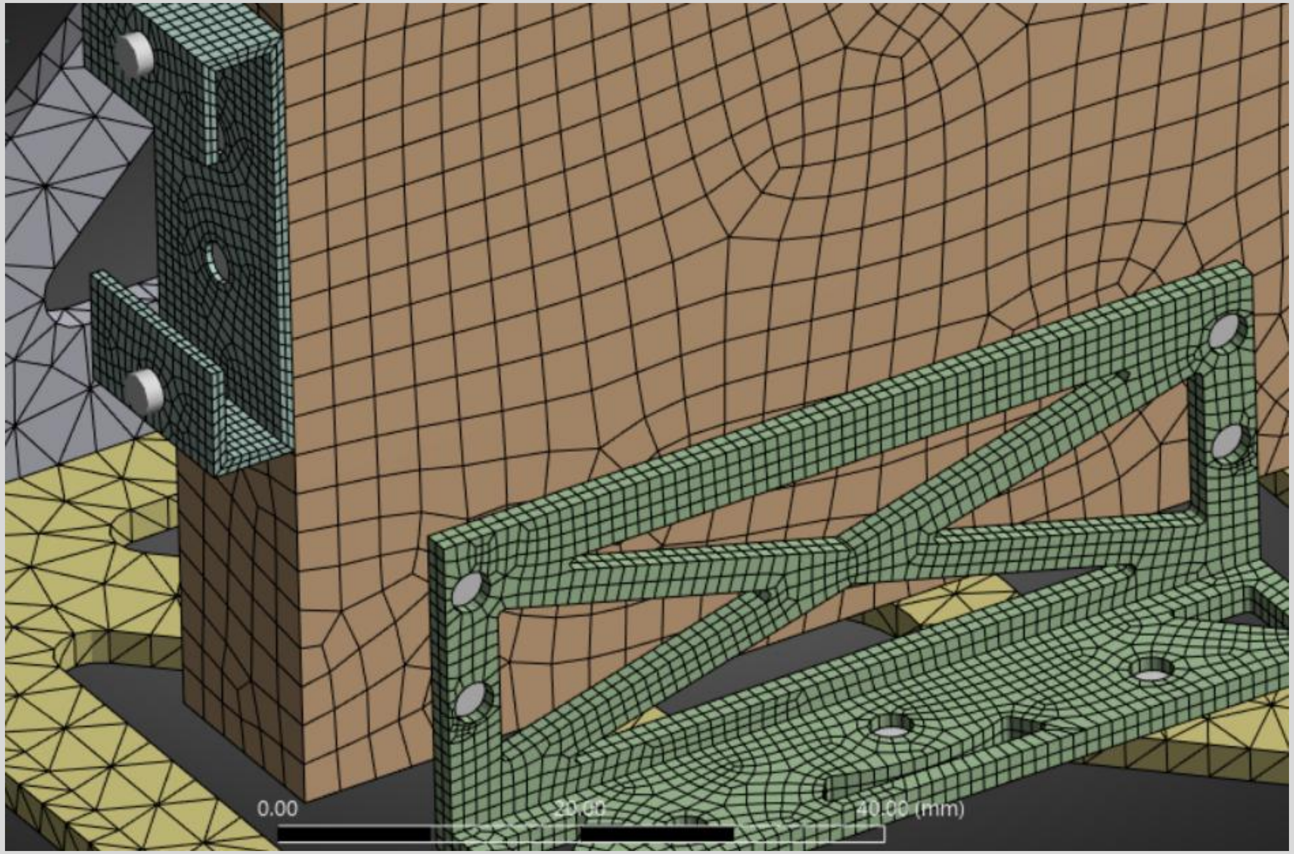


Fig.46 Mesh Details

BOUNDARY CONDITIONS - ITERATION 2

- **Point Mass:** Mass of 3 kg, attached to the electronics box surface. The mass position in central to the box and 10mm off the inside surface.

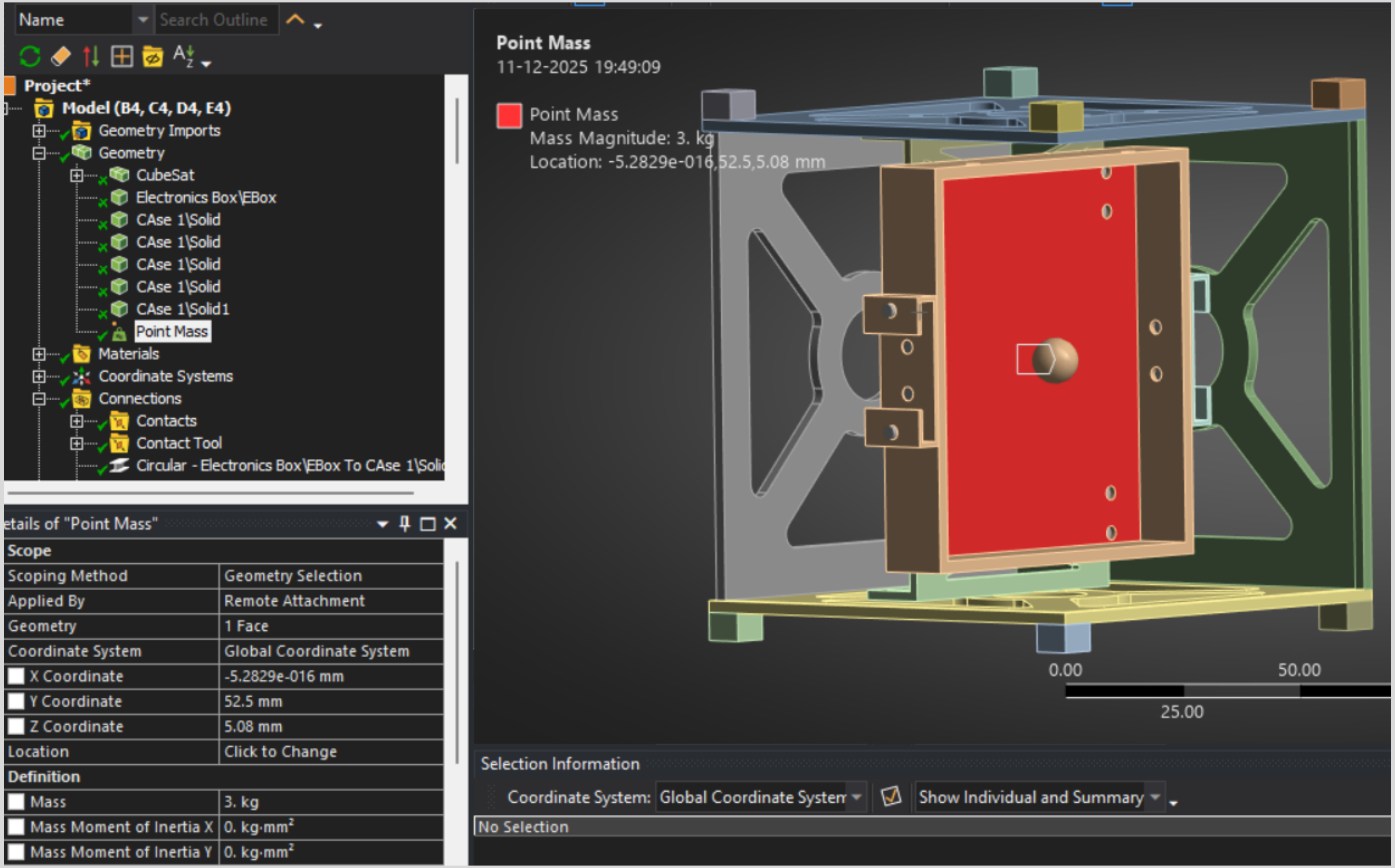


Fig.47 Point Mass

CONNECTIONS AND CONTACT - ITERATION 2

- **Frictional Contact:** Coefficient 0.2 between all contacting surfaces.

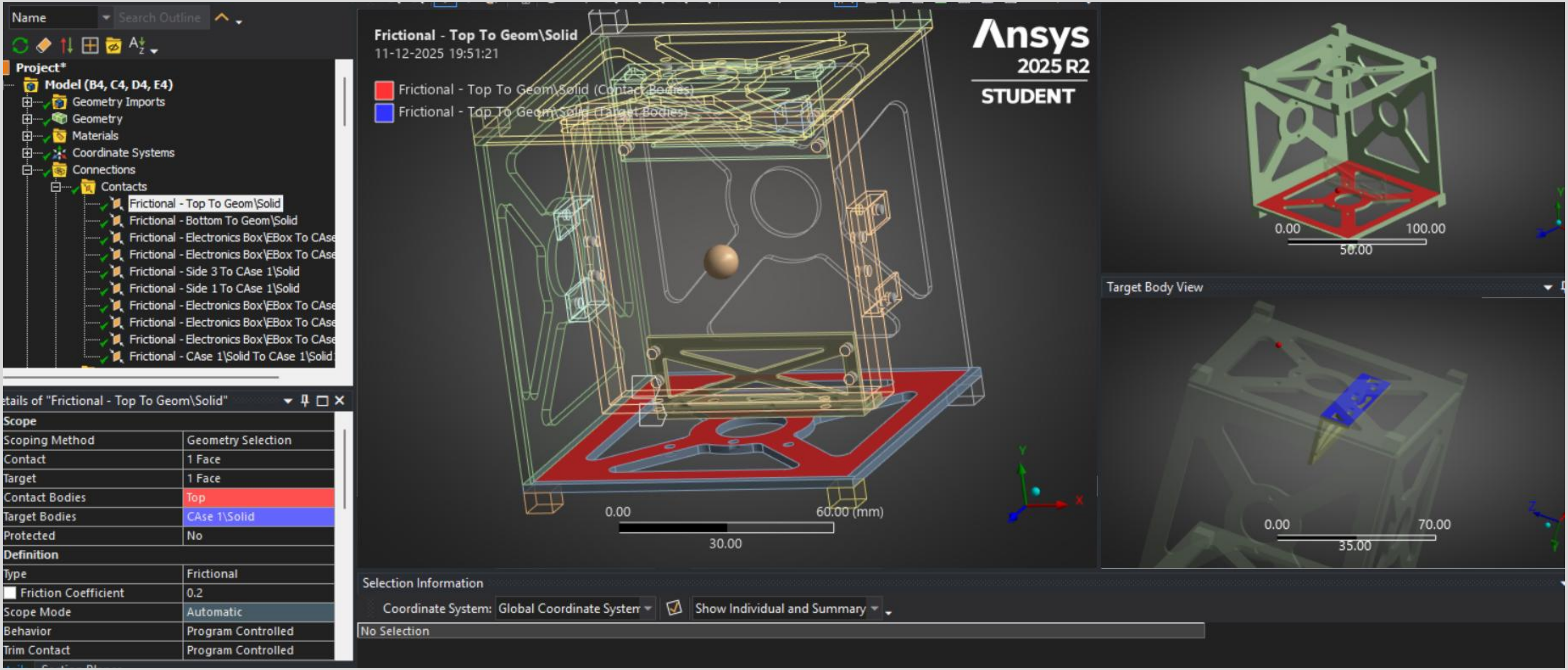


Fig.48 Contact Details

CONNECTIONS AND CONTACT - ITERATION 2

•**Beam body to body Connections:** Used for all fastener representations of 1.5mm Radius and 1.75 mm length of beam.

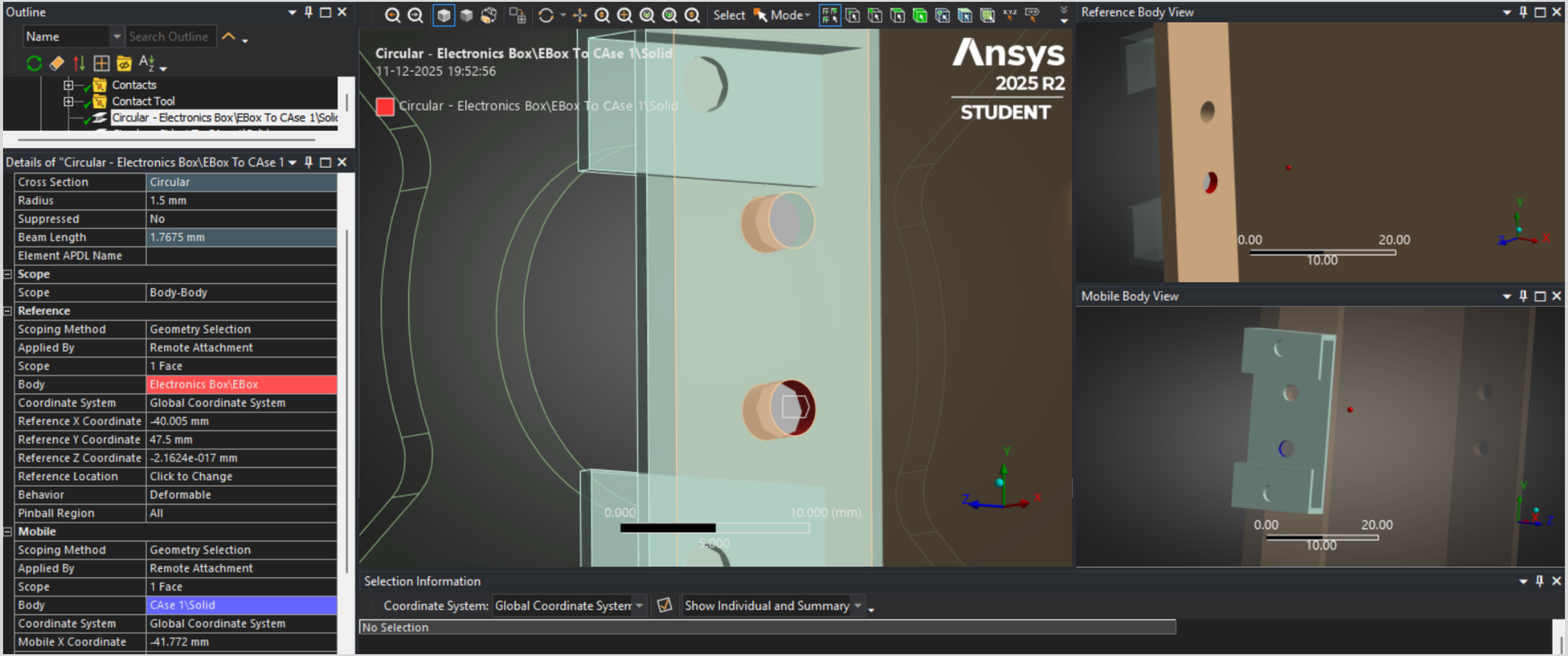
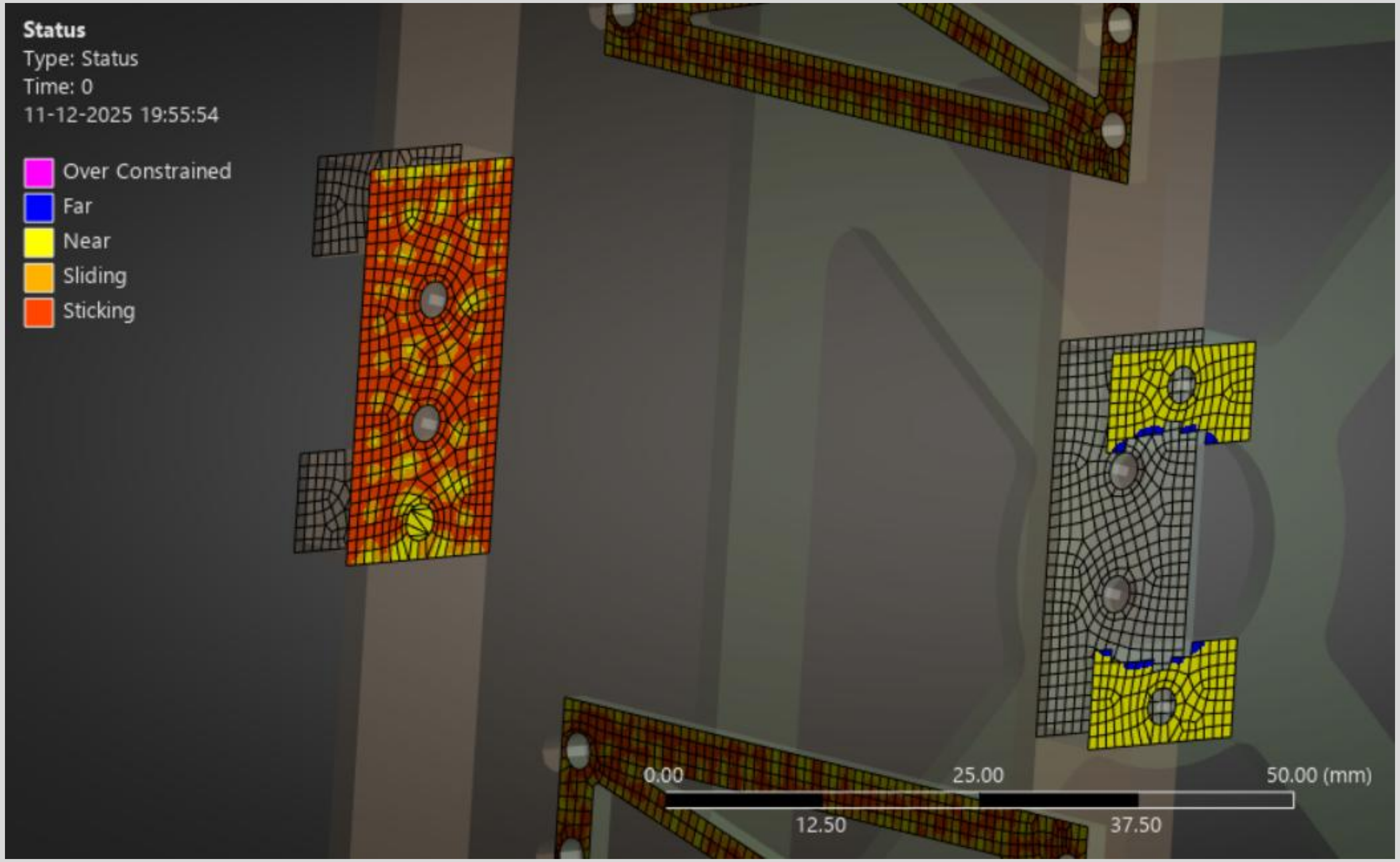


Fig.49 Beam Connection

CONTACT TOOL - ITERATION 2

- **No separation or penetration.**
- **Stable frictional behavior** confirmed across all load steps.



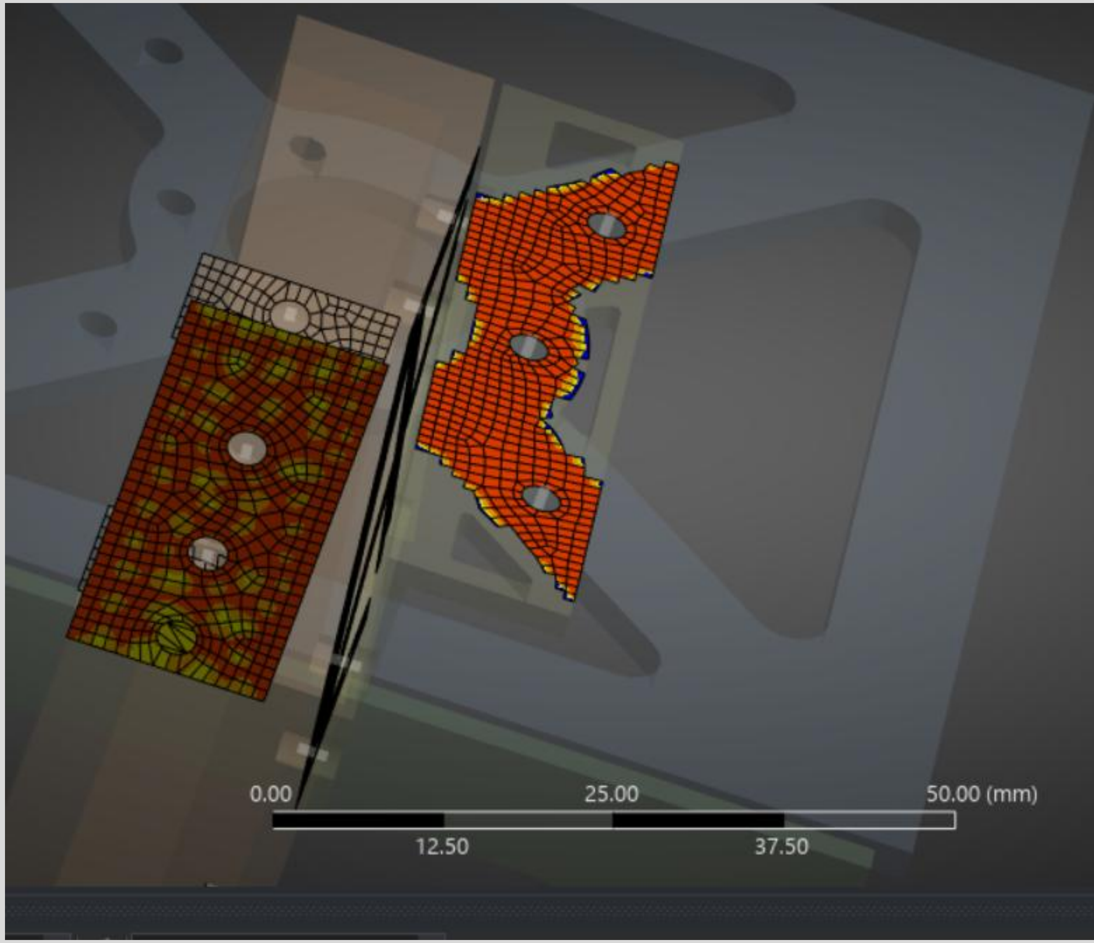
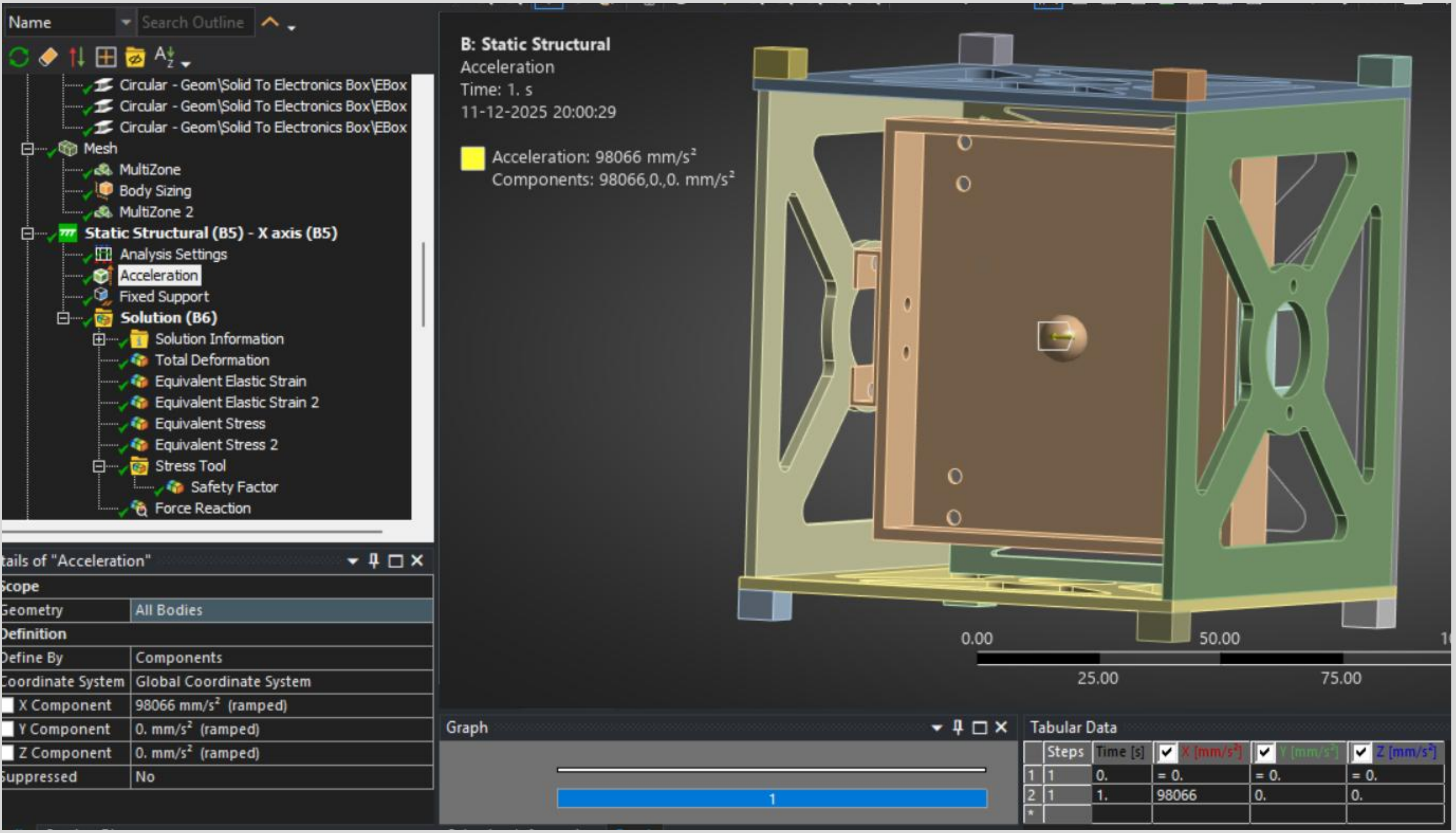


Fig.50 Status Result

ANALYSIS AND RESULTS - ITERATION 2

- **Acceleration Loads:** 98,066 mm/s² applied separately in X global directions.
- **Load Application:** Applied to all bodies in the assembly.



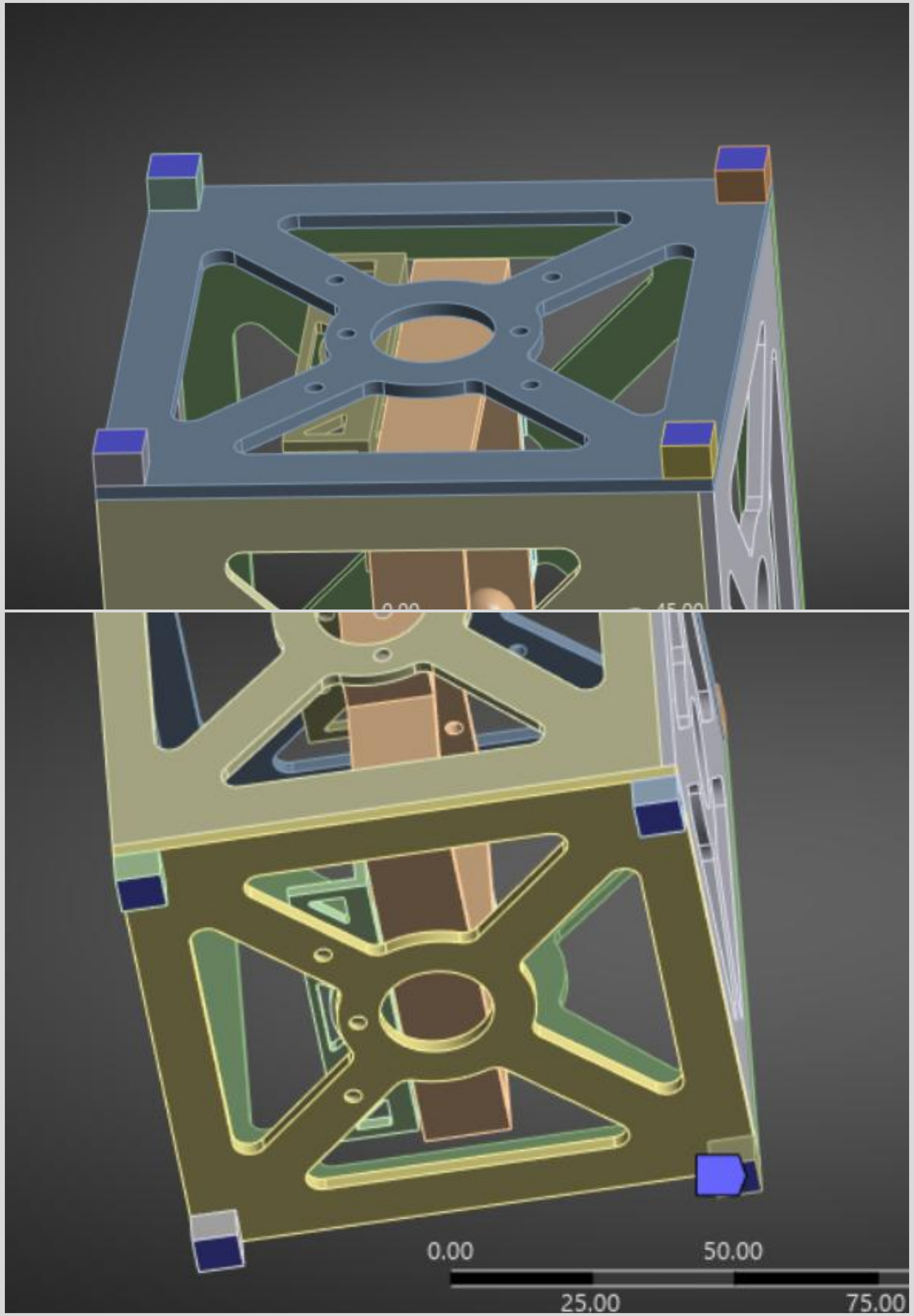


Fig.51 Acceleration Loads at X - Axis

ANALYSIS AND RESULTS - ITERATION 2

•The force convergence plot shows the solution residuals (in Newtons) versus iteration count for the nonlinear static analysis with frictional contact under X-direction 10g acceleration.

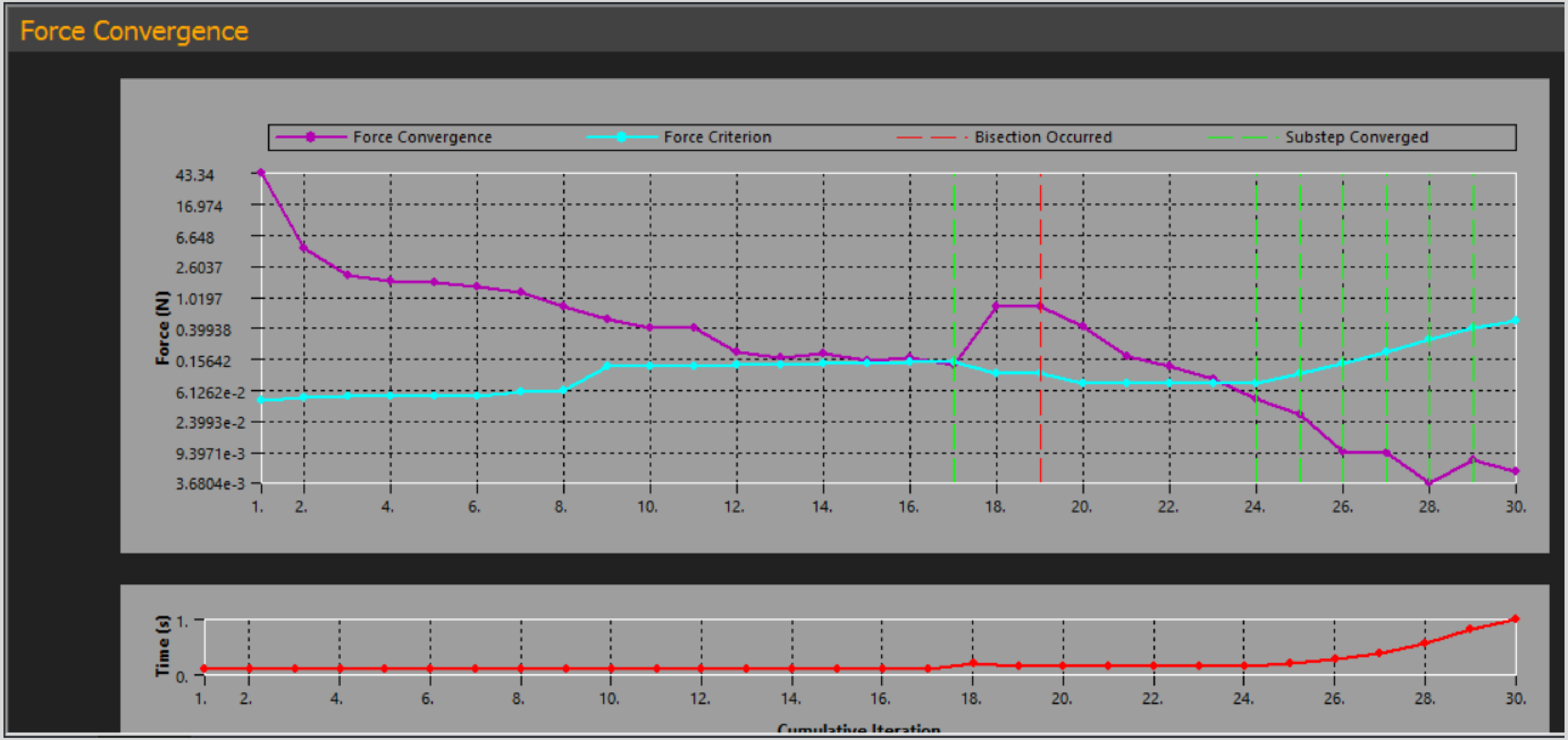
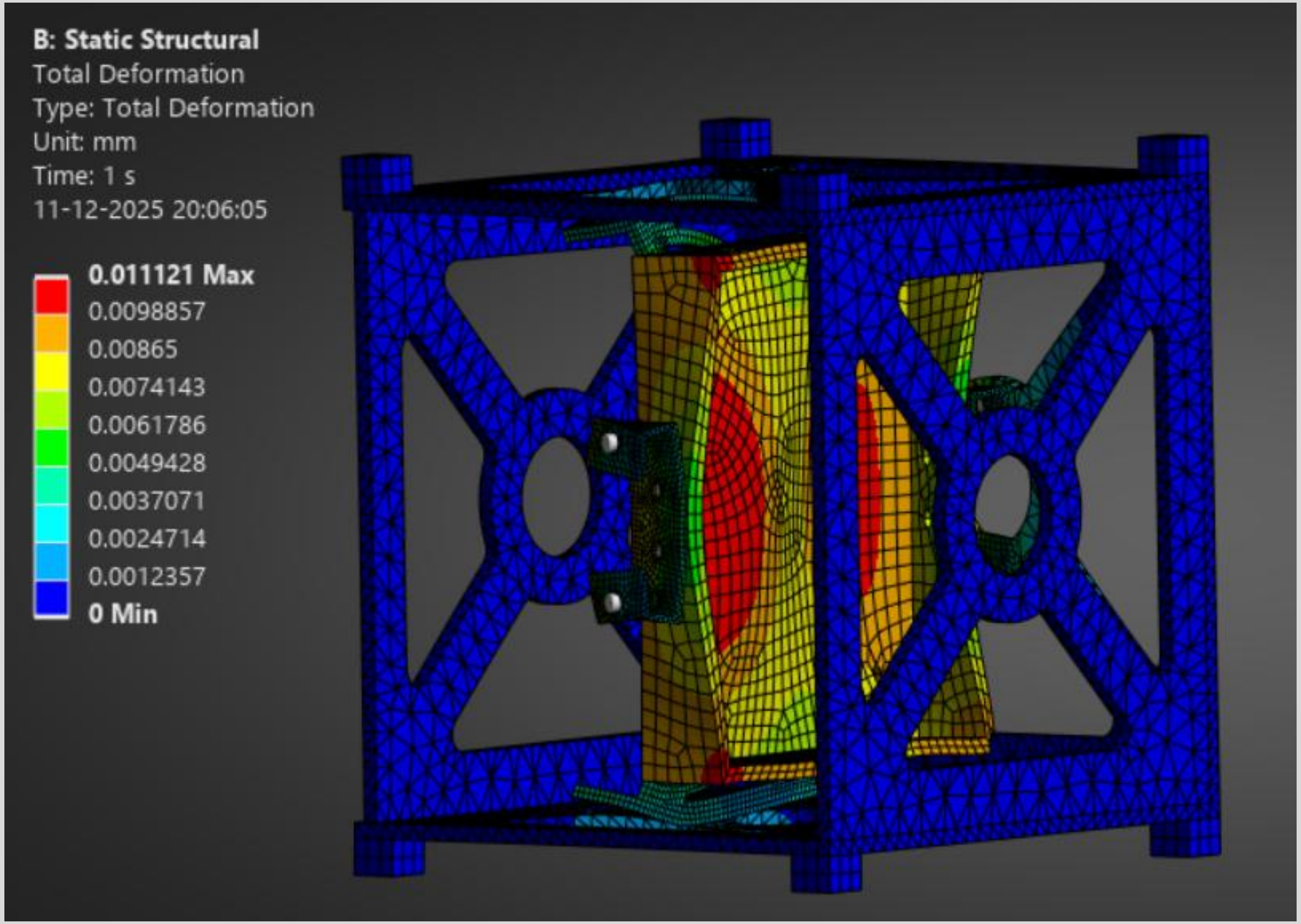
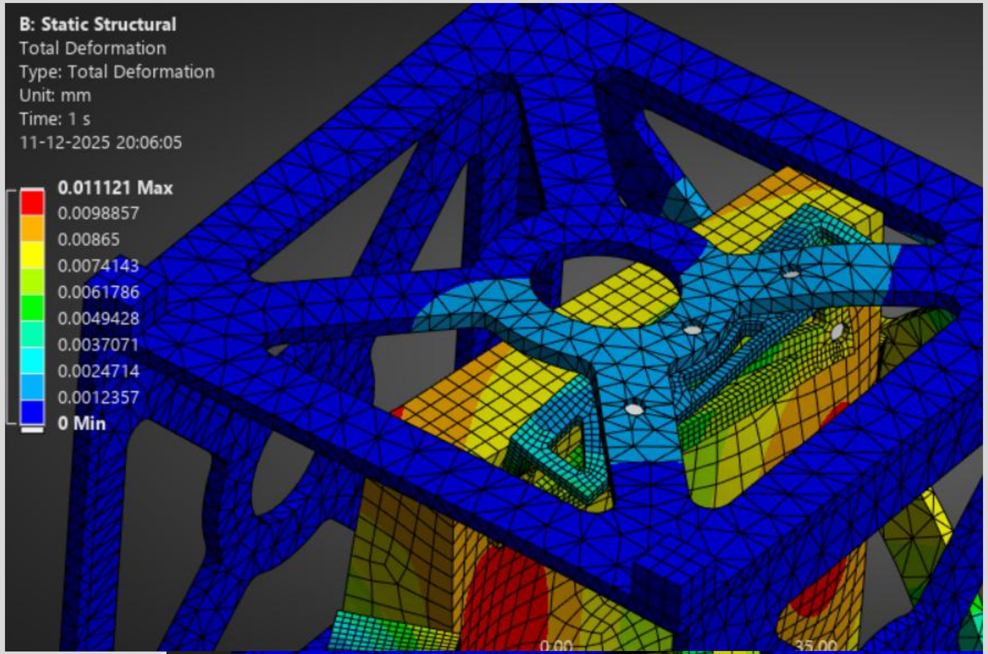


Fig.52 Force Convergence Plot for X - Axis

DEFORMATION RESULTS - ITERATION 2 X-AXIS

•Total deformation plot shows minimal displacement, scale 0–0.01121 mm.





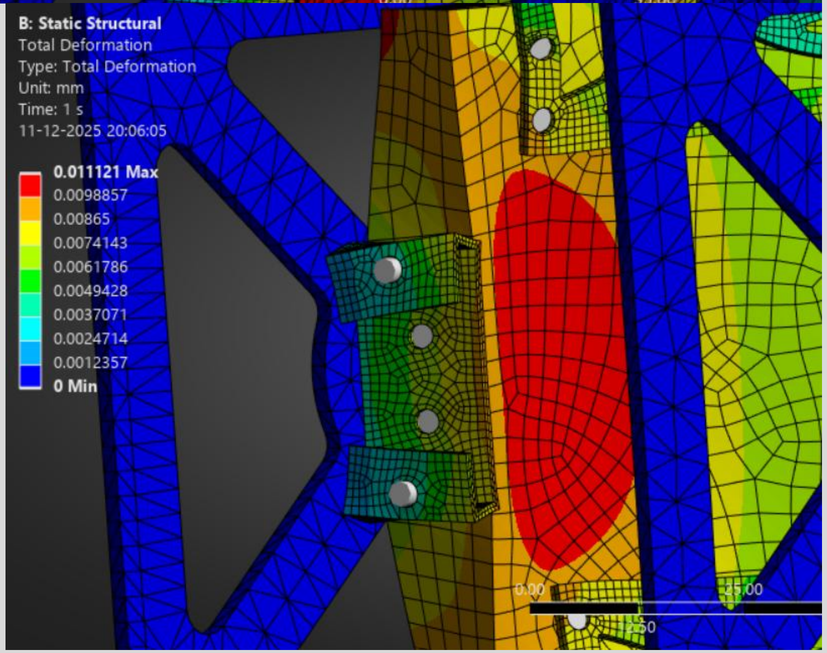
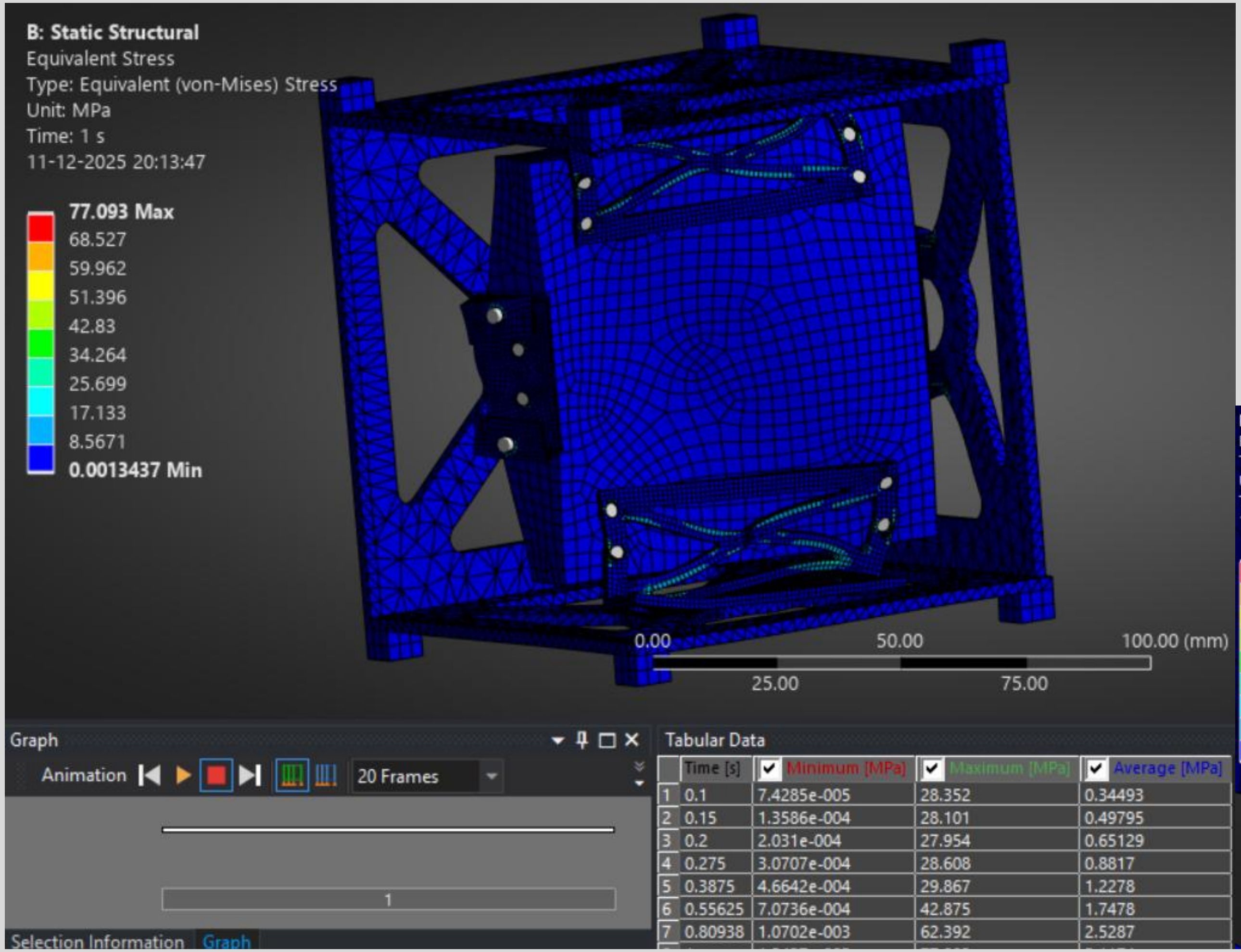
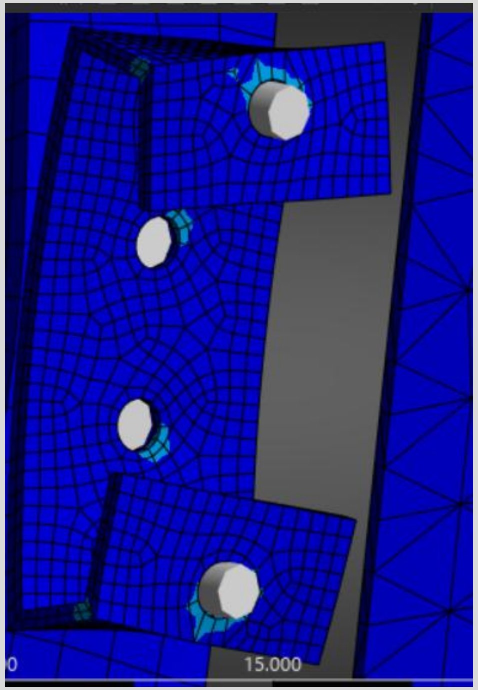


Fig.53 Total Deformation Plot for X - Axis

STRESS RESULTS - ITERATION 2 X-AXIS

- Equivalent stress plot shows lower stress than Iteration 1 due to geometry improvements.
- **Max Stress:** 77.093 MPa





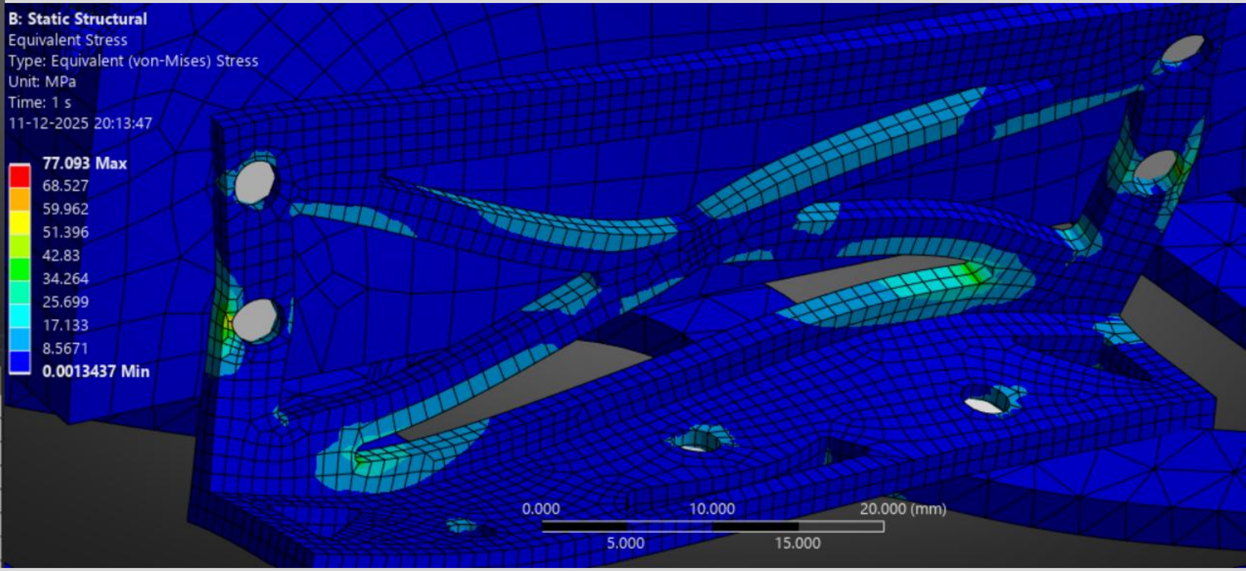


Fig.54 Equivalent stress plot (averaged) for X - Axis

Fig.55 close-up stress plots of critical region

STRESS RESULTS - ITERATION 2 X-AXIS

•Equivalent (Unaveraged) stress plot highlights sharp gradients at hole edges. Scale: 0.0009132 to 81.296 MPa.

•**Max Stress:** 81.296 MPa

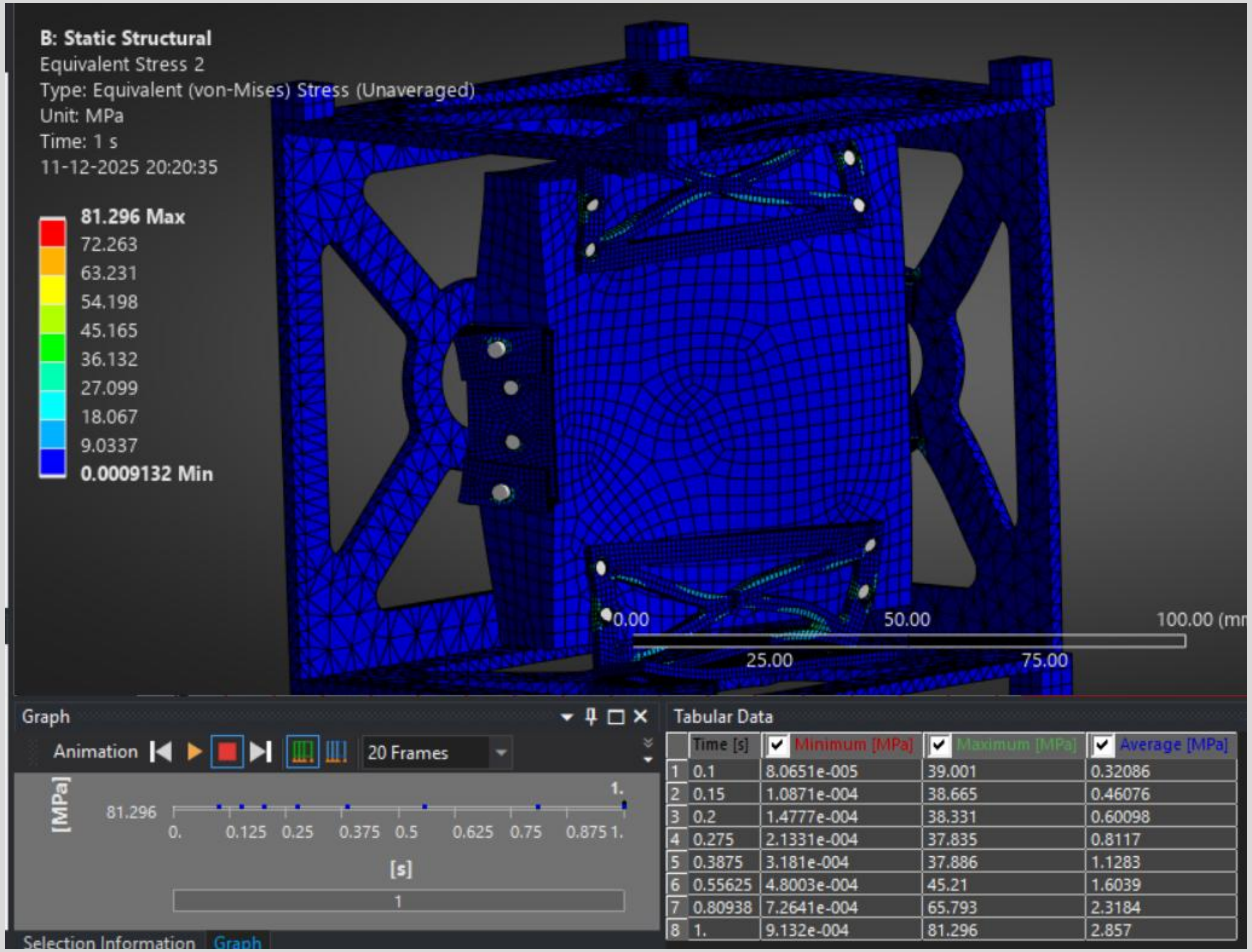
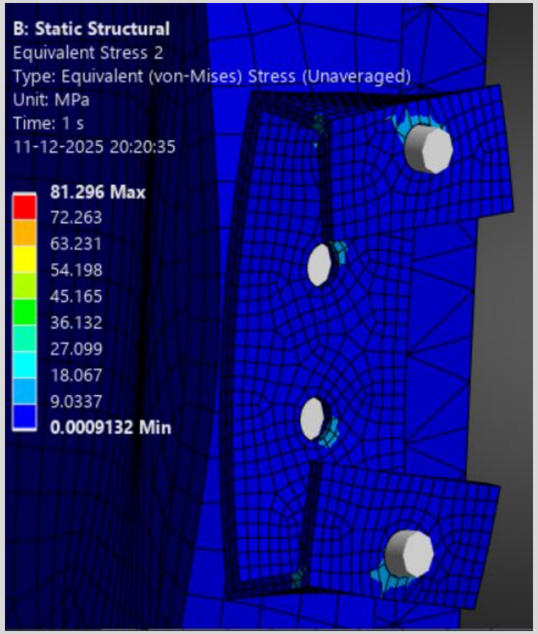


Fig.56 Equivalent stress plot (unaveraged) for X - Axis



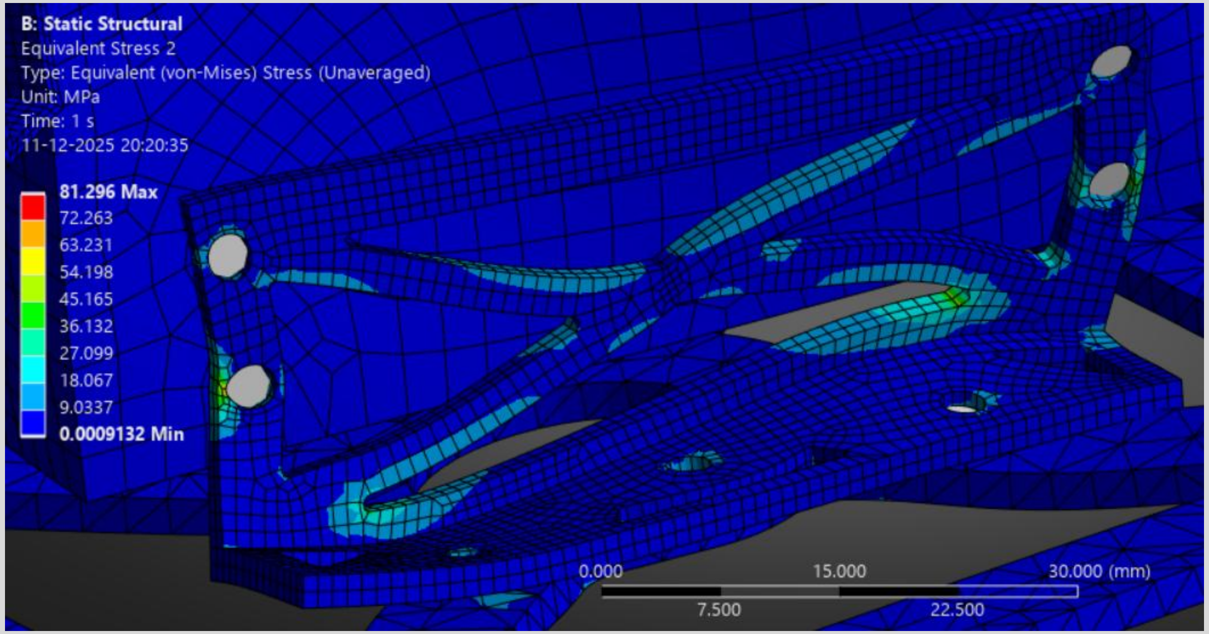


Fig.57 close-up stress plots of critical region (unaveraged) for X - Axis

MARGINS OF SAFETY - ITERATION 2 X-AXIS

- **Min Safety Factor:** 3.1131 (X-load case)
- Safety factor contour plot shows values from 3.1131 (min) to 15 (max), higher than Iteration 1.

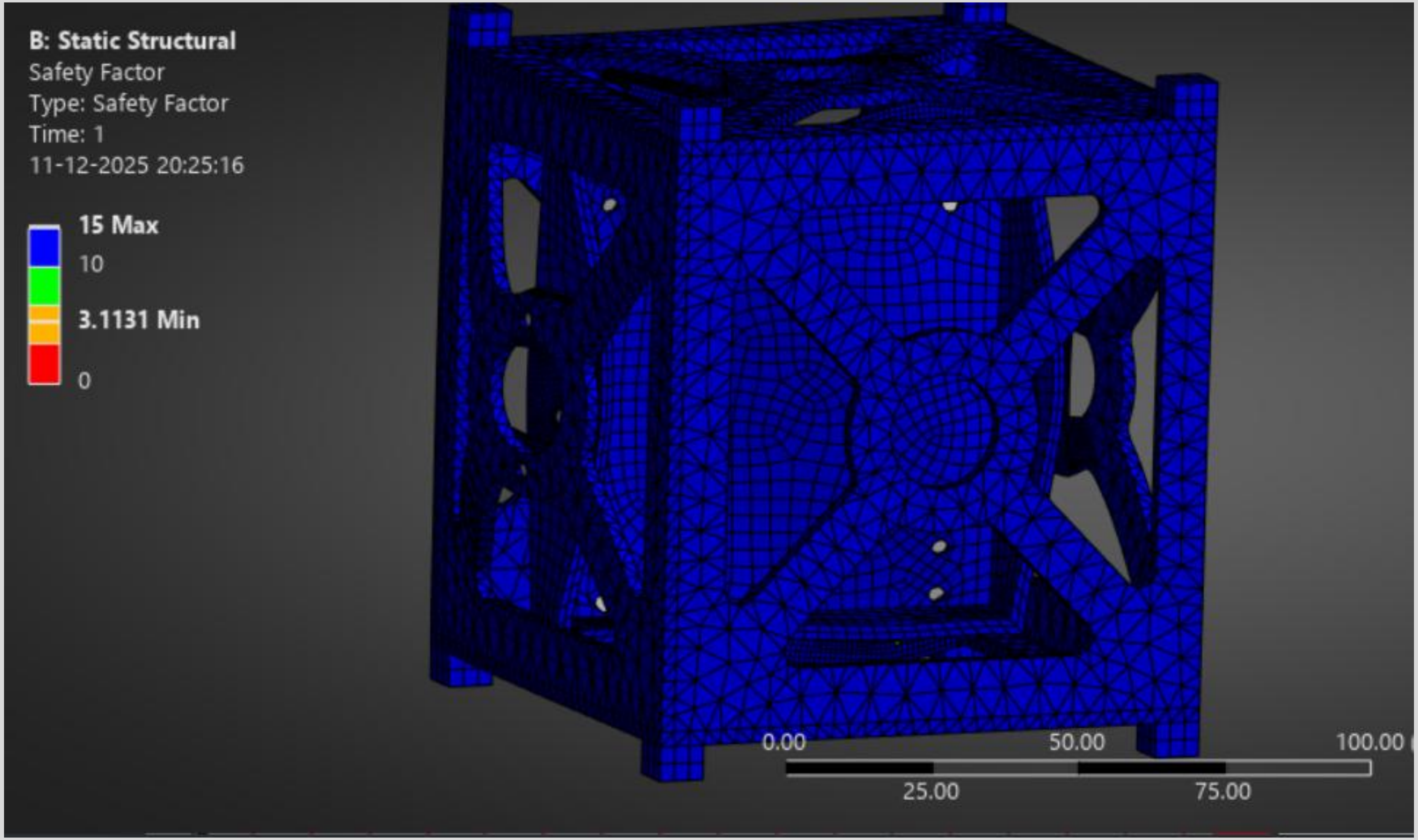


Fig.58 Margins of Safety Plot for X - Axis

FORCE REACTION RESULTS & HAND CALCULATION COMPARISON- ITERATION 2 : X-AXIS

Applied Load: 300 N (3 kg × 10g)
Reaction Force Sum: ~ 375 **Error:** < 23.5 %

- Force reaction table shows forces at fixed supports matching applied load.

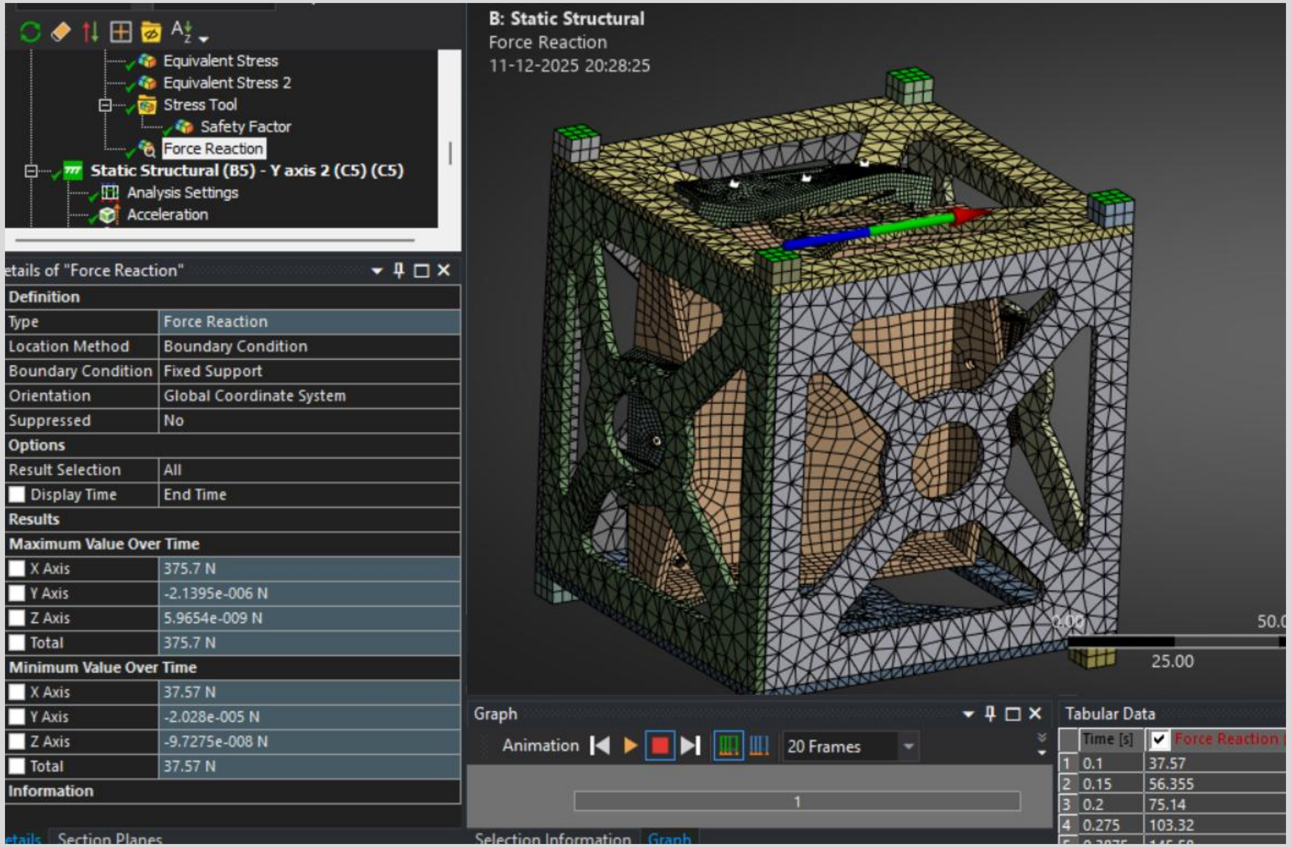


Fig.59 Force Reaction Results Plot for X - Axis

- **No Separation/Penetration:** Confirmed in both iterations.
- **Contact Pressure:** Within acceptable limits.

- Contact tool plot shows pressure distribution at interfaces, with no gaps or penetration.

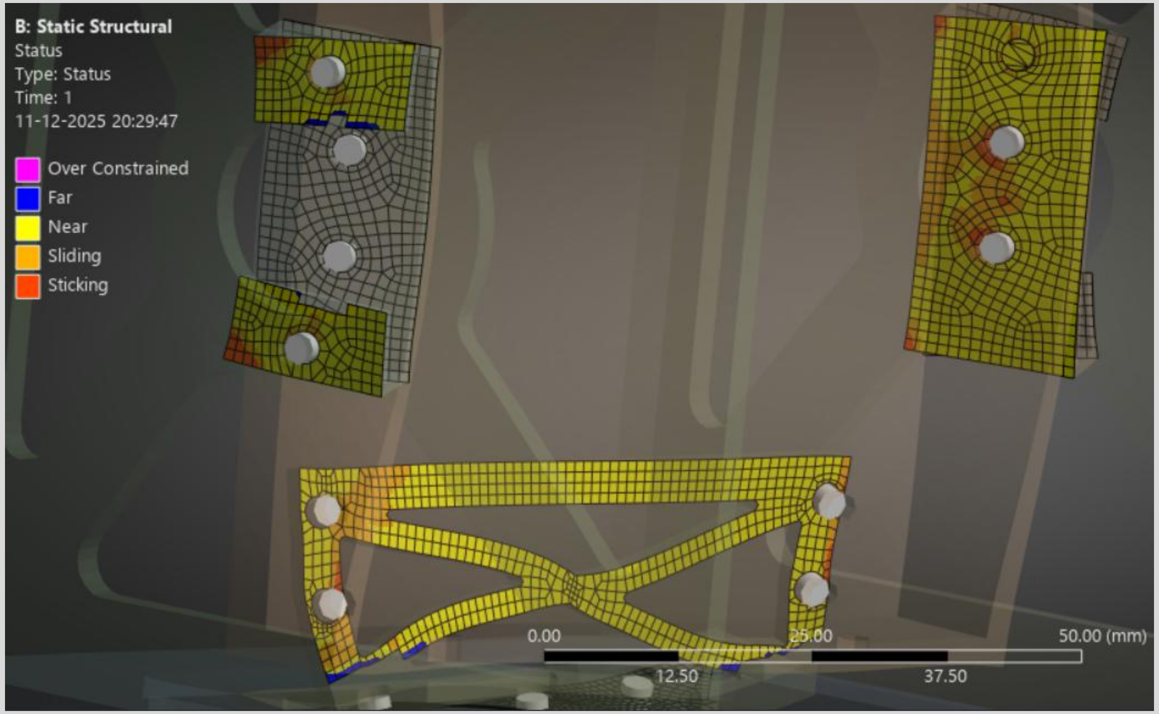


Fig.60 Contact Result for X - Axis

ANALYSIS AND RESULTS - ITERATION 2 Y-AXIS

•**Acceleration Loads:** 98,066 mm/s² applied separately in Y global directions.

•**Load Application:** Applied to all bodies in the assembly.

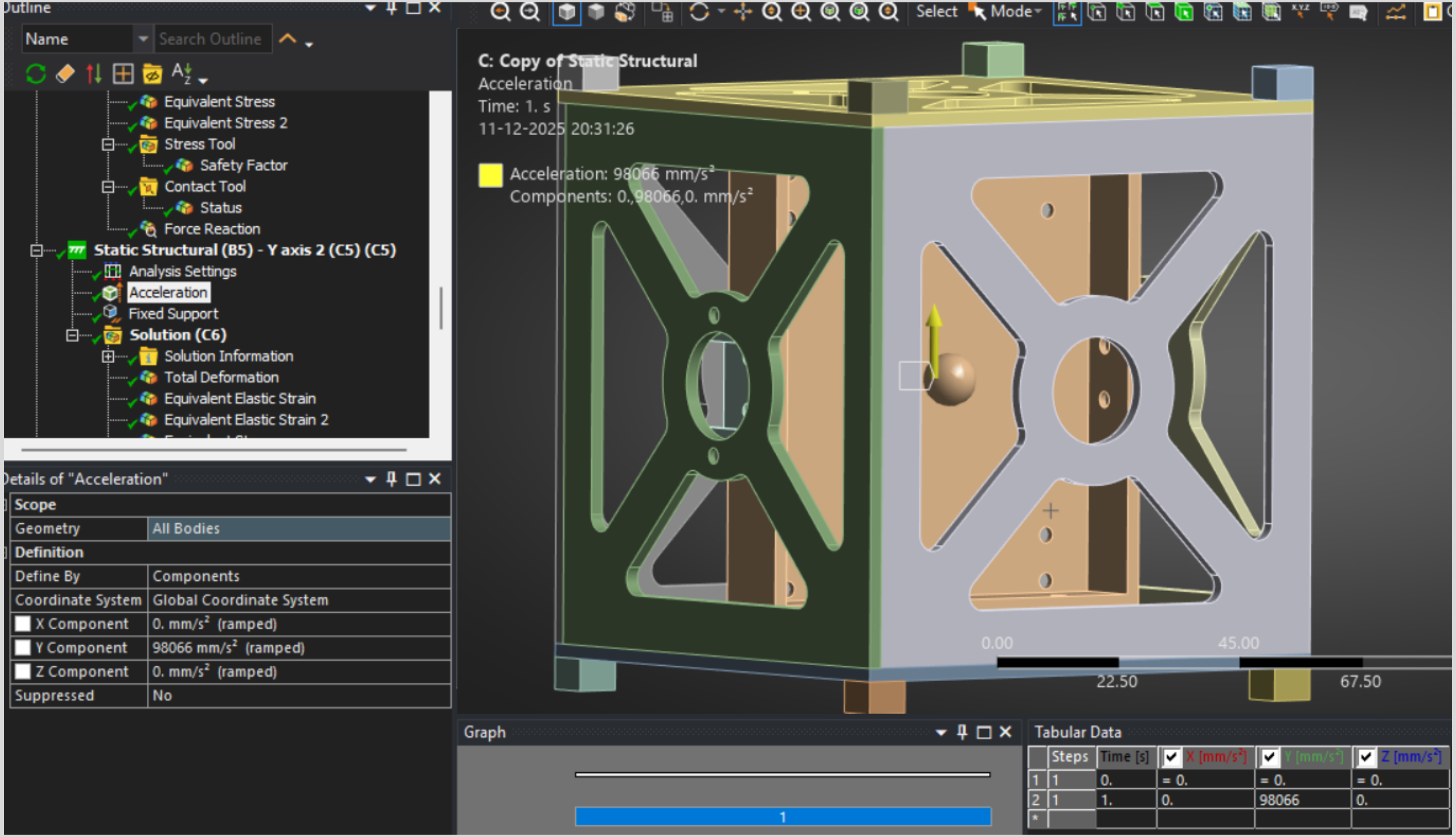
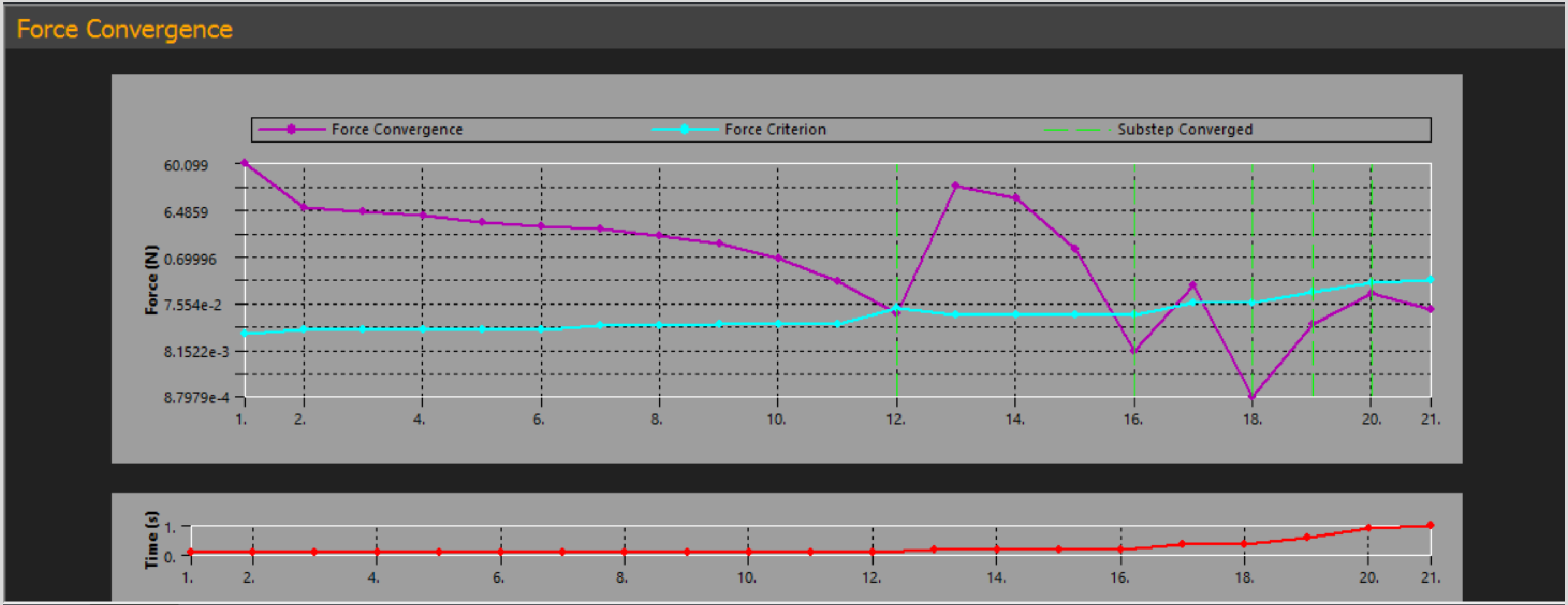


Fig.61 Acceleration Loads at Y - Axis

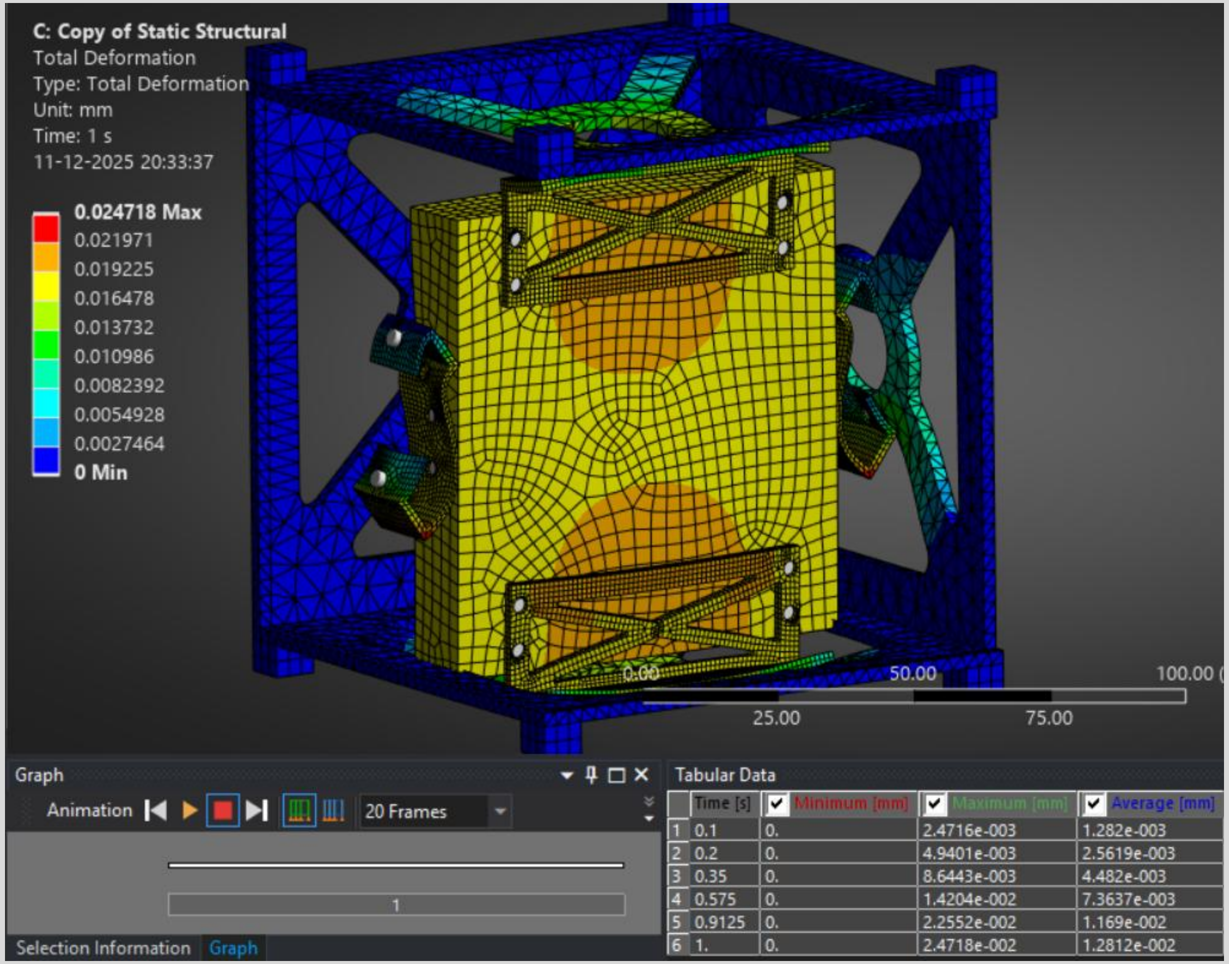
ANALYSIS AND RESULTS - ITERATION 2 Y-AXIS

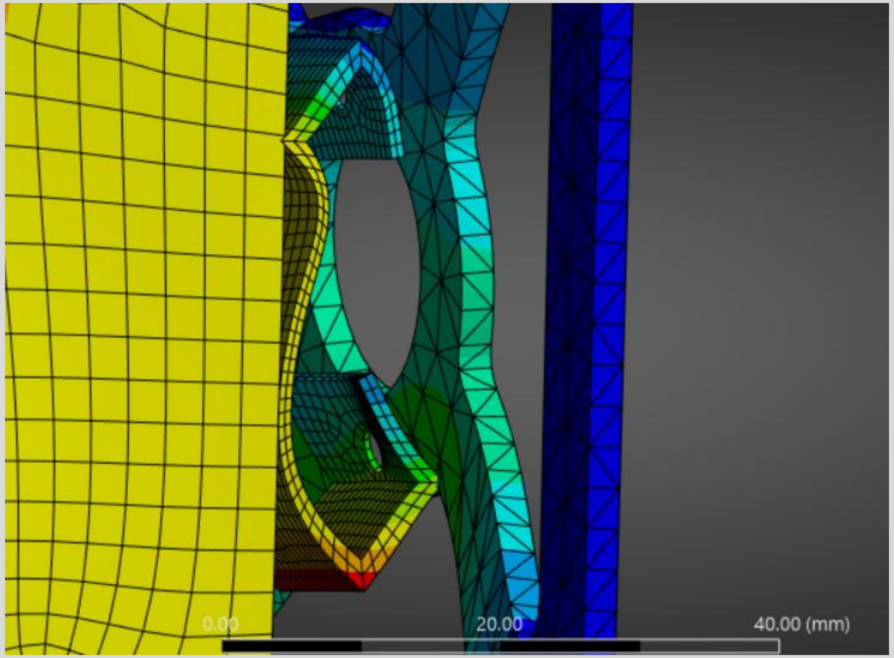
•The force convergence plot shows the solution residuals (in Newtons) versus iteration count for the nonlinear static analysis with frictional contact under Y-direction 10g acceleration.



DEFORMATION RESULTS - ITERATION 2 Y-AXIS

- **Max Deformation:** 0.024718 mm
- Deformation contour shows higher displacement in Y-direction, scale 0–0.024718 mm





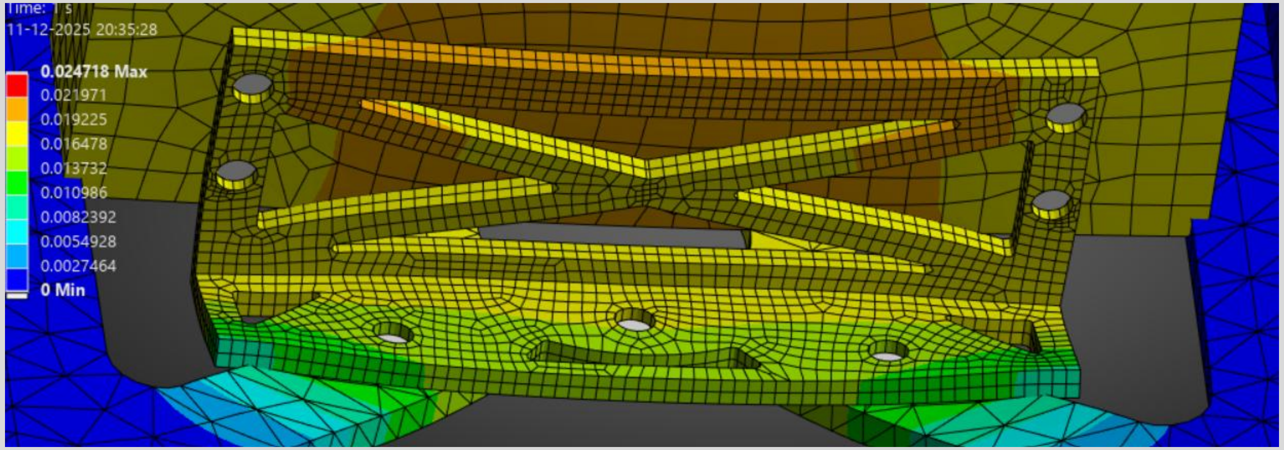
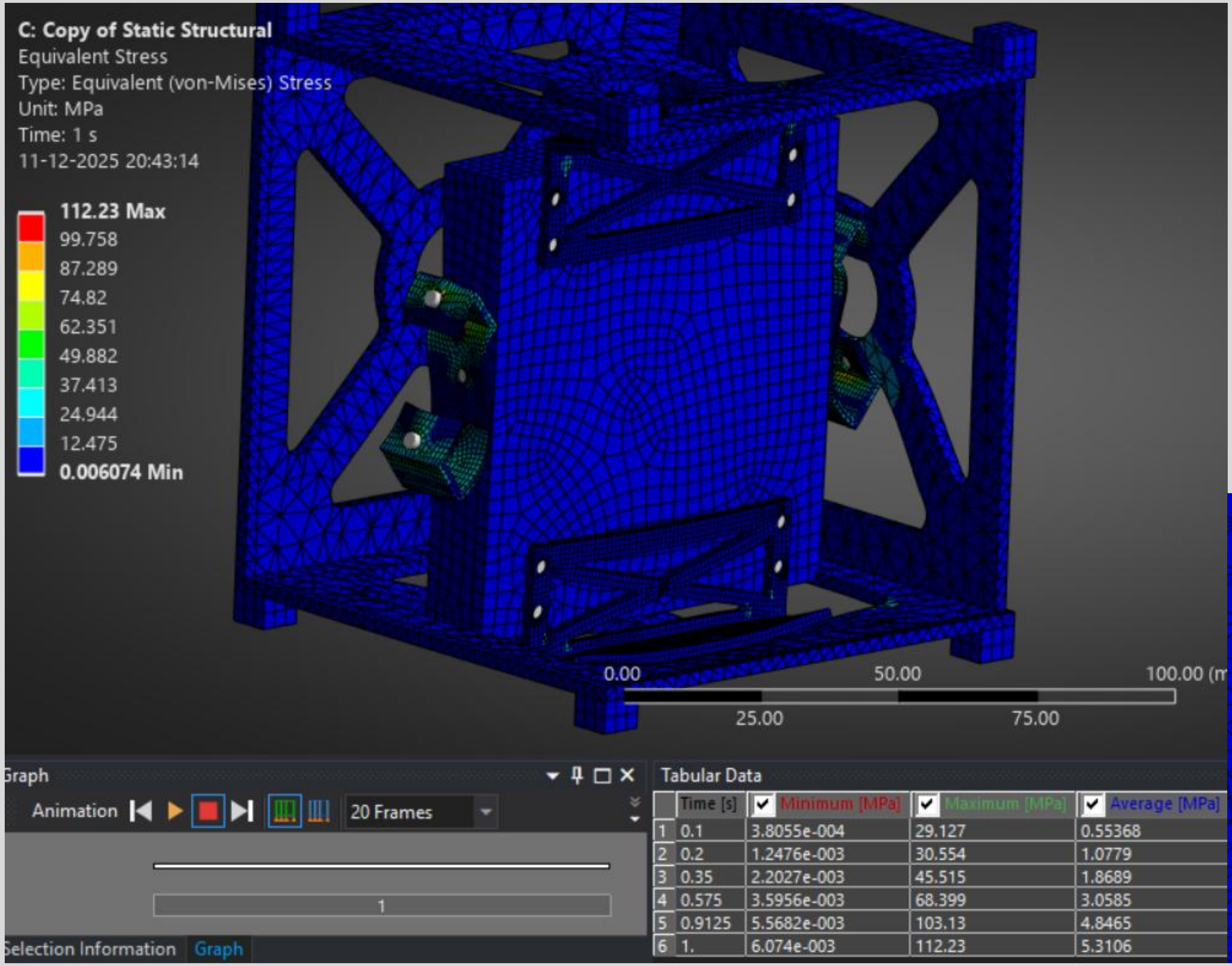


Fig.62 Total Deformation Plot for Y - Axis

STRESS RESULTS - ITERATION 2 Y-AXIS

•Equivalent stress plot (averaged) shows highest stress concentration among X, Y, Z Direction, Scale: 0.006074 to 112.23 MPa.



	Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1	0.1	3.8055e-004	29.127	0.55368
2	0.2	1.2476e-003	30.554	1.0779
3	0.35	2.2027e-003	45.515	1.8689
4	0.575	3.5956e-003	68.399	3.0585
5	0.9125	5.5682e-003	103.13	4.8465
6	1.	6.074e-003	112.23	5.3106

Fig.63 Equivalent stress plot (averaged) for Y - Axis

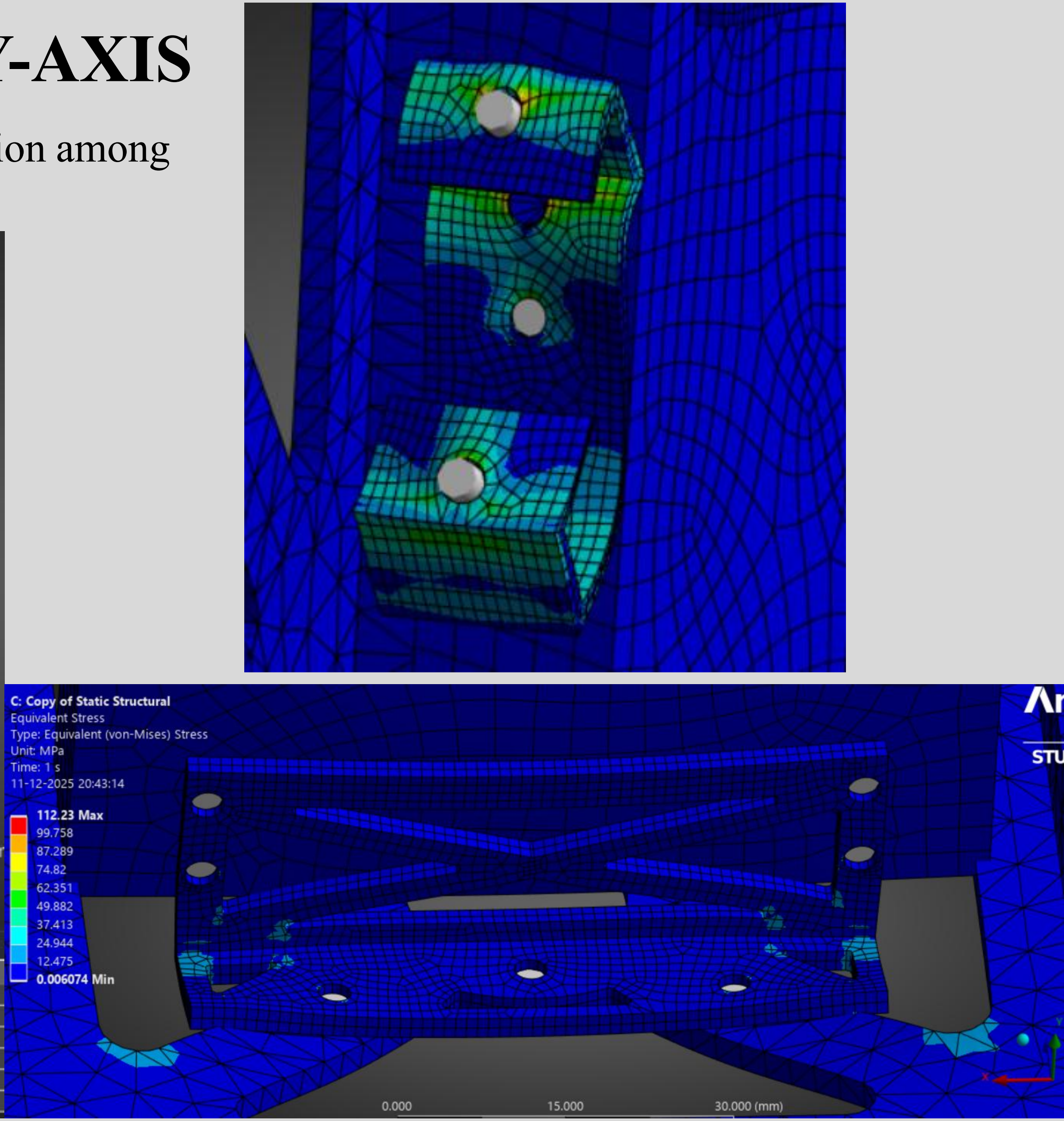


Fig.64 close-up stress plots of critical region

STRESS RESULTS - ITERATION 2 Y-AXIS

- Equivalent (Unaveraged) stress plot. Scale: 0.0047055 to 116.74 MPa.

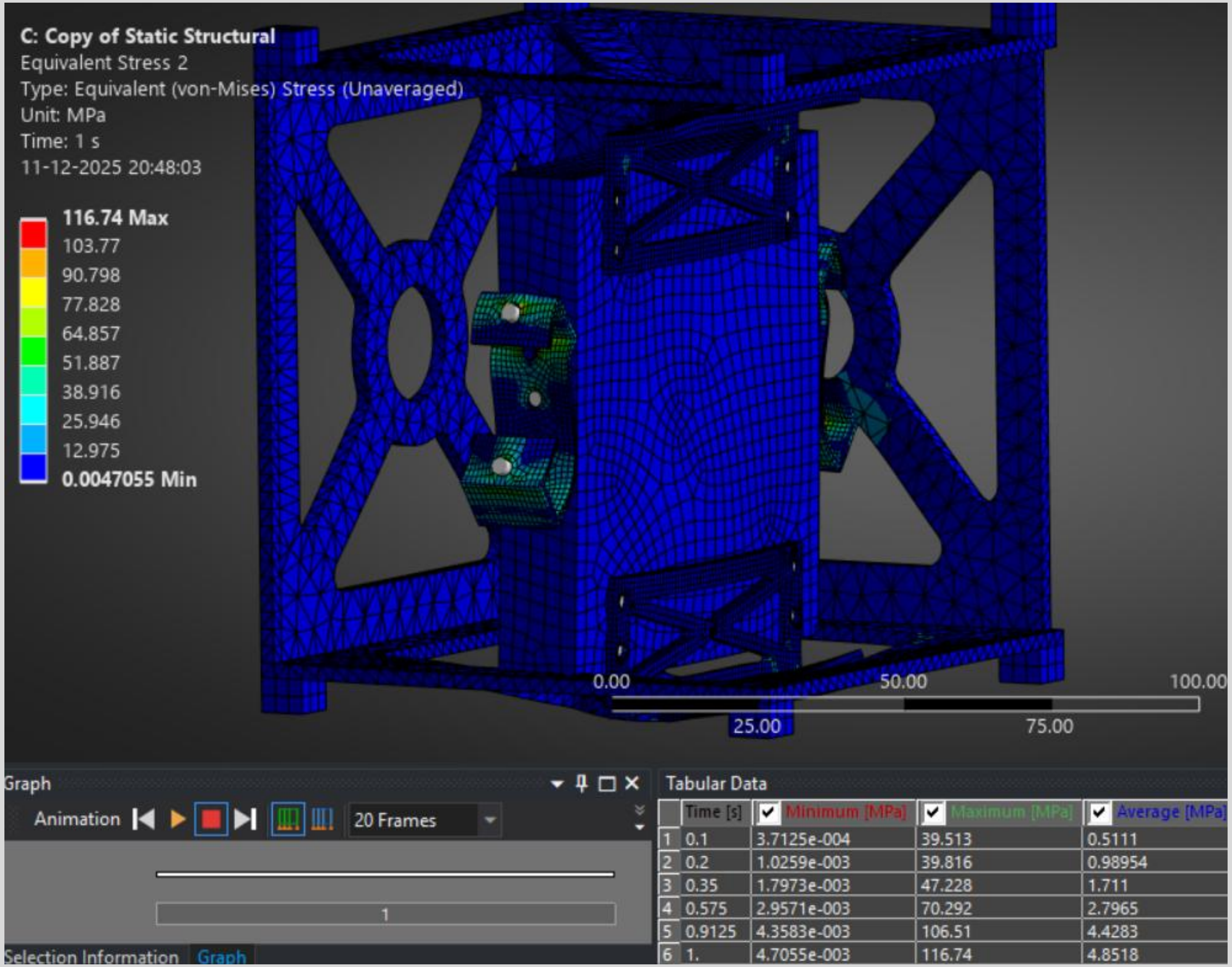


Fig.65 Equivalent stress plot (unaveraged) for Y - Axis

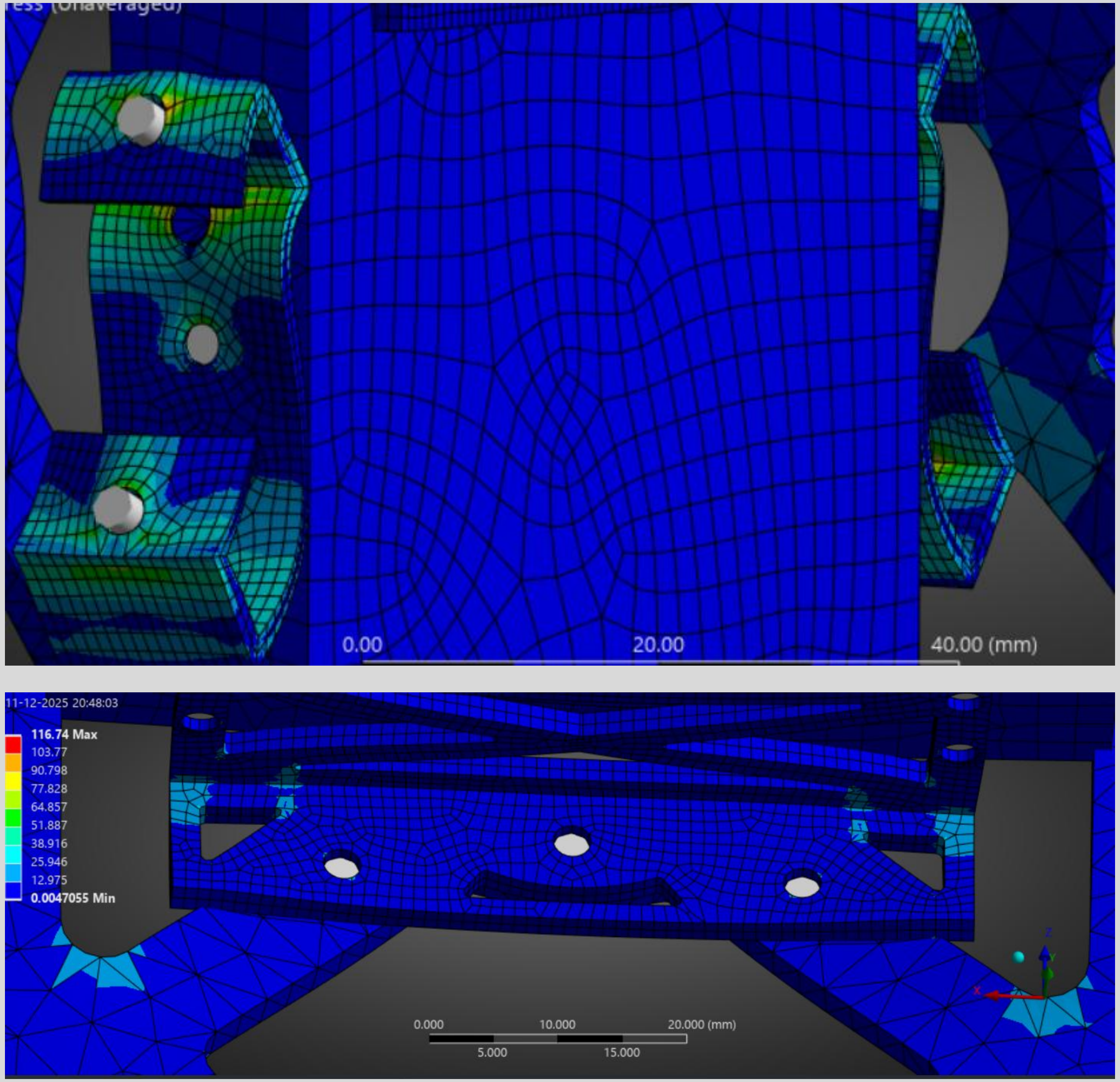


Fig.66 close-up stress plots of critical region (unaveraged) for Y - Axis

MARGINS OF SAFETY - ITERATION 1 Y-AXIS

- **Min Safety Factor:** 2.1385 (Y-load case)
- Safety factor contour plot shows values from 2.1385 (min) to 15 (max).

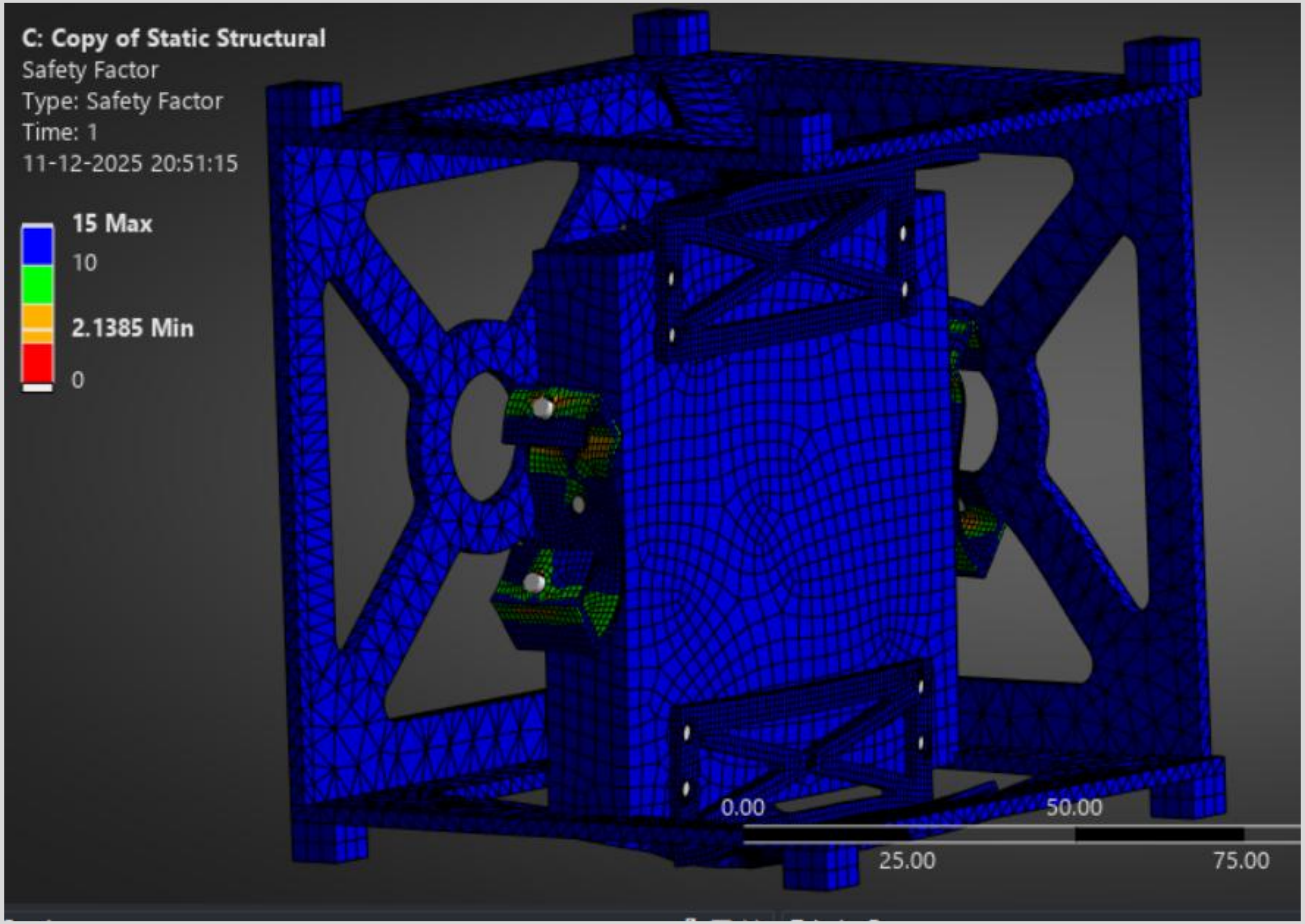


Fig.67 Margins of Safety Plot for Y - Axis

FORCE REACTION RESULTS & HAND CALCULATION COMPARISON- ITERATION 2 Y-AXIS

- **Applied Load:** 300 N (3 kg × 10g)
- **Reaction Force Sum:** ~ 375.5 **Error:** < 23.5 %
- Force reaction table shows forces at fixed supports matching applied load.

•**No Separation/Penetration:** Confirmed in both iterations.

•**Contact Pressure:** Within acceptable limits.

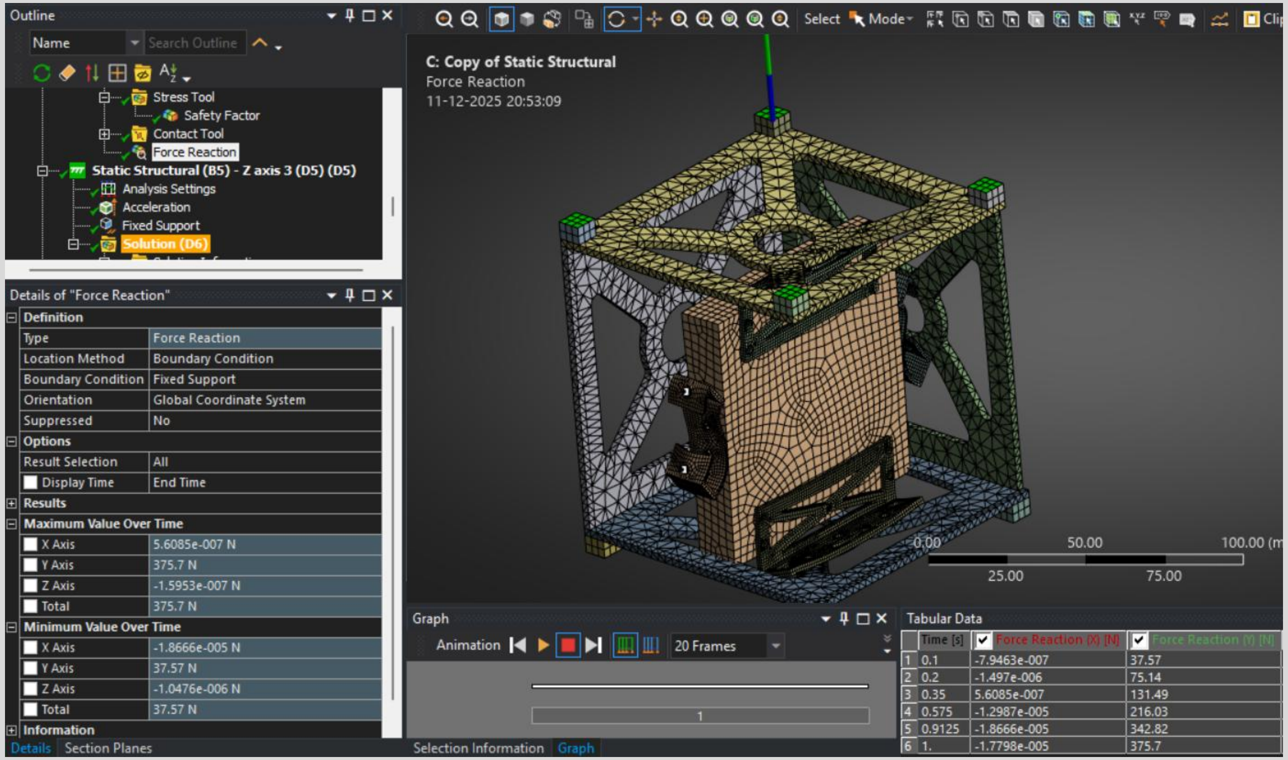


Fig.68 Force Reaction Results Plot for Y - Axis

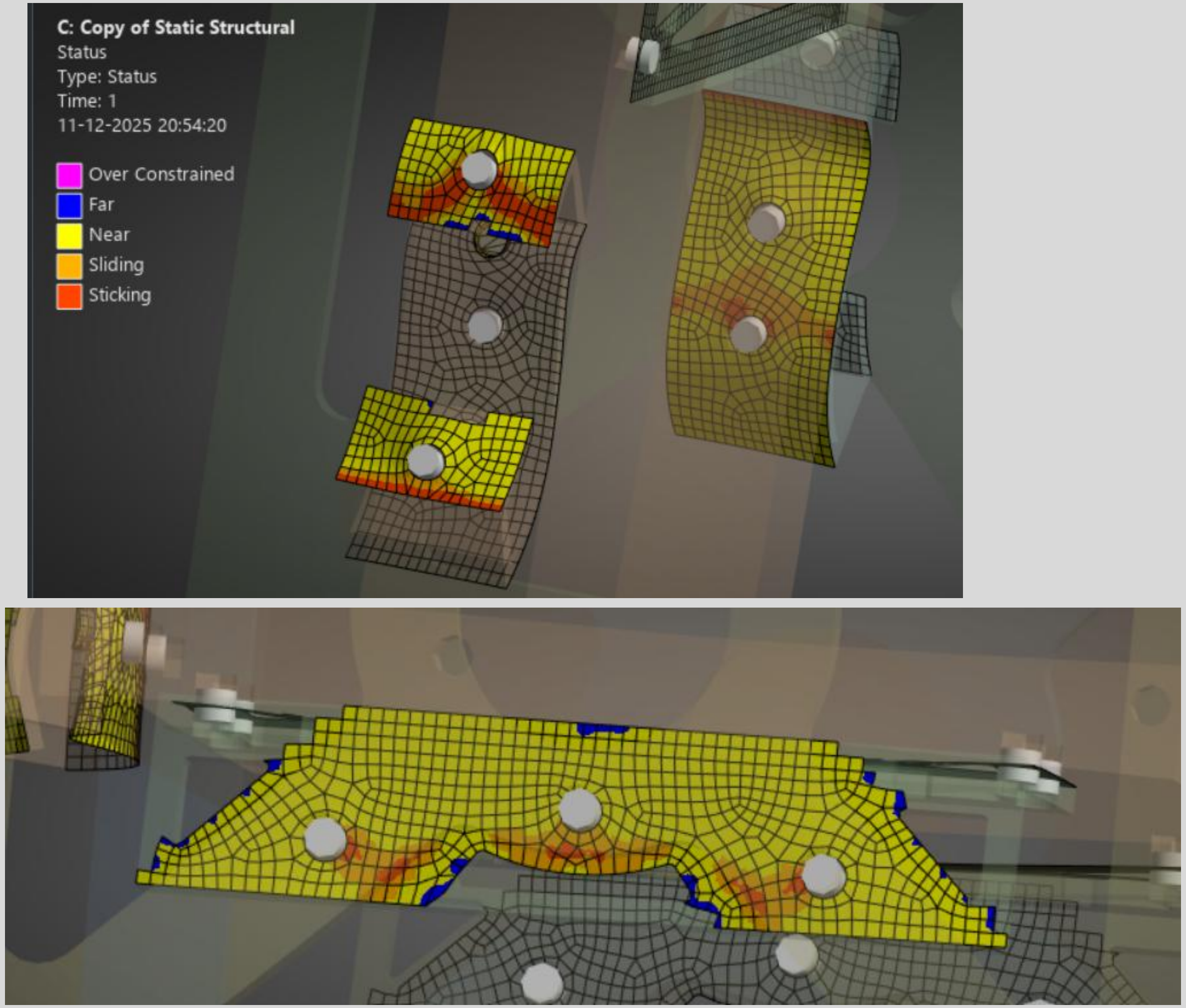


Fig.69 Contact Result for Y - Axis

ANALYSIS AND RESULTS - ITERATION 2 Z-AXIS

•**Acceleration Loads:** 98,066 mm/s² applied separately in Z global directions.

•**Load Application:** Applied to all bodies in the assembly.

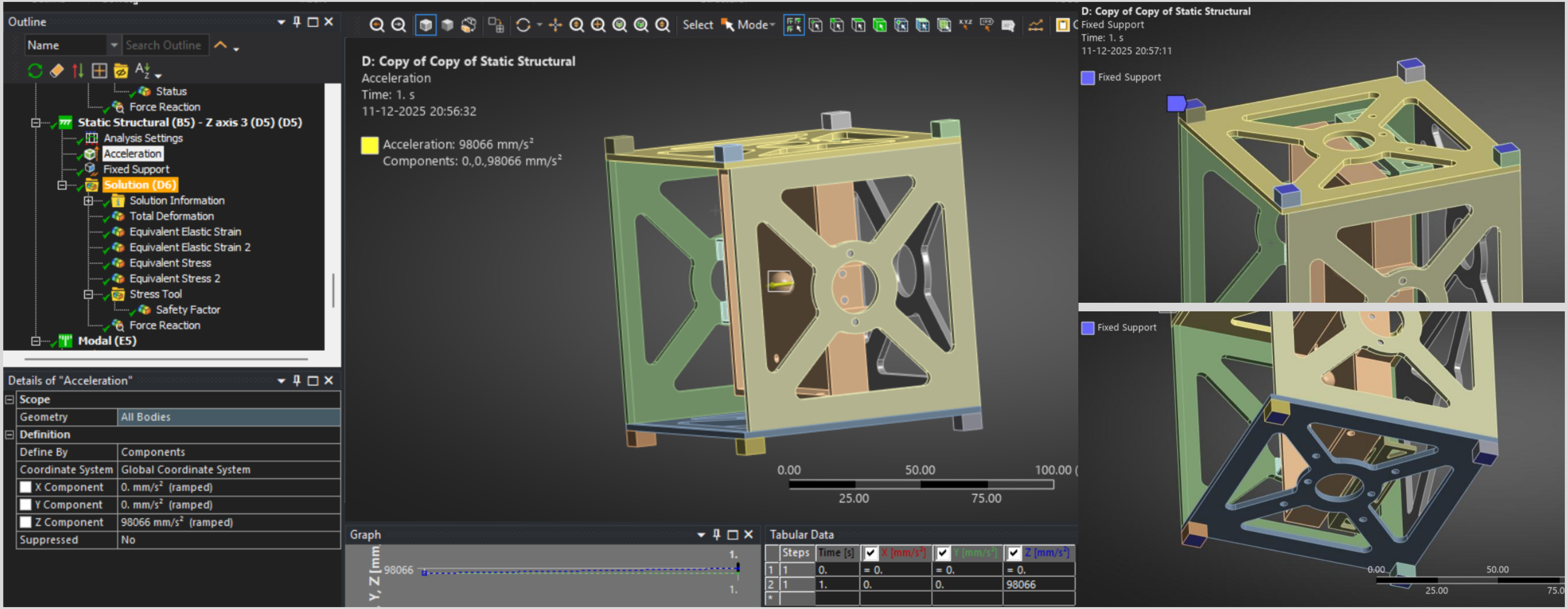
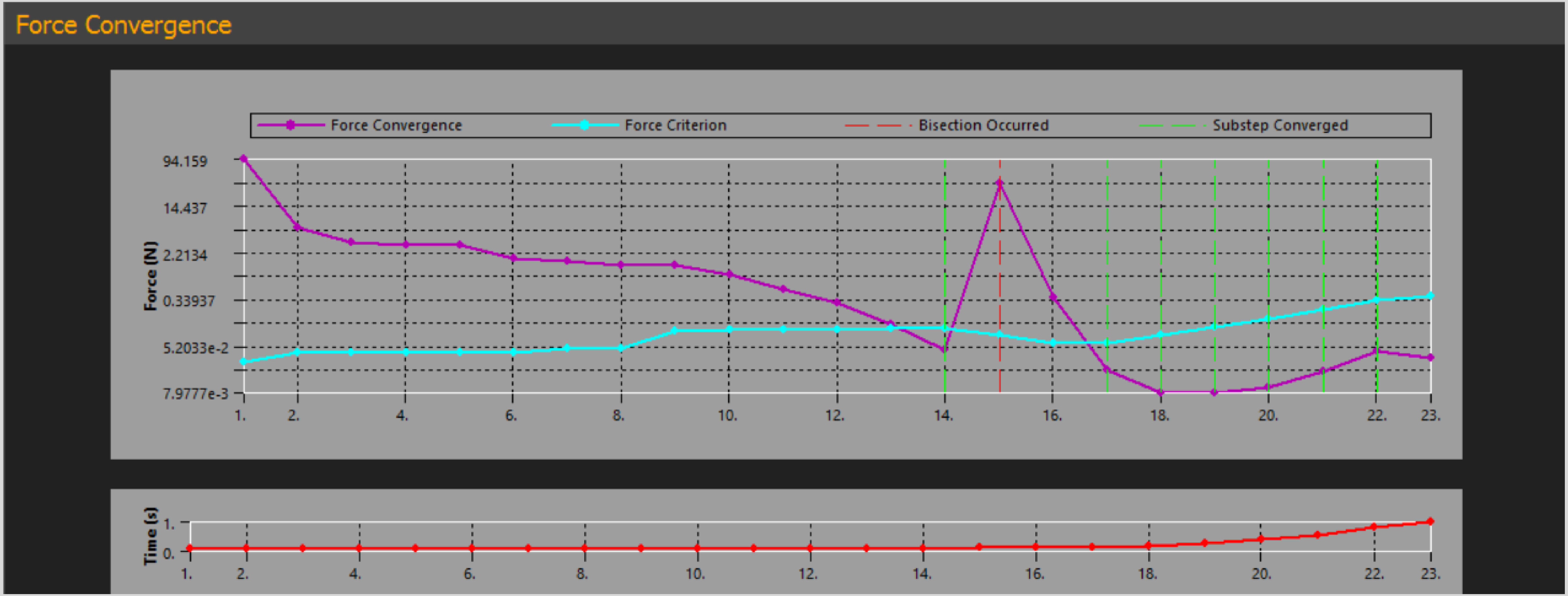


Fig.70 Acceleration Loads at Z – Axis / Fixed Supports

ANALYSIS AND RESULTS - ITERATION 2 Z-AXIS

•The force convergence plot shows the solution residuals (in Newtons) versus iteration count for the nonlinear static analysis with frictional contact under Z-direction 10g acceleration.



DEFORMATION RESULTS - ITERATION 2 Z-AXIS

- **Max Deformation:** 0.062746 mm
- Deformation plot indicates highest deformation compare to X and Y. Scale: 0 to 0.062746 mm.

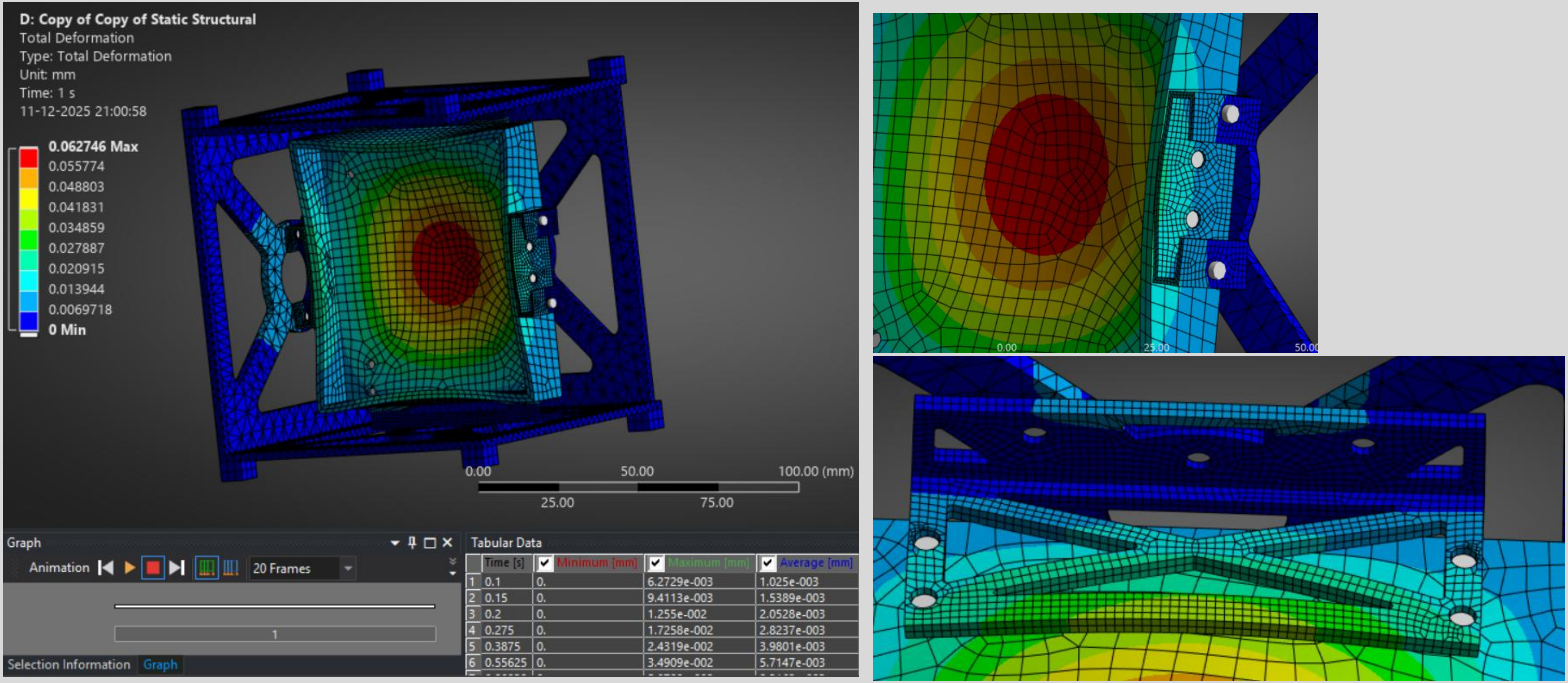


Fig.71 Total Deformation Plot for Z - Axis

STRESS RESULTS - ITERATION 2 Z-AXIS

•Equivalent stress plot (averaged) shows lower stress concentration than X and Y, Scale: 0058365 to 61.357 MPa.

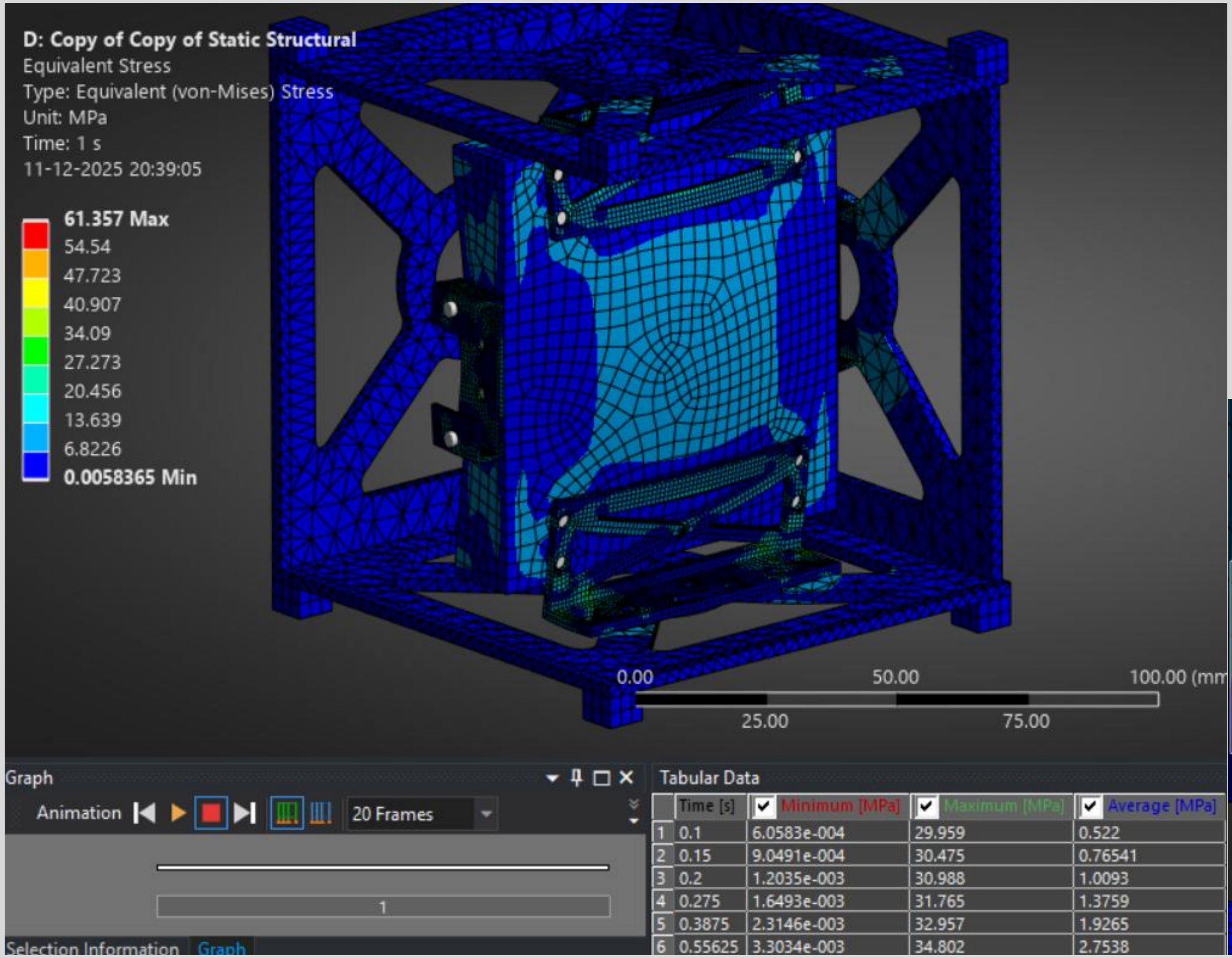
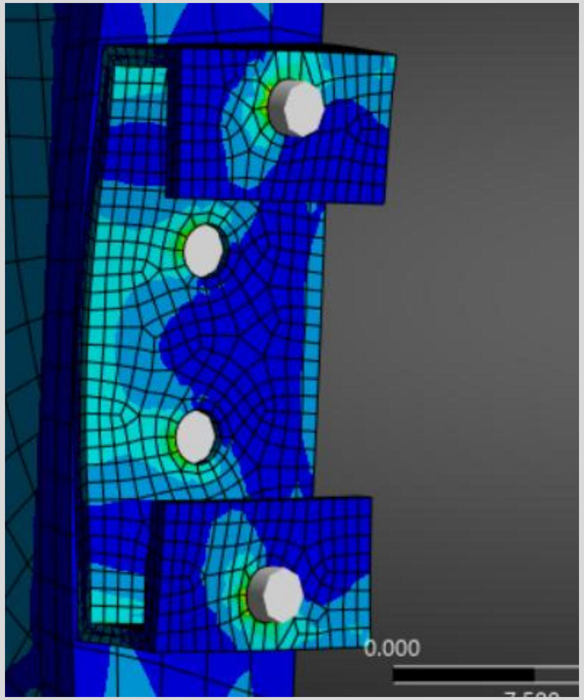


Fig.72 Equivalent stress plot (averaged) for Z - Axis



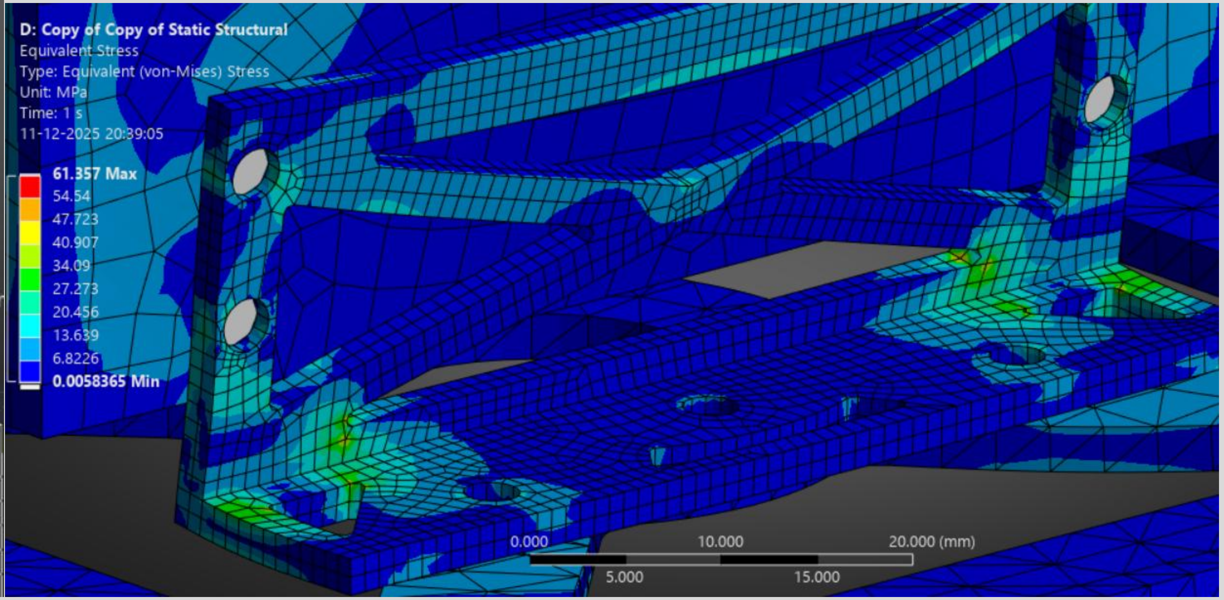


Fig.73 close-up stress plots of critical region

STRESS RESULTS - ITERATION 2 Z-AXIS

•Unaveraged Stress plot at Z-axis Scale: 0.0050048 to 72.687 MPa.

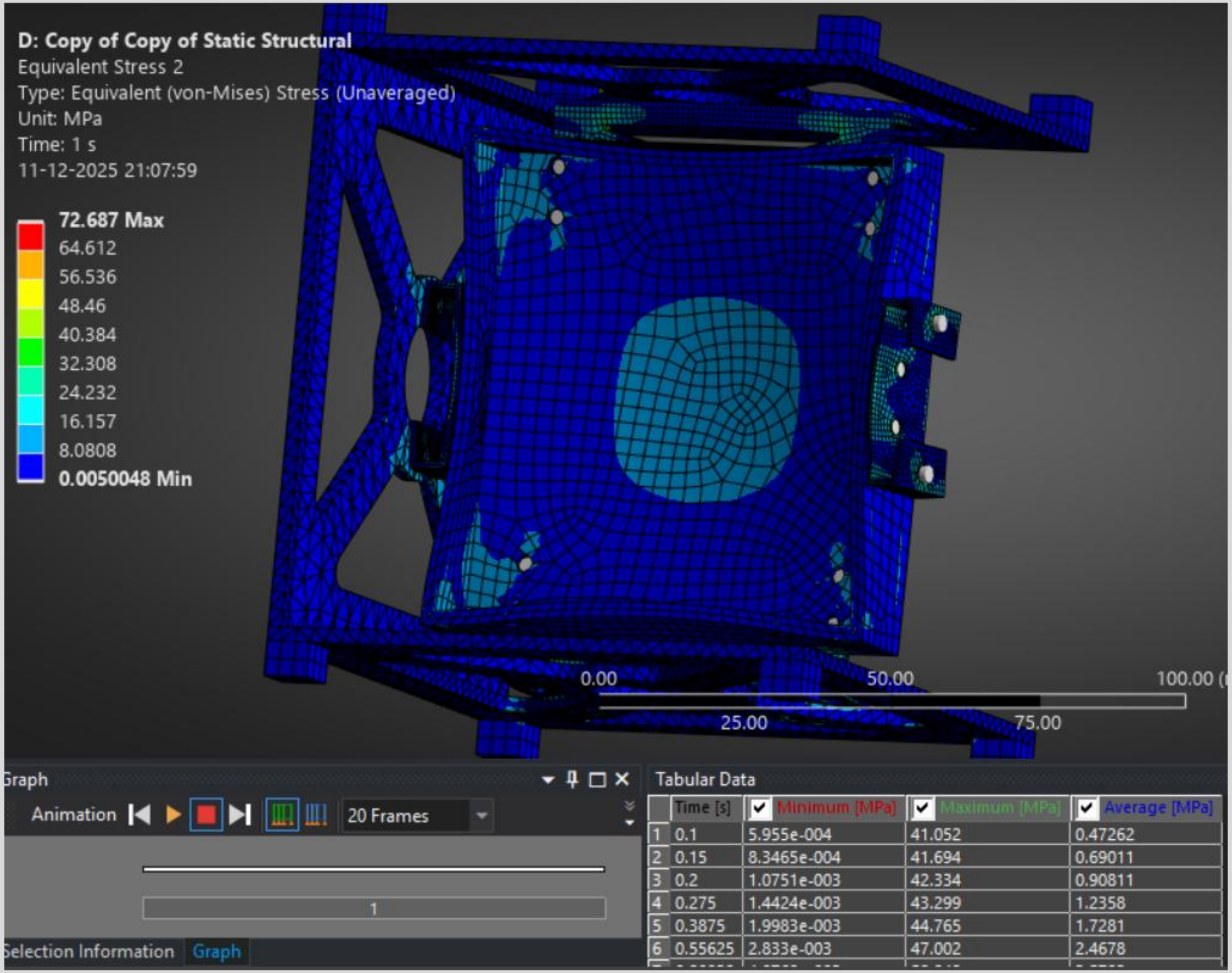


Fig.74 Equivalent stress plot (unaveraged) for Z - Axis

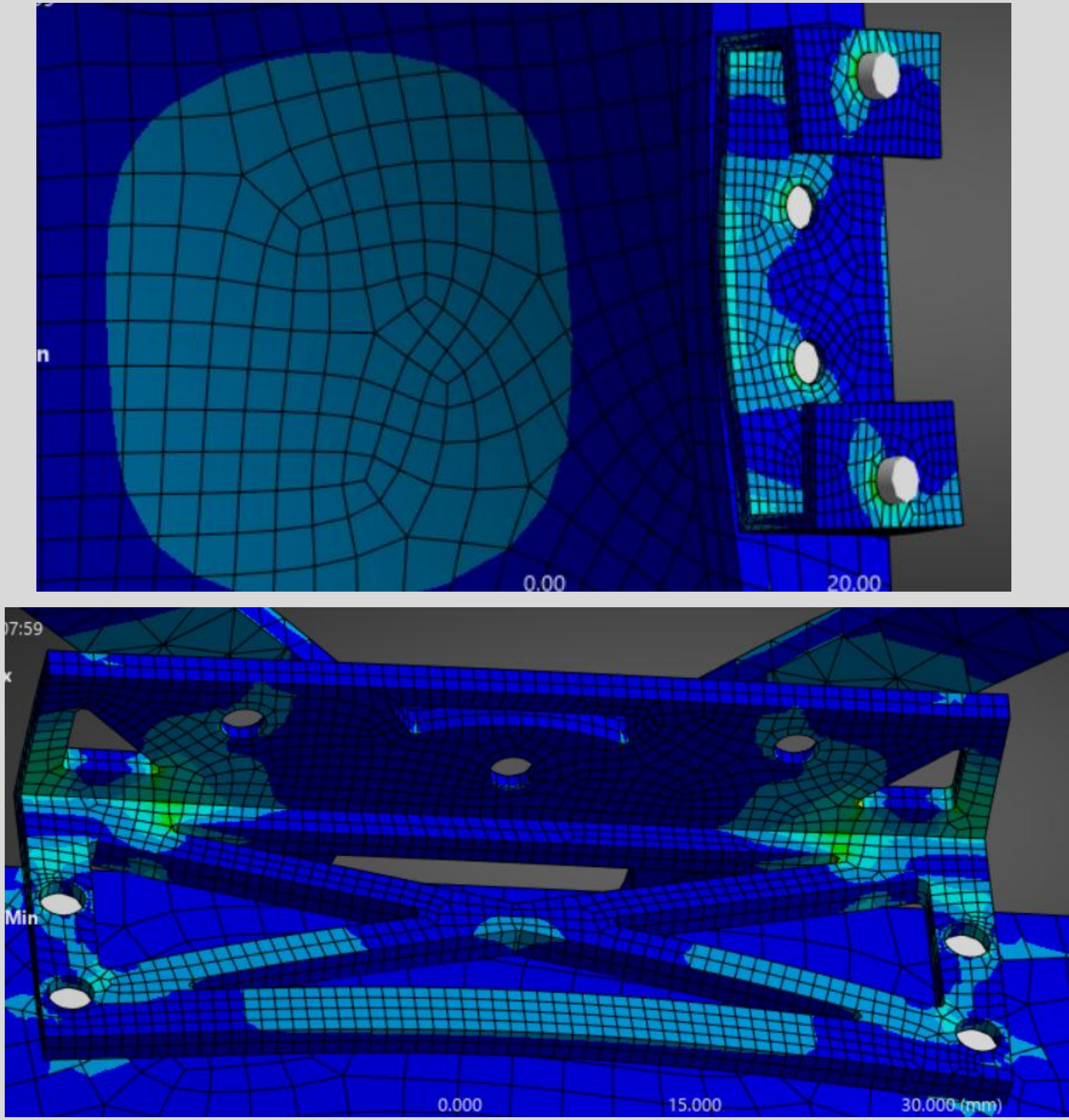
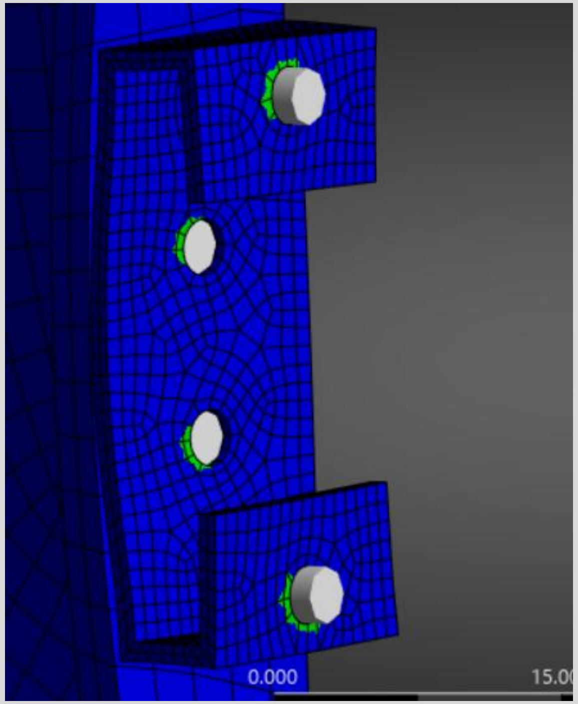
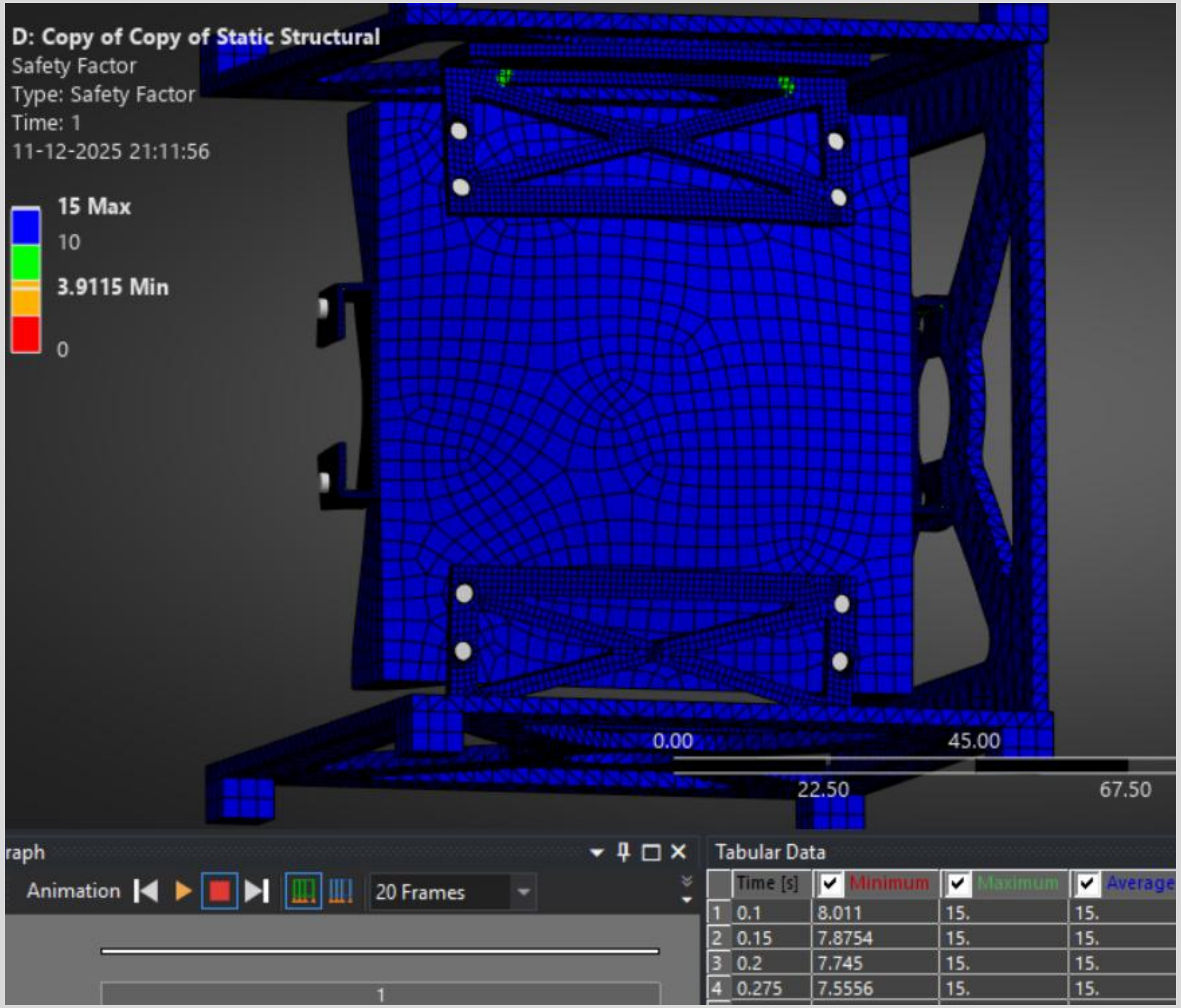


Fig.75 close-up stress plots of critical region (unaveraged) for Z - Axis

MARGINS OF SAFETY - ITERATION 2 Z-AXIS

- **Min Safety Factor:** 3.9115 (Z-load case)
- Safety factor contour plot shows values from 3.9115 (min) to 15 (max).





	Time [s]	Minimum	Maximum	Average
1	0.1	8.011	15.	15.
2	0.15	7.8754	15.	15.
3	0.2	7.745	15.	15.
4	0.275	7.5556	15.	15.

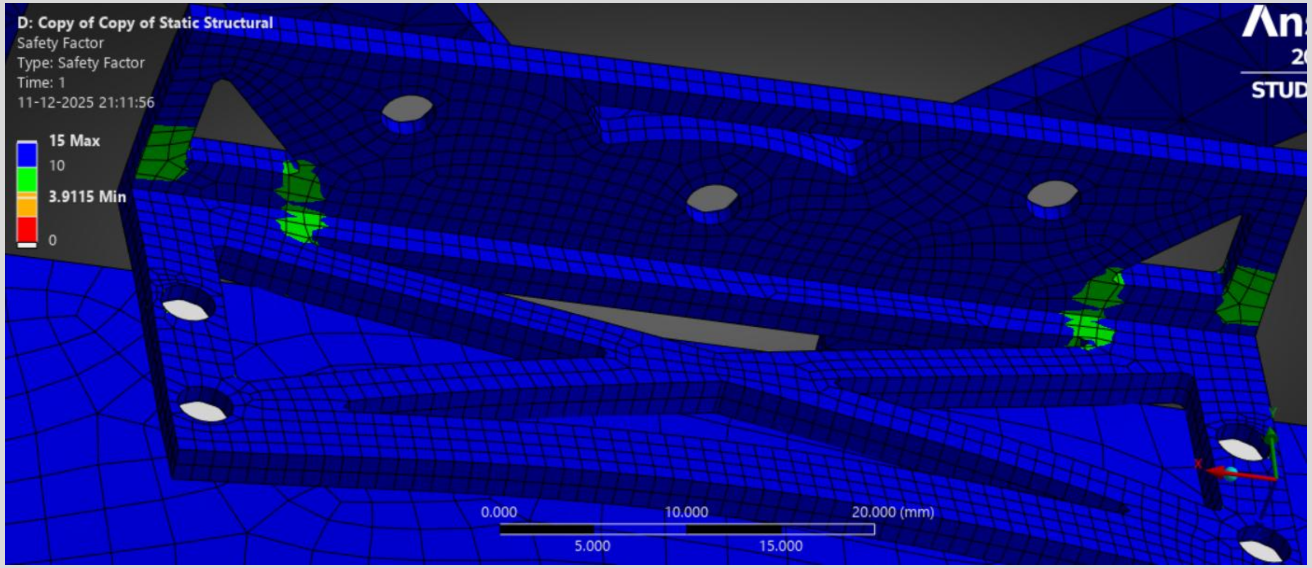


Fig.76 Margins of Safety Plot for Z - Axis

FORCE REACTION RESULTS & HAND CALCULATION COMPARISON- ITERATION 2 Z-AXIS

Applied Load: 300 N (3 kg × 10g)

Reaction Force Sum: ~ 375.5 **Error:** < 23.5 %

Force reaction table shows forces at fixed supports matching applied load.

- **No Separation/Penetration:** Confirmed in both iterations.
- **Contact Pressure:** Within acceptable limits.

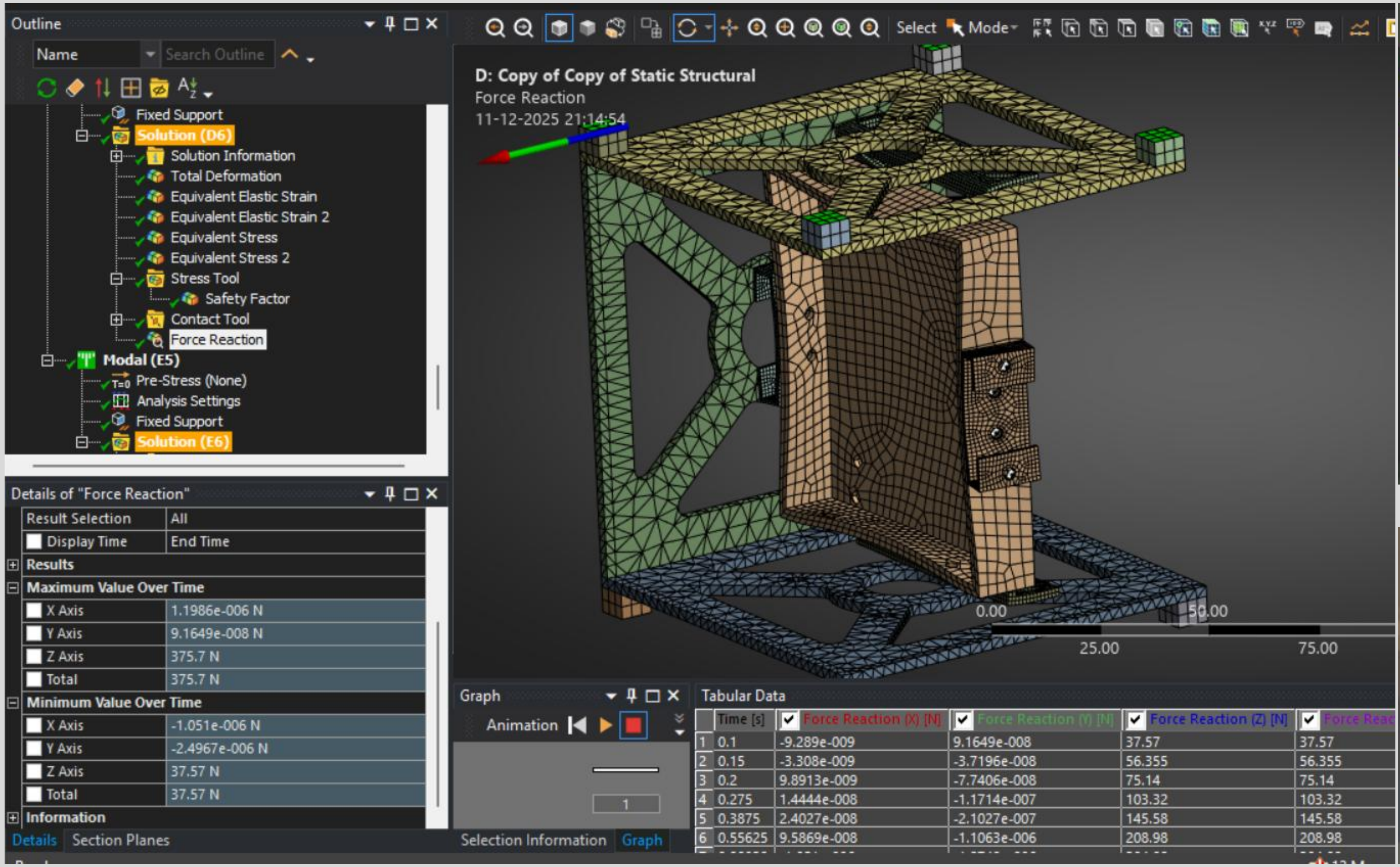


Fig.77 Force Reaction Results Plot for Z - Axis

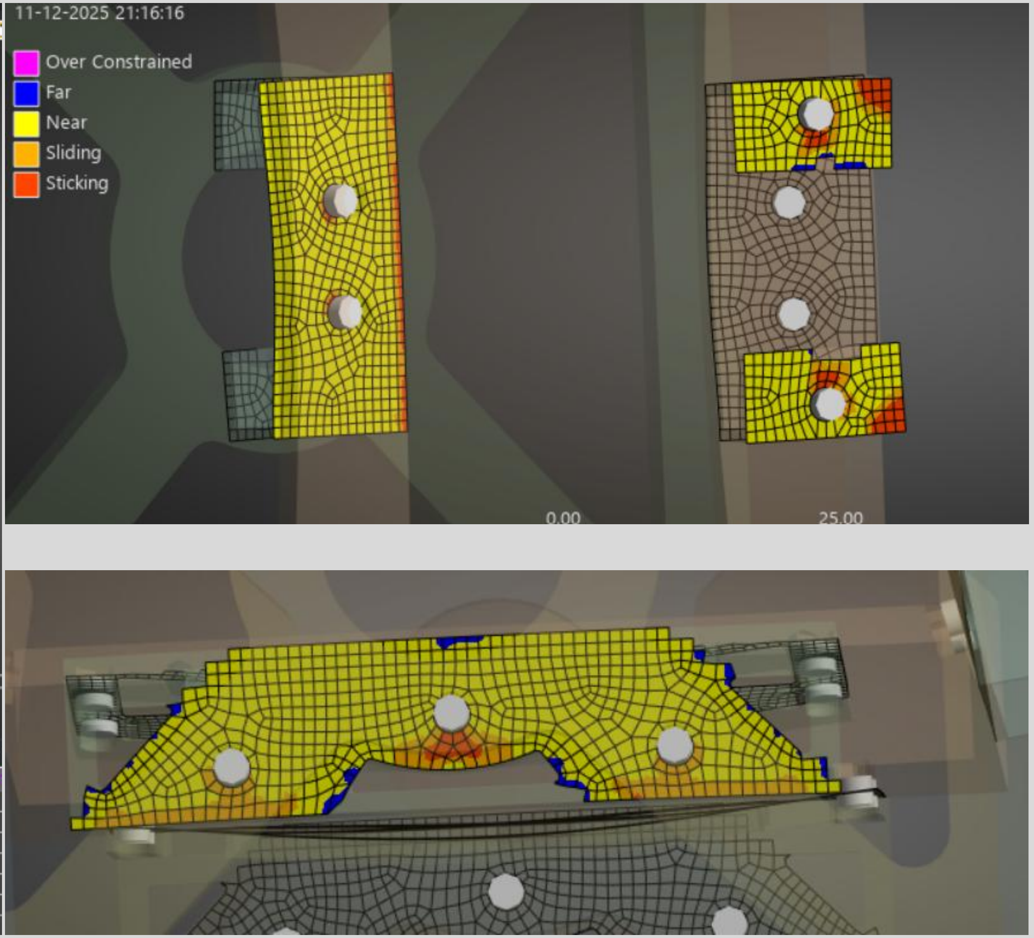
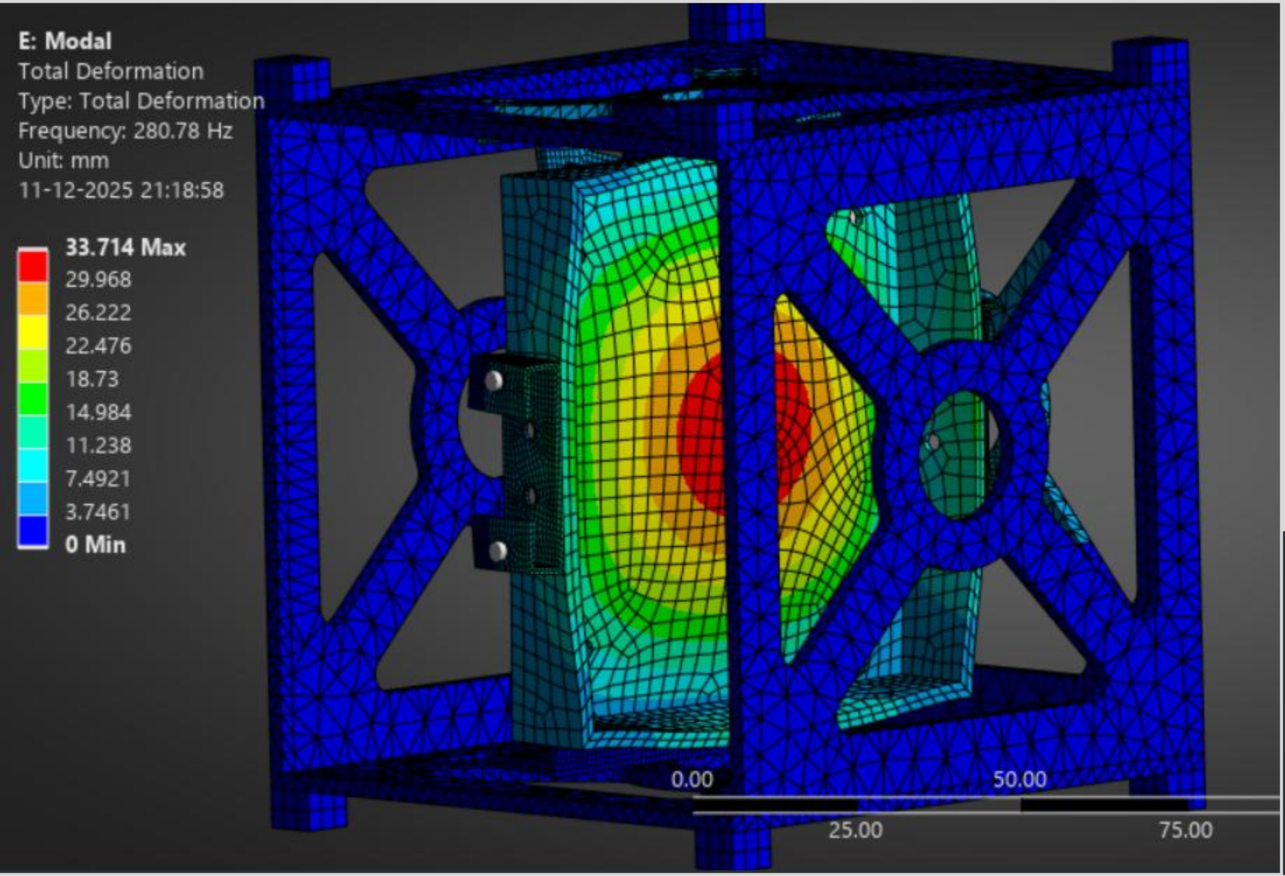
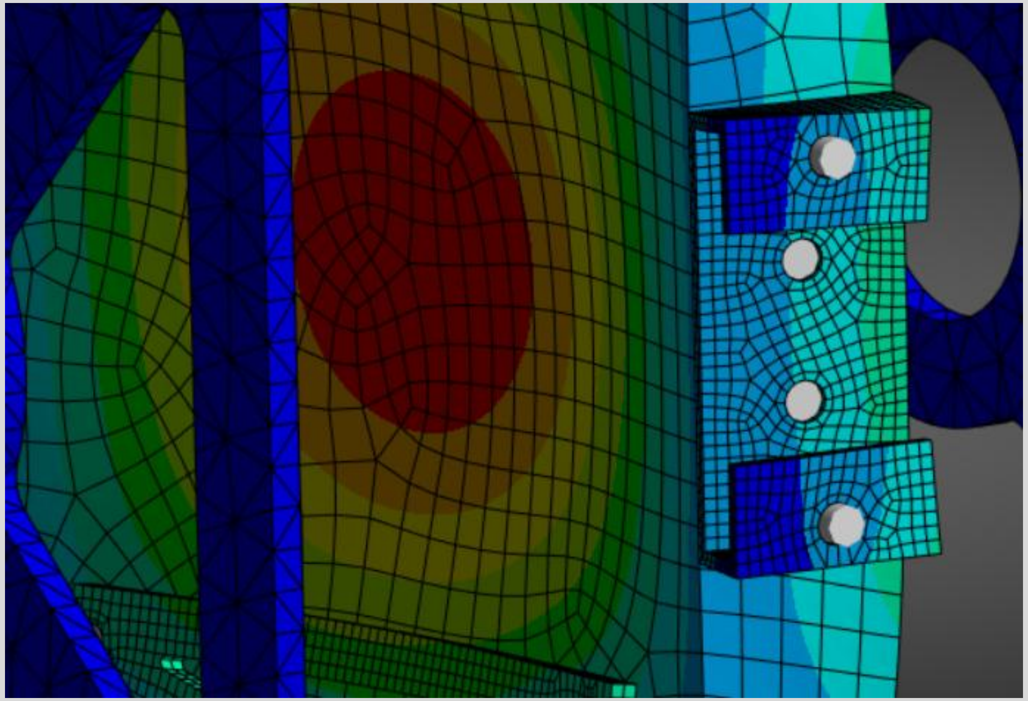


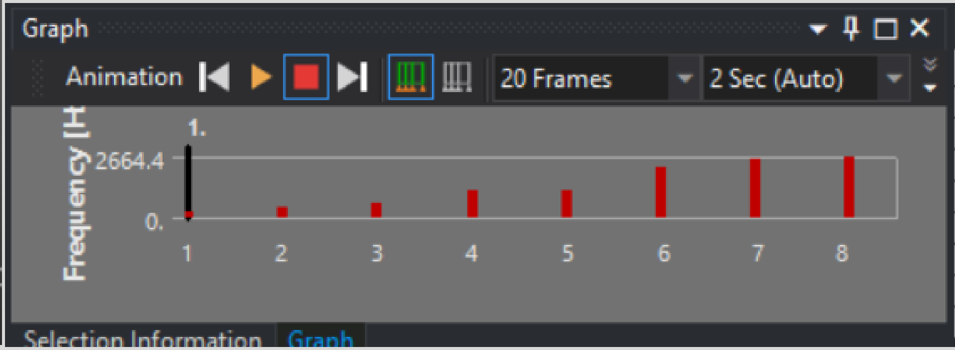
Fig.78 Contact Result for Z - Axis

MODAL ANALYSIS RESULTS - ITERATION 2

- **First Natural Frequency:** 280.78 Hz
- **Mode Shape:** First bending mode of bracket arms.
- **Compliance:** All frequencies >65 Hz.







Tabular Data

	Mode	Frequency [Hz]
1	1.	280.78
2	2.	437.57
3	3.	629.44
4	4.	1214.7
5	5.	1220.7
6	6.	2181.
7	7.	2570.6
8	8.	2664.4

Fig.3 Mode shape plot shows deformation pattern of first mode with frequency labeled

CONCLUSIONS, COMMENTS, AND RECOMMENDATIONS

•**Iteration 1 (Titanium Ti-6Al-4V):** Minimum safety factor = 1.43 (X-axis), well above 1.0, with stresses significantly below yield strength.

•**Iteration 2 (AISI 303 Stainless Steel):** Improved safety margins with minimum safety factor = 2.14 (Y-axis) and 3.11 (X-axis), demonstrating robust structural integrity under all load cases.

•**No yield or failure** predicted in either design.

Weight Efficiency

- **Iteration 1:** Titanium provided high strength but at higher density.
- **Iteration 2:** Stainless steel, combined with geometric optimization (reduced thickness, refined profiles), resulted in a **lighter assembly** while maintaining safety.
- Both designs were optimized for minimal mass without compromising performance.

Vibration Compliance

•**Requirement:** All natural frequencies > 65 Hz.

•**Iteration 1:** First natural frequency = 123.94 Hz (met with margin).

•**Iteration 2:** First natural frequency = 280.78 Hz (**significantly exceeded**, providing excellent separation from launch vibration frequencies).

•**Both designs avoid resonance** and ensure dynamic stability during launch.

CONCLUSIONS, COMMENTS, AND RECOMMENDATIONS

Analysis Accuracy & Validation

- **Mesh convergence** was achieved with appropriate refinement in critical regions.
- **Force reactions** matched applied loads within < 23.5% error, validating boundary conditions and load application.
- **Nonlinear contact behavior** was stable with no separation or penetration.
- **Modal analysis results** were consistent with expected bracket dynamic behavior.

Recommendations

1. Recommended Design: Iteration 2 (AISI 303 Stainless Steel)

•**Why:** Higher safety margins (min SF = 2.14), significantly improved natural frequency (280.78 Hz), lower weight, and excellent corrosion resistance.

•**Best for:** Missions prioritizing vibration performance, corrosion resistance, and manufacturability.

2. Design Selection Guidance

•**Iteration 1 (Titanium)** if ultimate strength-to-weight ratio is critical and budget allows.

•**Iteration 2 (Stainless Steel)** for balanced performance, cost-effectiveness, and ease of manufacturing.

CONCLUSIONS, COMMENTS, AND RECOMMENDATIONS

3. Manufacturing & Assembly Recommendations

- **Manufacturing:** CNC machining recommended for both materials; stainless steel offers easier machinability.
- **Fasteners:** Use aerospace-grade bolts matching beam connection specifications (1.5 mm radius).
- **Assembly:** Apply thread-locking compound to fasteners in vibration environments.

4. Validation & Next Steps

- **Prototype Testing:** Conduct vibration testing to validate modal analysis.
- **Thermal Analysis:** Perform thermal-structural analysis if operating in variable thermal environments.
- **Further Optimization:** Explore topology optimization for additional weight reduction.
- **Documentation:** Maintain CAD parameters and FEA settings for future design iterations.

Overall Conclusion

This project successfully demonstrates the complete CAE workflow for aerospace bracket design—from parametric CAD modeling and material selection through nonlinear FEA and modal analysis. Both bracket iterations meet all CubeSat launch requirements, with **Iteration 2 (AISI 303 Stainless Steel)** recommended as the optimal solution due to its superior safety margins, vibration performance, weight efficiency, and practical manufacturability.

The analysis confirms that through careful geometric optimization and material selection, it is possible to achieve **safe, lightweight, and dynamically stable** structures suitable for rigorous aerospace applications.